Creating One-Click Installers for Open Object Rexx and the Bean Scripting Framework for Open Object Rexx for Microsoft Windows, Linux and Apple Mac OS X

Erstellung von One-Click Installern für Open Object Rexx und das Bean Scripting Framework für Open Object Rexx für Microsoft Windows, Linux und Apple Mac OS X

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Contents

1 Introduction **7**

1.1 ooRexx . . . 7

1.2 BSF4ooRexx . . . 7

1.3 Installers . . . 8

1.3.1 Advantages . . . 9

1.3.2 Disadvantages . . . 9

1.3.3 Research Question . . . 9

2 NSIS Installer for Windows **10**

2.1 The Installer . . . 10

2.2 Creating the Install Package for Windows with NSIS . . . 11

2.2.1 General Introduction . . . 11

2.2.2 BSF4ooRexx NSIS Script . . . 13

2.2.3 BSF4ooRexx Reinstaller NSIS Script . . . 22

2.2.4 Compile the package . . . 23

2.3 Use Cases . . . 23

2.3.1 Install . . . 24

2.3.2 Reinstall . . . 26

2.3.3 Uninstall . . . 28

3 DEB Package for Debian-based Linux Distributions **30**

3.1 The DEB Package System . . . 30

3.2 Creating the DEB Package for Debian-based Linux Distributions . . . 30

3.2.1 Install the required software . . . 30

3.2.2 Root folder . . . 31

3.2.3 Meta data . . . 32

3.2.4 File system structure . . . 34

3.2.5 Install scripts . . . 35
3.2.6 Building the package . . . 36
3.3 Use Cases . . . 37
3.3.1 Install . . . 37
3.3.2 Reinstall . . . 40
3.3.3 Uninstall . . . 42

4 RPM Package for Red Hat-based Linux Distributions 45
4.1 The RPM Package Management System . . . 45
4.2 Creating the RPM Package for Red Hat-based Linux Distributions . . . 46
4.2.1 Installing the required programs . . . 46
4.2.2 Creating the required directory structure . . . 47
4.2.3 Creating the Specifications (spec) File . . . 49
4.2.4 Building the Final RPM Package . . . 53
4.3 Use Cases . . . 54
4.3.1 Install . . . 54
4.3.2 Reinstall . . . 56
4.3.3 Uninstall . . . 57

5 Iceberg Installer for Mac OS X 60
5.1 Preliminary Works . . . 60
5.1.1 The Source Folder's Structure . . . 60
5.1.2 The Install Script . . . 64
5.1.3 The Uninstall and Reinstall Scripts . . . 65
5.2 The Installer . . . 66
5.3 Creating the Install Package for Mac OS X with Iceberg . . . 66
5.3.1 Settings . . . 68
5.3.2 Documents . . . 70
5.3.3 Scripts . . . 71

5.3.4 Plugins . . . 73
5.3.5 Files . . . 74
5.3.6 Building the Package . . . 76
5.4 Use Cases . . . 76
5.4.1 Install . . . 76
5.4.2 Reinstall . . . 79
5.4.3 Uninstall . . . 80

6 Roundup and Outlook 82

7 Appendix 83
7.1 Used Programs . . . 83
7.1.1 Windows . . . 83
7.1.2 Debian Linux . . . 83
7.1.3 Red Hat Linux . . . 84
7.1.4 Mac OS X . . . 84
7.2 Windows Source Codes . . . 85
7.2.1 The NSIS Installation Script for 32 bit Windows . . . 85
7.2.2 The NSIS Installation Script for 64 bit Windows . . . 100
7.2.3 The NSIS Reinstall Script for 32 bit Windows . . . 115
7.2.4 The NSIS Reinstall Script for 64 bit Windows . . . 130
7.2.5 Java Version Check Script . . . 144
7.2.6 ooRexx Bitness Check Script . . . 145
7.2.7 Get BSF Version Script . . . 146
7.2.8 cmd File for Automatically Creating the Windows Installers . . . 147
7.3 Debian Linux Source Codes . . . 148
7.3.1 Debian Package `control` File . . . 148
7.3.2 Debian Package `preinst` Script . . . 149
7.3.3 Debian Package `postinst` Script . . . 150

7.3.4 Debian Package `prerm` Script 151
7.3.5 Shell Script for Automatically Creating the Debian Package 152
7.4 Red Hat Linux Source Codes 154
7.4.1 `SPECS` File for Red Hat based Linux Distributions 154
7.4.2 Shell Script for Automatically Creating the RPM Package 156
7.5 Mac OS X Source Codes 158
7.5.1 The Mac OS X Menu Bar 158
7.5.2 The Mac OS X Installer's 32 bit Install Script 163
7.5.3 The Mac OS X Installer's 64 bit Install Script 168
7.5.4 The Mac OS X Installer's Uninstall Script 173
7.5.5 The Mac OS X Installer's 32 bit BSF4ooRexx Uninstall Script 174
7.5.6 The Mac OS X Installer's 64 bit BSF4ooRexx Uninstall Script 178
7.5.7 The Mac OS X Installer's ooRexx Uninstall Script 182
7.5.8 The Mac OS X Installer's Reinstall Script 184
7.5.9 Makefile for Automatically Creating the Mac OS X Packages 186
7.6 List of Figures 189
7.7 List of Tables 191
7.8 List of Listings 192
7.9 References 195

8 License **198**

Abstract

The main purpose of this thesis is to create “one-click installers” for BSF4ooRexx for the operating systems Microsoft Windows, Linux, and Apple Mac OS X. For each operating system an appropriate installer is chosen, with which the corresponding installation package is created. This package has to have the ability to install, reinstall and uninstall. Each installation package is also shown in action. Finally, an outlook for future improvements and open issues is given.

Kurzfassung

Der Zweck dieser Arbeit ist die Erstellung von “One-Click Installern” für BSF4ooRexx für die Betriebssysteme Microsoft Windows, Linux und Apple Mac OS X. Dabei wird für jedes Betriebssystem ein geeigneter Installer ausgewählt, mit dessen Hilfe das jeweilige Installationspaket, mit der Fähigkeit zur Installation, Reinstallation und Deinstallation, erstellt wird. Jedes Installationspaket ist dabei auch in Aktion zu sehen. Abschließend wird ein Ausblick auf zukünftige Verbesserungsmöglichkeiten und offene Punkte gegeben.

1 Introduction

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The following chapters first give the reader a short introduction to the world of ooRexx and BSF4ooRexx and will later discuss the advantages and disadvantages of installers. Chapters 2, 3, 4 and 5 will take a closer look at the creation of installers for Microsoft Windows, Debian-based Linux, Red-Hat-based Linux and Apple Mac OS X. For understanding reasons the Mac OS X chapter requires the knowledge of [1].

1.1 ooRexx

"The programming language Open Object Rexx (ooRexx) is an object-oriented scripting language that was developed by IBM for its operating system OS/2.
It is an enhancement of the classic programming language Rexx and was initially developed for the operating system VM/SP, and later ported to the platforms MVS, OS/2 and PC-DOS. In October 2004, IBM gave the ooRexx source code for a couple of operating systems, including Microsoft Windows, Sun Solaris and Linux, to the Rexx Language Association (RexxLA), which released it under the Common Public License (CPL). The clear advantage of ooRexx is, that its syntax has large similarities to the English language and therefore it is very easy to learn and to understand. Besides its simplicity, it is a very mighty programming language and fully supports object oriented programming. Additionally, ooRexx offers full downward compatibility to the classic REXX language." [1]

1.2 BSF4ooRexx

"The 'Bean Scripting Framework for Open Object Rexx' (in short BSF4ooRexx) is a project that enables a developer to access ooRexx from Java and vice versa. What started as proof-of-concept-project in the context of a seminar at the University of Essen, Germany, in winter

semester 2000/2001, rapidly developed into a highly interesting project with a large number of followers. In 2003, the 'Augsburg' version of BSF4ooRexx was released, which made it possible to load Java from ooRexx and to interact with Java objects. The access itself was quite complex at this time and constantly required the developer to explicitly define the Java data types when calling a Java method. Because ooRexx, and therefore also BSF4ooRexx, are built upon the principle of simplicity, this problem had to be solved. Only three years later, the 'Vienna' version was released which not only improved the access to Java drastically, but also implemented support for OpenOffice.
BSF4ooRexx makes it, for example, possible to create graphical user interfaces (in short GUIs) or to automate Java applications from within ooRexx scripts. Especially the last point becomes very important in connection with the automation of OpenOffice and lets one sense the high value of this project." [1]

1.3 Installers

Installers are software programs that install applications on a user's operating system. Such installers may have a graphical interface and typically include procedures for installing and uninstalling software programs. Sophisticated installers also have a procedure for reinstalling, in order to add additional features afterwards or to repair a defective installation. Depending on their capabilities, a large number of installers can be distinguished.
The simplest form of an installer only copies files from the installer package to a destination directory on the user's hard disk. In this case, no scripts of any kind are executed before or after copying.
On the other hand, there are installers that can execute scripts before or after the copy process. Depending on the time of execution, such scripts are called preflight or postflight.
Additionally, installers can also check if specific preconditions are met. For example, if the required Java version for the program to be installed, is present.
The advantages and disadvantages of such installers will be discussed in further detail in

chapter 1.3.1 and 1.3.2.

1.3.1 Advantages

The main advantage of an installer is its simplicity. The user can install a sophisticated software program by simply executing the installer. He or she does not have to worry about where files and folders have to be copied in order to make the program work. Most installers will also have an uninstall function, which allows the user to cleanly reverse all changes made during the installation without negatively influencing other installed applications.

1.3.2 Disadvantages

The primary disadvantage is, that in most installers the users possibilities to intervene is limited. Although this may not be a problem for inexperienced users, it could be problematic for advanced users, such as developers, that want to change the behavior of the installer. Due to their large number of features, the initial creation of such installers is quite complex, as it will be shown in the following chapters.

1.3.3 Research Question

Based on the advantages of installers the main goal of this thesis is to create graphical installer packages for Windows, Linux and Mac OS X. Such installers are easier to use than the existing installation methods which have to be executed from the operating system's command line. This fact is daunting for inexperienced users that normally don't work this way. To make it more attractive for such persons, as well as more experienced users, this thesis' research question is, which possibilities for creating an installer for a specific operating system exist, which the most appropriate one for the given requirements is, how an installer is created using the chosen technology and what the resulting benefits are.

2 NSIS Installer for Windows

Author: Manuel Paar

For the creation of the installer package for Windows, we used a program called Nullsoft Scriptable Install System, short NSIS [2].
Chapter 2.1 will take a closer look on the features of NSIS, followed by chapter 2.2, which shows how an installer will be created. Last but not least, chapter 2.3 will show the installer in action.

2.1 The Installer

Nullsoft Scriptable Install System (NSIS) is a script based Installer for Windows *to create the logic to handle even the most complex installation tasks* [3]. It was developed by Nullsoft. NSIS is open source software and licensed under different free software licences [3].

It has no graphical user interface, so for a more comfortable development we used the HM NIS Editor [4]. Using this Editor, or the corresponding command from the command line (as shown in chapter 2.2.4), the NSIS scripts can be compiled to create the executable installer packages.

The advantages of NSIS are its wide spreading and that it is freely available. Its biggest competitor is the Windows Installer (MSI), which is a proprietary system by Microsoft. Due to Microsoft's version compatibility policy, different installers for different Windows versions would be required [5]. Because of this issue and its closed-source principle we decided to use NSIS instead.

2.2 Creating the Install Package for Windows with NSIS

For an easier handling, the installation files have been copied in one folder, which has the same structure like on the target system. This step is optional. The files can also be spread over the hard disk. Then the directory structure of the installation folder has to be created manually via script commands.

Chapter 2.2.1 will give a general overview of the NSIS script, followed by chapter 2.2.2, which describes the creation of the BSF4ooRexx Installer for Windows. The whole scripts will be found in chapter 7.2 in the Appendix.

2.2.1 General Introduction

A NSIS script has the extension `.nsi` and is a simple text document.

The script consists of

- Attributes,
- Sections,
- Functions and
- Variables.

Attributes, like shown in listing 1, set general properties of the installer. For example, `OutFile` sets the name of the package file and `InstallDir` the default installation directory. There are many other attributes, which can be set in the script.

```
1 OutFile "${PRODUCT_NAME}_Setup.exe"
2 InstallDir "$PROGRAMFILES\${PRODUCT_NAME}"
```

Listing 1: Usage of `OutFile` and `InstallDir` in NSIS.

A section contains commands, which logically belong together. The section `Uninstall` in listing 2 will be automatically called by the uninstaller. It removes the install directory with all the subdirectories.

```
1 Section Uninstall
2     RMDir /r "$INSTDIR"
3 SectionEnd
```

Listing 2: The section `Uninstall`.

There are two types of functions. A callback-function, which is shown in listing 3, will be called by the installer. In this example, a message box will be shown when the uninstallation was successful. Another function type is the user function. They can be called in sections or other functions by using the `Call` instruction [6][1].

```
1 Function un.onUninstSuccess
2     HideWindow
3     MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully uninstalled."
4 FunctionEnd
```

Listing 3: A NSIS callback-function.

Sections and functions for the uninstaller have to start with the letters “un” and a dot (`un.`). Uninstall functions and sections cannot be used by install functions and sections and vice versa.

Global variables are defined outside of sections and functions, as shown in listing 4.

```
1 Var JAVA_HOME
2 Var JAVA_VER
3 Var JAVA_INSTALLATION_MSG
```

Listing 4: Global variables in NSIS.

Auxiliary variables are only valid in the same function or section. These variables are stored in a stack that follows the Last In First Out (LIFO) principle [7]. An example of putting some

[1] The “↪”-symbol indicates a continued line and is not part of the actual source code. It will be used in upcoming source codes as well.

values onto the stack is shown in listing 5.

```
1 Push $0
2 Push $1
3 Push $2
```

Listing 5: Storing auxiliary variables in NSIS.

To remove a stored value from the stack the `Pop` command is used. It removes the values from the stack in reversed order, which is shown in listing 6 [7].

```
1 Pop $2
2 Pop $1
3 Pop $0
```

Listing 6: Removing auxiliary variables in NSIS.

These variables are register variables, which don't have to be declared, but can be used identically to the global variables [6, 8]. The usage of variables is shown in listing 13, for example.

2.2.2 BSF4ooRexx NSIS Script

To create a new script, the user has to start the HM NIS Editor or another text editor. After starting the program, the window, depicted in figure 1, will be displayed.

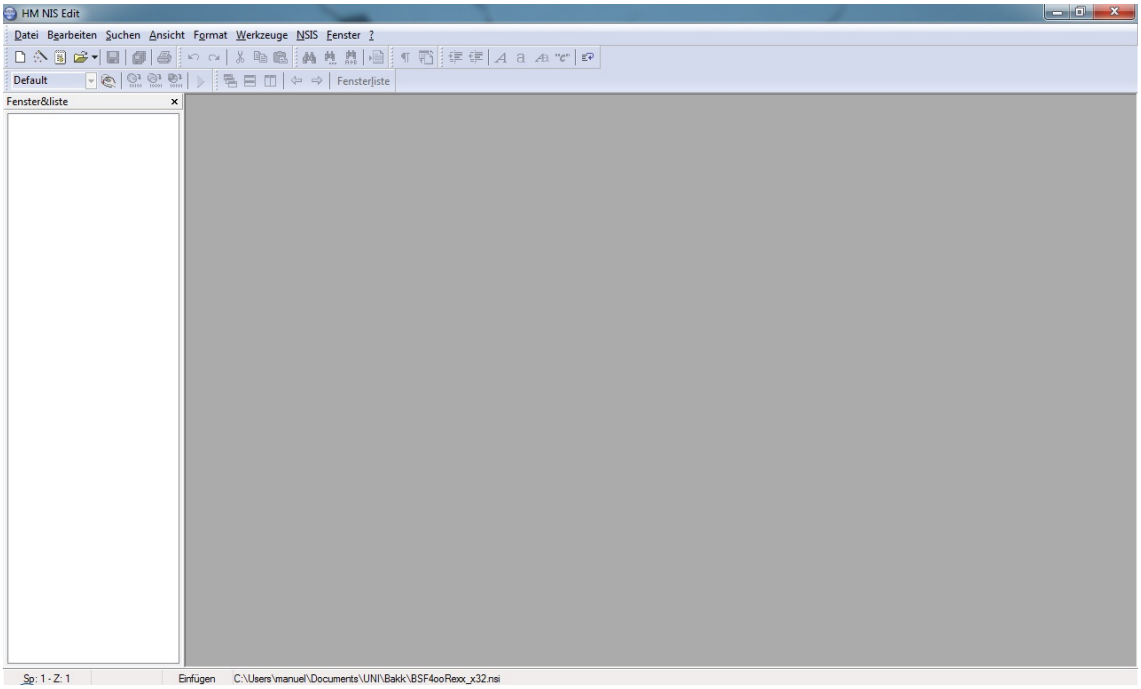


Figure 1: HM NSIS Editor.

First of all, we include the `MUI.nsh` and `EnvVarUpdate.nsh` header files as shown in listing 7. `MUI.nsh` gives the installer a modern user interface and the `EnvVarUpdate.nsh` header file allocates functions to manipulate the `PATH` environment variable. These macros come with every NSIS installation.

```
1 !include "MUI.nsh"
2 !include "EnvVarUpdate.nsh"
3
4 SetCompressor lzma
```

Listing 7: NSIS include- and compressor settings.

The command `SetCompressor lzma` defines that the Lempel–Ziv–Markov chain algorithm will be used to compress the installation package [9]. This algorithm was chosen because of its high compression ratio and its high decompression speed[2] [8].

Then some constants, as shown in listing 8, have to be defined.

[2]Other available options are `zlib` and `bzip2`.

```
1 !define PRODUCT_NAME "BSF4ooRexx"
2 !define PRODUCT_UNINST_KEY "Software\Microsoft\Windows\...."
3 !define MUI_ICON "bsf4oorexx.ico"
4 !define MUI_UNICON "bsf4oorexx.ico"
5 ...
```

Listing 8: Definition of constants in NSIS.

Defines are compile time constants [6]. They can be defined at the beginning and then used in the script. A define starts with `!define`, followed by the constant name and the value. For example, `MUI_ICON` defines the icon of the installer, depicted in figure 2.



Figure 2: The BSF4ooRexx setup icon.

After the constants, the pages, which the installation wizard will have, will be defined.

```
1 !insertmacro MUI_PAGE_WELCOME
2 !define MUI_LICENSEPAGE_CHECKBOX
3 !insertmacro MUI_PAGE_LICENSE "License.txt"
4 !insertmacro MUI_PAGE_INSTFILES
5 !insertmacro MUI_PAGE_FINISH
```

Listing 9: Defining the installation wizard's pages.

The BSF4ooRexx installer, as shown in listing 9, has a welcome, license, installation progress and finish page. To accept the terms of the license agreement, we want a checkbox, which is set with the `!define MUI_LICENSEPAGE_CHECKBOX` command. This setup was designed to match the layout of nearly every standard installer.

The uninstaller, as shown in listing 10, will only have a progress page, because in this case no further pages are necessary.

```
1 !insertmacro MUI_UNPAGE_INSTFILES
```

Listing 10: Defining the uninstaller's progress page.

The language of the installer will be set with `MUI_LANGUAGE`. More than one language can be defined. As shown in listing 11, we only use English, because the primary language of BSF4ooRexx, including its documentation, is English.

```
1 !insertmacro MUI_LANGUAGE "English"
```

Listing 11: Setting the installer's language.

With the attributes, shown in listing 12, the name of the installer, the file name and the installation directory will be defined. During the installation and uninstallation process, details will be displayed.

```
1 Name "${PRODUCT_NAME} ${PRODUCT_VERSION}"
2 OutFile "${PRODUCT_NAME}_Setup.exe"
3 InstallDir "$PROGRAMFILES\${PRODUCT_NAME}"
4 ShowInstDetails show
5 ShowUnInstDetails show
```

Listing 12: Defining additional attributes.

The function `.onInit`, as shown in listing 13, is a callback-function which will be called by the installer before the first installation window will be displayed. In this function, we check if the correct version of Windows[3], Java and ooRexx are installed and get the OpenOffice and/or LibreOffice installation path if it is installed. With the command `Call`, we start some functions to check the requirements. Such a function is shown in listing 13. If the function returns an error, a message box is displayed and the installation process is cancelled.

```
1  Function .onInit
2      Push $0
3      Push $1
4      Push $2
5      InitPluginsDir
6
7      GetVersion::WindowsPlatformArchitecture
8      Pop $WINDOWS_ARCHITECTURE ;32 oder 64
9
10     Call getBSFVersion
11     Call FindooRexx
12     StrCmp "" $REXX_INSTALLATION_MSG RexxSuccess RexxNoSuccess
```

[3]To check if the correct Windows architecture is installed, we use the GetVersion plugin [10].

```
13
14      RexxSuccess:
15          Call FindJava
16          StrCmp "" $JAVA_INSTALLATION_MSG JavaSuccess JavaNoSuccess
17
18      JavaSuccess:
19          Call FindOpenOffice
20          Call FindLibreOffice
21          ReadRegStr $0 ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
                ↪ DisplayName"
22          ReadRegStr $1 ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
                ↪ DisplayVersion"
23          StrCmp $0 "$(^Name)" FoundBSF
24          Goto Done
25
26      JavaNoSuccess:
27          MessageBox MB_OK|MB_ICONEXCLAMATION $JAVA_INSTALLATION_MSG
28          Goto Error
29
30      RexxNoSuccess:
31          MessageBox MB_OK|MB_ICONEXCLAMATION $REXX_INSTALLATION_MSG
32          Goto Error
33
34      FoundBSF:
35          IfSilent jump_if_silent jump_if_not_silent
36
37          jump_if_not_silent:
38              MessageBox MB_ICONQUESTION|MB_YESNO "Version $1 of ${PRODUCT_NAME}
                    ↪ is installed. You have to uninstall it before installing this
                    ↪ version. Would you like to uninstall it?" IDNO err
39
40          jump_if_silent:
41              ExecWait '"$INSTDIR\uninst.exe" /S _?=$INSTDIR'
42              Goto Done
43      err:
44          MessageBox MB_OK|MB_ICONEXCLAMATION "The installation was aborted."
45          Abort
46
47      Error:
48          RMDir /r "$INSTDIR"
49          Abort
50
51      Done:
52          Pop $2
53          Pop $1
54          Pop $0
55  FunctionEnd
```

Listing 13: The `onInit` function.

The function `FindJava`, which is shown in listing 14, executes an ooRexx program (shown in listing 40 in the Appendix), which returns the Java version and -bitness and checks if the correct one is installed.

```
Function FindJava
    Push $0
    Push $1

    SetOutPath "$INSTDIR"
    File ".\java_version.rex"
    ExecWait "rexx java_version.rex bitness" $0
    ExecWait "rexx java_version.rex version" $1
    Delete "$INSTDIR\java_version.rex"
    StrCmp $0 "64" CheckVersion JavaVerNotCorrect

    CheckVersion:
        StrCmp $1 "1" Done JavaVerNotCorrect

    JavaVerNotCorrect:
        StrCpy $JAVA_INSTALLATION_MSG "You need Java version 1.4 (64 bit) or
           ↪ newer to run this program."

    Done:
        Pop $1
        Pop $0
FunctionEnd
```

Listing 14: The `FindJava` function.

The other functions are shown in chapter 7.2.1 in the Appendix.

The main procedure is the `"BSF4ooRexx" SEC01` section. The most important parts are shown in listing 15. The whole procedure is also shown in chapter 7.2.1 in the Appendix.

```
Section "BSF4ooRexx" SEC01
    SetOverwrite ifnewer

    SetOutPath "$INSTDIR"
    File "ApacheLicense2.0.txt"
    File "asm_LICENSE.TXT"
    File "bsf-rexx-engine.jar"
    File "bsf-v400-20090910.jar"

    ...

    File /r ".\information"
    File /r ".\install"
    File /r ".\samples"
    File /r ".\Templates"
    File /r ".\utilities"

    CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}"
    CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\GUI RexxTry Program (ooRexxTry.
       ↪ rxj).lnk" "$INSTDIR\utilities\ooRexxTry.rxj" "" "$INSTDIR\install\
       ↪ images\windows\oorexxtry.ico"
```

```
20      CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Information (ReleaseNotes,
            ↪ introductions to ooRexx, ${PRODUCT_NAME} overview).lnk" "$INSTDIR\
            ↪ information"
21
22      CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}\Installation"
23      CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Download Page.lnk"
            ↪ "http://..."
24
25      CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Samples.lnk" "$INSTDIR\samples"
26      CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Utilities.lnk" "$INSTDIR\
            ↪ utilities"
27
28      WriteRegStr HKCR ".RXJ" "" "RexxJavaScript"
29      WriteRegStr HKCR ".RXJ" "Content Type" "application/x-rexx-java"
30      WriteRegStr HKCR ".RXJ" "PerceivedType" "text"
31      WriteRegStr HKCR ".RXJ\ShellNew" "FileName" "$INSTDIR\Templates\
            ↪ RexxJavaScript.rxj"
32
33      WriteRegStr HKCR "RexxJavaScript" "" "RexxJavaScript (Rexx script using Java
            ↪ ) executed by ooRexx"
34      WriteRegStr HKCR "RexxJavaScript\DefaultIcon" "" "$INSTDIR\install\images\
            ↪ windows\bsf4oorexx.ico,0"
35      WriteRegStr HKCR "RexxJavaScript\shell" "" "open"
36      WriteRegStr HKCR "RexxJavaScript\shell\edit" "" "Edit"
37      WriteRegStr HKCR "RexxJavaScript\shell\edit\command" "" 'notepad.exe "%1"'
38      WriteRegStr HKCR "RexxJavaScript\shell\open" "" "Run"
39      WriteRegStr HKCR "RexxJavaScript\shell\open\command" "" '"$REXX_HOME\rexx.
            ↪ exe" "%1"
40
41      ...
42
43      StrCmp "" $OO_HOME SearchLibreOffice OOFound
44
45      OOFound:
46          KillProcDLL::KillProc "soffice.exe"
47          KillProcDLL::KillProc "soffice.bin"
48          ...
49          ; copy files
50          ...
51          ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEprogram\"
52          ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEBasis\program\classes
                ↪ \unoil.jar"
53          ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\ridl.jar"
54          ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\jurt.jar"
55          ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\juh.jar"
56          ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$OO_HOMEprogram"
57          SetOutPath "$INSTDIR\install"
58          ExecWait '"$OO_HOMEprogram\unopkg.exe" add --shared
                ↪ ScriptProviderForooRexx.oxt'
59          StrCmp "" $LO_HOME AddOfficeReg LOFound
60
61      SearchLibreOffice:
62          ...
63
64      LOFound:
65          ...
```

```
66
67      AddOfficeReg:
68          ...
69
70      Done:
71          ExecWait "$INSTDIR\install\associationChanged.exe"
72  SectionEnd
```

Listing 15: The "BSF4ooRexx" SEC01 section (selection).

The attribute SetOverwrite determines whether an existing file should be overwritten or not [8]. SetOutputPath sets and (if it is necessary) creates the destination folder where the File commands will copy the files. The path next to the File command is the path to the file on the system where the installer will be created. The /r parameter specifies that all files and subdirectories of the folder will be copied recursively[11].

The function CreateDirectory creates a new folder by using the path in the quotes. CreateShortcut creates a new shortcut to the path in the first quotes. The second parameter is the path to the destination [8].

The command WriteRegStr creates a new entry in the registry. File type definitions and file extensions are registered by creating a key in HKEY_CLASSES_ROOT (HKCR). The first parameter is the root directory (in our case HKCR), the second one defines the subfolder. The third parameter defines the name of a key in the subfolder and the last one is the value of this key [12].

With StrCmp, two strings can be compared. If equal then jump to the SearchLibreOffice mark, otherwise to the OOFound mark [6].

Before the installation of the OpenOffice or LibreOffice support can be started, OpenOffice/LibreOffice has to be closed. So we use the KillProcDLL::KillProc [13] plugin function to kill all sorts of OpenOffice/LibreOffice processes.
The function EnvVarUpdate manipulates environment variables (e.g. the CLASSPATH) and is only available if the macro EnvVarUpdate.nsh (it comes with every NSIS installation) is included [14].

To execute `unopkg.exe`[4] we use the function `ExecWait`. It waits *for the executed process to quit* [8] before the next command in the script will be executed. In the `Done` section, we start the `associationChanged.exe`, which is located in the `install` folder of the BSF4ooRexx installation directory. It refreshes all file association settings.

```
1  Section Uninstall
2      ...
3      RMDir /r "$INSTDIR"
4      RMDir /r "$SMPROGRAMS\${PRODUCT_NAME}"
5
6      ...
7
8      DeleteRegKey HKCR ".JREXX"
9      DeleteRegKey HKCR ".ORX"
10     DeleteRegKey HKCR ".REXX"
11     DeleteRegKey HKCR ".REXXJ"
12     DeleteRegKey HKCR ".RXJ"
13
14     ...
15
16     ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
17     ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-v400-20090910.jar
           ↪ "
18     ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$INSTDIR"
19     ...
20 SectionEnd
```

Listing 16: The `Uninstall` section.

The section `Uninstall` will be automatically executed by the uninstaller. The most important parts are shown in listing 16. The whole procedure is shown in chapter 7.2.1 in the Appendix.

It removes the installation folder and start menu folder by using the function `RMDir`. The registry entries can be completely removed with the `DeleteRegKey` command. Entries in the environment variables will be deleted with the `un.EnvVarUpdate` function by using the parameter "`R`", which stands for "remove"[5] [14]. The function has the prefix `un.`, because only such functions can be executed in sections of the uninstaller.

[4] `unopkg.exe` is used to add the BSF4ooRexx extension to OpenOffice/LibreOffice.
[5] In contrast to the parameter "`A`", which stands for "append".

2.2.3 BSF4ooRexx Reinstaller NSIS Script

The reinstaller script looks pretty much the same as the installer script. The only differences are, that first, the reinstaller saves itself from the installation folder to avoid being deleted in the next step. Secondly, it starts the uninstall process, which completely removes the BSF4ooRexx installation. Finally, the reinstaller acts just like the installer and installs BSF4ooRexx again. Last but not least, it copies itself back to the installation folder, because it cannot install itself. The necessary changes to produce this reinstaller from the installer script are described in this chapter.

The reinstaller doesn't have an user interface, because in this case it is not necessary, so the lines in listing 9 can be removed.
In addition, line 2 in listing 17 has to be added to the `.onInit` function. It copies the reinstaller from the installation folder to the root folder of the system drive to save it. To start the uninstaller, the command in listing 18 has to be added to the top of the `"BSF4ooRexx" SEC01` section.

```
1 JavaSuccess:
2     CopyFiles "$INSTDIR\${PRODUCT_NAME}_Setup_x64_reinstall.exe" "$SYSDRIVE\"
3     Call FindOpenOffice
4     Call FindLibreOffice
5     Goto Done
```

Listing 17: The `JavaSuccess` function.

```
1 ExecWait '"$INSTDIR\uninst.exe" /S _?=$INSTDIR'
```

Listing 18: Uninstalling the existing version of BSF4ooRexx.

In the `"BSF4ooRexx" SEC01` section, the line in listing 19 has to be replaced by listing 20, because the reinstaller cannot install itself and has to be copied back to the installation folder instead.

```
1 File "${PRODUCT_NAME}_Setup_x64_reinstall.exe"
```

Listing 19: The original line.

```
1 CopyFiles "$SYSDRIVE\${PRODUCT_NAME}_Setup_x64_reinstall.exe" "$INSTDIR"
2 Delete "$SYSDRIVE\${PRODUCT_NAME}_Setup_x64_reinstall.exe"
```

Listing 20: The replacement for the original line.

2.2.4 Compile the package

There are different ways to compile the script:

- HM NIS Editor: Only one click on the “compile script” button (shown in figure 3),



Figure 3: HM NIS Editor’s “compile script” button.

- Command-Line: `makensis.exe script.nsi`,
- MakeNSIS Windows Wrapper, a windows GUI interface (called `makensisw.exe` and is stored in the NSIS installation directory) or
- executing the script in chapter 7.2.8 in the Appendix, which creates all Windows Installers in one run.

It is very important, that the reinstaller is compiled before the installer, otherwise the reinstaller cannot be added to the installation package.

2.3 Use Cases

The usage of the installer, reinstaller and uninstaller for Windows is explained in the following chapters.

2.3.1 Install

After double-clicking on the installer file, the installer starts and shows a dialog as depicted in figure 4.

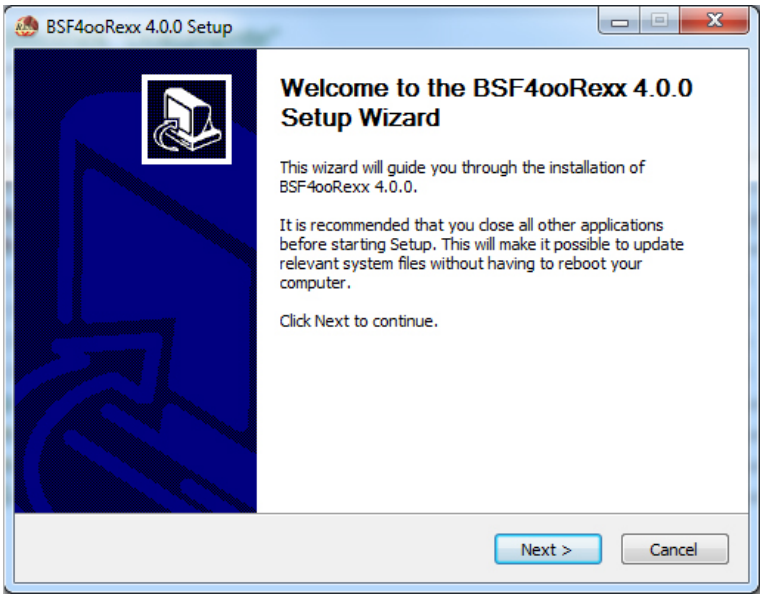


Figure 4: Welcome dialog of the BSF4ooRexx installer for Windows.

When the “Next” button will be clicked, the licence agreement, as shown in figure 5, will be displayed.

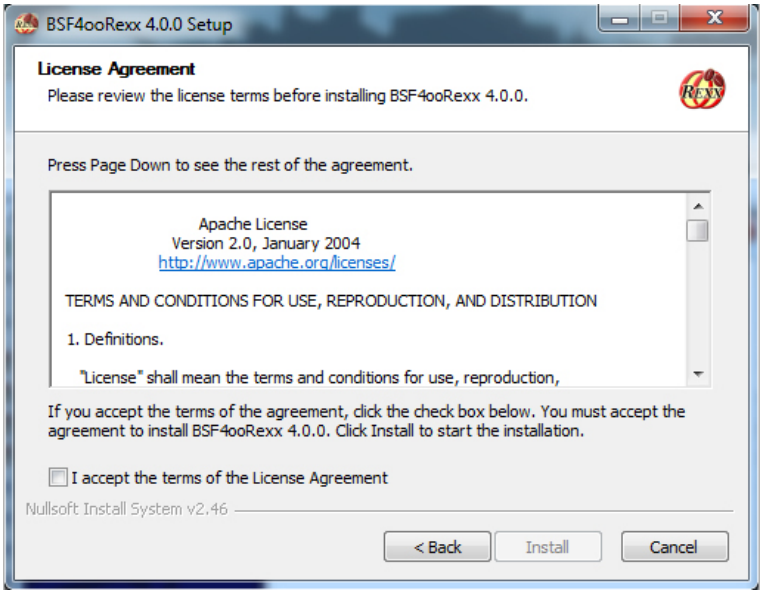


Figure 5: Licence agreement of the BSF4ooRexx installer for Windows.

After accepting the licence, the installation, as shown in figure 6, begins.

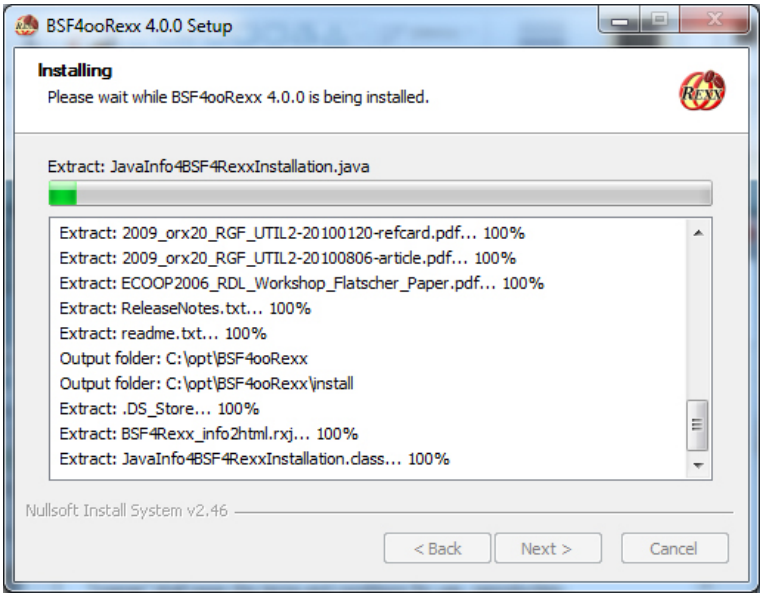


Figure 6: Installing window of the BSF4ooRexx installer for Windows.

The installation was successful, when the dialog, depicted in figure 7, is displayed.

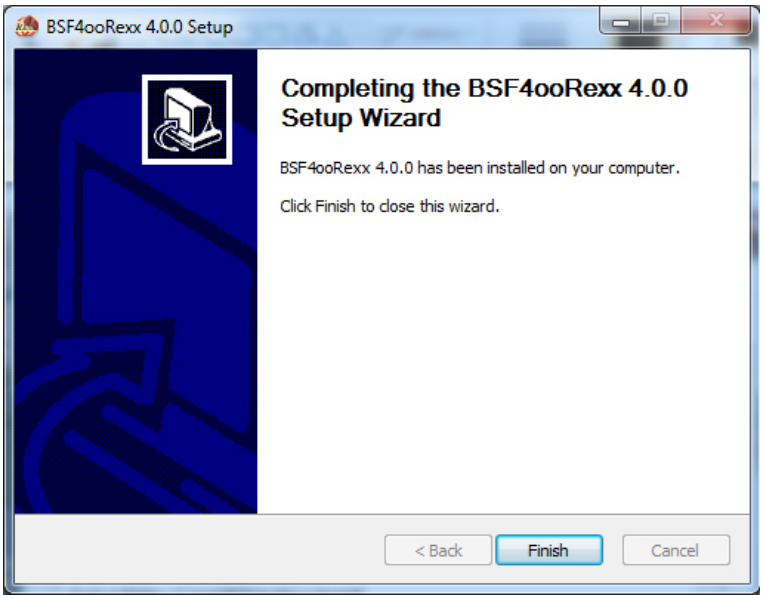


Figure 7: Finish dialog of the BSF4ooRexx installer for Windows.

Alternatively, the installation can be run in silent mode from the command line, which doesn't require user interaction, by invoking the installer using the command given in listing 21. The

`/S` parameter is case sensitive and has to be upper case.

```
1 BSF4ooRexx_setup.exe /S
```

Listing 21: Installing BSF4ooRexx in silent mode.

2.3.2 Reinstall

The reinstaller is located in the `Installation` folder of the BSF4ooRexx start menu directory. After clicking on the menu item, the message shown in figure 11 will be displayed.

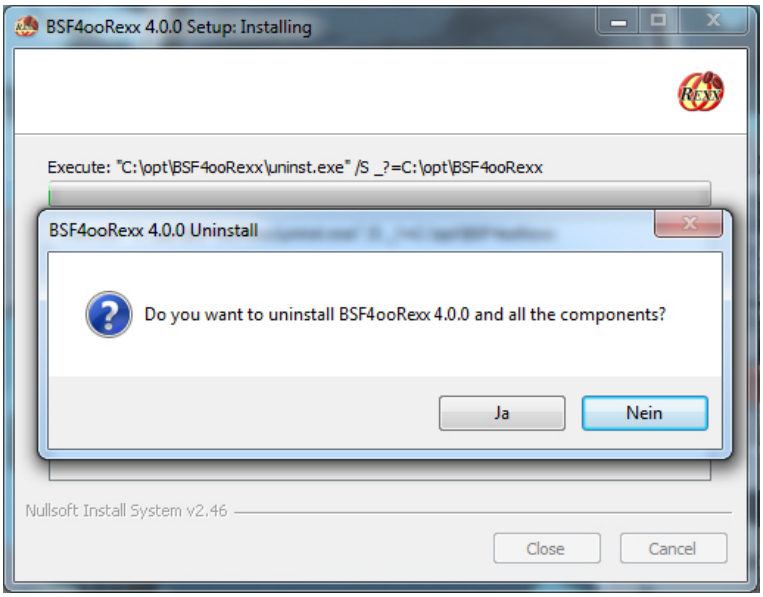


Figure 8: Uninstall message of the BSF4ooRexx reinstaller for Windows.

After accepting the message, the existing version of BSF4ooRexx will be uninstalled. When the uninstallation was successful, the installation starts, as depicted in figure 9.

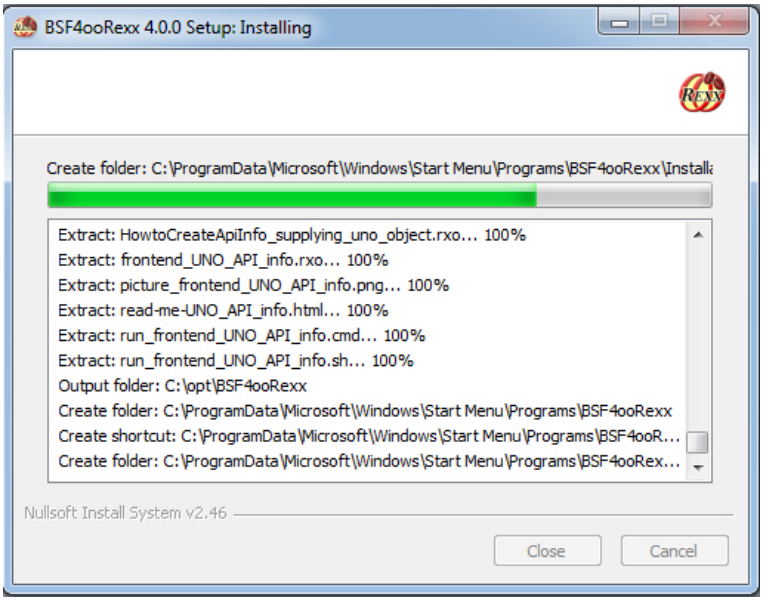


Figure 9: Installing window of the BSF4ooRexx reinstaller for Windows.

The reinstallation was successful, when the dialog, shown in figure 10, is displayed.

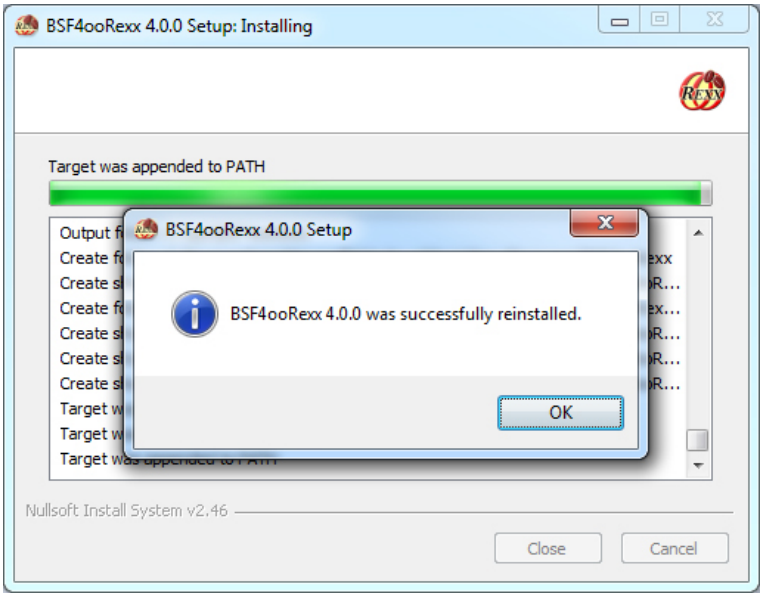


Figure 10: Finish dialog of the BSF4ooRexx reinstaller for Windows.

Alternatively, the reinstallation can be run in silent mode from the command line, which doesn't require user interaction, by invoking the reinstaller using the command given in listing 22. The `/S` parameter is case sensitive and has to be upper case.

```
1 BSF4ooRexx_reinstall.exe /S
```

Listing 22: Reinstalling BSF4ooRexx in silent mode.

2.3.3 Uninstall

The uninstaller can also be found in the `Installation` folder of the BSF4ooRexx start menu directory. After clicking on the menu item, the message shown in figure 11 will be displayed.

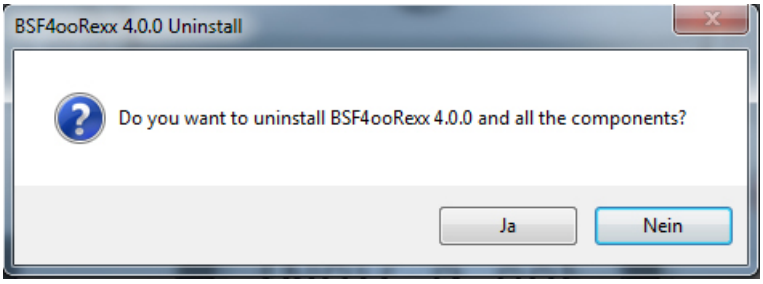


Figure 11: Uninstall message of the BSF4ooRexx uninstaller for Windows.

After accepting the message, the existing version of BSF4ooRexx will be uninstalled. During the uninstallation process, the dialog shown in figure 12, will be shown.

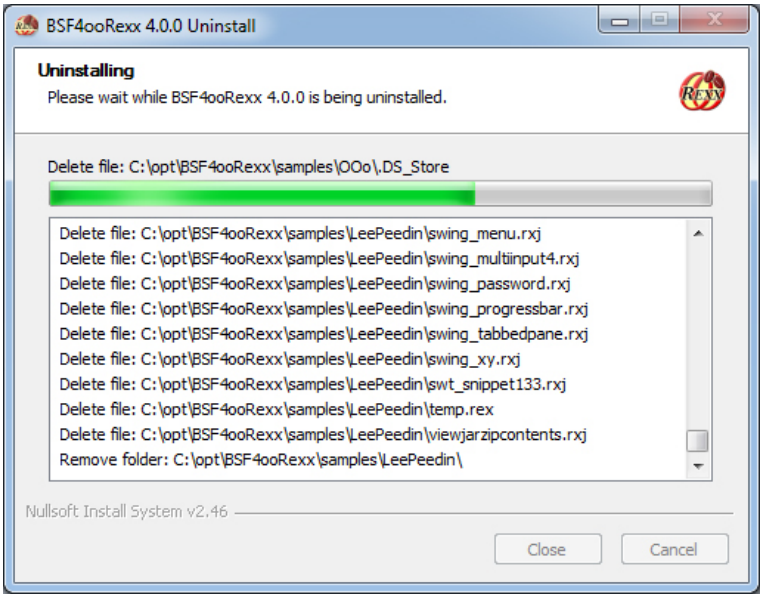


Figure 12: Uninstalling window of the BSF4ooRexx uninstaller for Windows.

The uninstallation was successful, when the message shown in figure 13 is displayed.

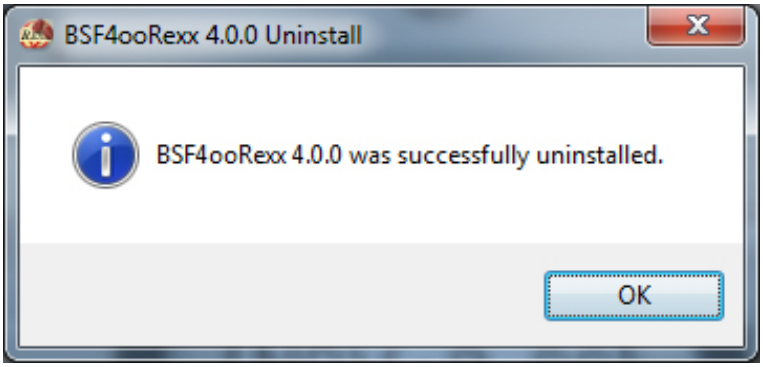


Figure 13: Finish message of the BSF4ooRexx uninstaller for Windows.

Alternatively, the uninstallation can be run in silent mode from the command line, which doesn't require user interaction, by invoking the uninstaller using the command given in listing 23. The `/S` parameter is case sensitive and has to be upper case.

```
1 uninst.exe /S
```

Listing 23: Uninstalling BSF4ooRexx in silent mode.

3 DEB Package for Debian-based Linux Distributions

Author: Jürgen Hesse

For the creation of the installer package for Debian-based Linux distributions, we used the Debian Package Management System (`dpkg`) [15].
Chapter 3.1 will take a closer look on the features of this system, followed by chapter 3.2, which shows how a package will be created. Last but not least, chapter 3.3 will show the package in action.

3.1 The DEB Package System

A Debian package, or a Debian archive file, contains the executable files, libraries, and documentation associated with a particular suite of program or set of related programs. Normally, a Debian archive file has a filename that ends in `.deb` [16].

The following chapters will show how a Debian package for BSF4ooRexx can be built.

3.2 Creating the DEB Package for Debian-based Linux Distributions

The following chapters describe the required steps to create a Debian (DEB) package.

3.2.1 Install the required software

Actually, there are no special requirements that have to be met to create packages on Debian-based Linux systems. It is not necessary to install additional software components, because the required components are already present in every standard installation.

Nevertheless, it is highly recommended to install a good text editor for editing the packages install scripts and the meta data.

For installing the newly created package, on the other hand, it will be required to have the appropriate versions of Java and ooRexx installed. We will take a closer look at this issue in chapter 3.2.3.

3.2.2 Root folder

The root folder contains the files that will be installed on the target system. While installation, the directory structure given in the root folder will be copied to the target file system and additional scripts will be executed. These additional scripts are contained in a folder called `DEBIAN` within the root folder. To be executed correctly at the right time during installation, the additional scripts must follow explicit naming conventions [17]. We will take a look at this issue in chapter 3.2.5.

The root folder must have the same name as the resulting package. Generally, this name should follow the definition given in [18]: *"You must choose the package name to consist only of lower case letters (`a-z`), digits (`0-9`), plus (`+`) and minus (`-`) signs, and periods (`.`). It must be at least two characters long, must start with an alphanumeric character, and must not be the same as existing ones. It is a good idea to keep its length within 30 characters. ... If you need to invent a version string, use the `YYYYMMDD` format such as `20110429` as upstream*[6] *version. This ensures that `dpkg` interprets later versions correctly as upgrades."*

In our case, the project name is, of course, `bsf4oorexx`. The version and revision numbers follow the values given in BSF4ooRexx's `BSF.CLS`. At the moment of writing this document it was `406.20110822`. Therefore, the version numbers have been chosen accordingly and the root folder called `bsf4oorexx-406.20110822` has been created. As mentioned

[6] "Upstream" is an often used term for the software components a package is built for, e.g. the BSF4ooRexx files which are normally distributed as .zip-Archive.

above, the term `20110822` indicates that the package's content was built on August 22nd, 2011 and allows to give the user additional version information. After adding the installation files and the installation information, the package will finally have the structure depicted in figure 14. The necessary steps are described in the following chapters.

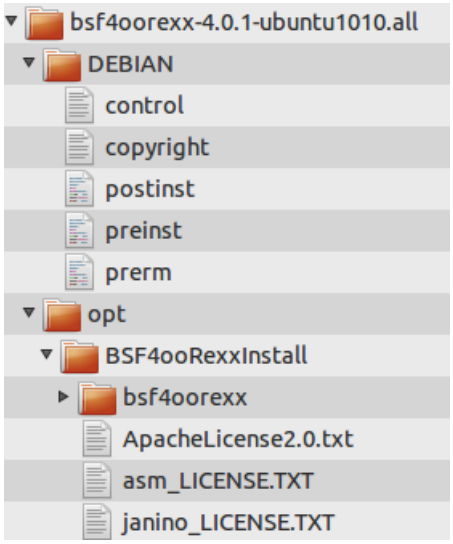


Figure 14: Debian package structure.

After creating the basic package structure, the next step is to create the package's meta data, which controls its behaviour while installation etc.

3.2.3 Meta data

The package's meta data (also called control information) is stored in a file called `control` inside the `DEBIAN` folder of the root directory [17]. Its content is shown in listing 24.

```
1 Package: bsf4oorexx
2 Version: 406.20110822
3 Section: Interpreted Computer Languages
4 Priority: optional
5 Homepage: http://sourceforge.net/projects/bsf4oorexx
6 Maintainer: BSF4ooRexx <bsf4oorexx-devel@lists.sourceforge.net>
7 Architecture: all
8 Description: Bean Scripting Framework for Open Object Rexx (BSF4ooRexx)
```

```
9  BSF4ooRexx version for ooRexx 4.1.0 and later exploiting the new ooRexx kernel
       ↪ (e.g. allowing implementing Java methods in ooRexx, callbacks from Java
       ↪ to ooRexx). [Camouflages Java as the dynamically typed, caseless and easy
       ↪  to use ooRexx.]
10 .
11 Special note for users of Ubuntu 11.04: If you want to use the LibreOffice
       ↪ version from the repository, you should invoke the command "sudo apt-get
       ↪ install libreoffice libreoffice-java-common" from the command line. If
       ↪ you then experience any problems working with LibreOffice, you need to
       ↪ remove the .libreoffice folder from your home directory. When the program
       ↪  starts up the next time, the folder is rebuilt and everything works well
       ↪  again.
12 Depends: oorexx (>= 4.1.0), openjdk-6-jre
13 Recommends: openoffice.org (>= 3.2.0), openoffice.org-calc (>= 3.2.0),
       ↪ openoffice.org-draw (>= 3.2.0), openoffice.org-impress (>= 3.2.0),
       ↪ openoffice.org-writer (>= 3.2.0)
```

Listing 24: The control file.

`Package` simply describes the package's name as it is defined in the root folder's name. Please note that the version numbers etc. are omitted in the package name. Because the version information cannot be omitted completely it gets it's own tag called `Version`. The version information is the second part of the root folder's name, to be exact, everything between the hyphens [18].

Software packages for Debian-based Linux distributions are organized in sections. These sections describe what kind of software the user is dealing with. For example, the section `admin` contains programs for administering the system. `net`, on the other hand, contains software for networking, e.g. email clients etc. Because ooRexx and BSF4ooRexx are programming languages we are using the same section `as ooRexx, namely "Interpreted Computer Languages"`, here. This is done to ensure that users will find the BSF4ooRexx package in the same place as the ooRexx package. Nevertheless it would also be possible to use the predefined keyword "`devel`" here. The tag `priority` describes how important this package is for an average user. For new packages the value `optional` is the best choice. `Homepage` and `Maintainer` point to the developers of this package and are set accordingly. The architecture in this case is `all` and refers therefore to both bitnesses of 32 bit and 64 bit. [18].

Two very important settings are `Depends` and `Recommends`. `Depends` defines a list of

other packages (and their versions) that have to be present before the actual package can be installed. In our case we defined, that Java (Version 1.6.0 or higher) and ooRexx (Version 4.1.0 or higher) have to be installed at least [18].

The possible settings for `Recommends` work quire similar. It doesn't list required packages, but packages that should be installed. In other words it lists packages that are recommended, as the tag name indicates. If they are not present the correct functionality of the actual package is not harmed. In our case, we recommend to install OpenOffice because BSF4ooRexx is great for OpenOffice automation. If the user doesn't need the OpenOffice integration, he or she is still able to install the package without the OpenOffice capabilities. Finally, the `Description` tag can be used to give a short summary about the package's contents [18].

3.2.4 File system structure

Besides the `DEBIAN` folder with package information, the root folder also contains the files that have to be installed to the file system of the target system. These files are organized in a directory structure that represents the target system.

For example, if a file called `info.txt` has to be installed to the `/opt` folder of the file system, simply a folder called `opt` has to be created inside the root folder and the `info.txt` file has to be put in there. While installing, the `opt` folder in the root of the target file system will be created if it doesn't exist and the text file will be copied there [19].

Following this easy example we created an `opt` folder and put the BSF4ooRexx install folder in it. During installation this folder will be copied into the `opt` directory of the target system. From there we will be able to execute the installation scripts (which is described in more detail in the following chapter), that configure BSF4ooRexx for the target system.

3.2.5 Install scripts

The file system structure is, as defined in chapter 3.2.4, a vital part of the installation package but its abilities are limited to simple copy-and-paste operations. In order to add some logic to the package and make it more intelligent, it is possible to define up to four scripts with predefined names that are executed on special conditions [17, 18]:

- `preinst` – This script defines operations that have to occur before the contents of the package are copied and the `postinst` script is executed. This could be to stop a specific service or create a user etc. We use this script to remove old versions of BSF4ooRexx from the `/opt` directory. The complete source code of the script is depicted in chapter 7.3.2 in the Appendix.

- `postinst` – Operations within this script are executed after the package's contents have been copied to their destination in the file system. Typical operations could be the restart of a service or the configuration of system-specific settings. In our case, we use the `postinst` script to run the `setupAllAndRun.rex` script from the BSF4ooRexx distribution to set up all required components. The script is shown in chapter 7.3.3 in the Appendix.

- `prerm` – As the name indicates, the `prerm`-script contains commands that are executed before the package is removed. We used it to remove all BSF4ooRexx files from the file system before the package itself is removed. The content of the postrm script is shown in chapter 7.3.4.

- `postrm` – This script contains commands that are executed after the package has been removed from the system. Typically, such commands restart a service, remove a user or clean up temporary data. This script type is not used in the package.

Please note that the scripts must use these predefined names. If they do not, the package will not work as intended.

As the first line indicates, these scripts are simple UNIX shell scripts that will be executed in a specific moment. Therefore they must be made executable using the UNIX `chmod` command, as shown in listing 25.

```
1 chmod +x postinst
```

Listing 25: Making the `postinst` script executable.

If, and only if the scripts have been made executable the package will work as intended. The lines in listing 46 in the Appendix, for example, start an ooRexx-script that configures the system to run ooRexx and BSF4ooRexx programs. Please note that it would be possible to include all the operations of this ooRexx-script into this file, but due to the fact that it works excellently where it is, it was decided to leave it there and just call the ooRexx-script from here [7]. Once executed, the configuration script will create file associations for ooRexx, will create the desktop shortcuts, etc.

3.2.6 Building the package

After carrying out the previous steps we are finally ready to actually build the BSF4ooRexx package. To do so, we change into the directory where the root folder is located and start the package creation using the command given in listing 26 [19].

```
1 dpkg-deb --build bsf4oorexx-406.20110822
```

Listing 26: Building the Debian package.

After a few moments the Debian-package has been created and is ready to be distributed and to be installed.

[7] A wise person once said: “Keep it simple, stupid.”

3.3 Use Cases

The install, reinstall and uninstall features of the BSF4ooRexx Debian package are explained in the following chapters[8].

3.3.1 Install

The installation of a Debian package can either be carried out graphically, e.g. by using the Ubuntu Software Center or manually by invoking the corresponding commands from the command line. To install the package graphically the user must simply double-click on the file created in the steps above. On Ubuntu, the Software Center will be lauched, showing some informations about the package and allowing the user to install the file. This case is shown in figure 15.

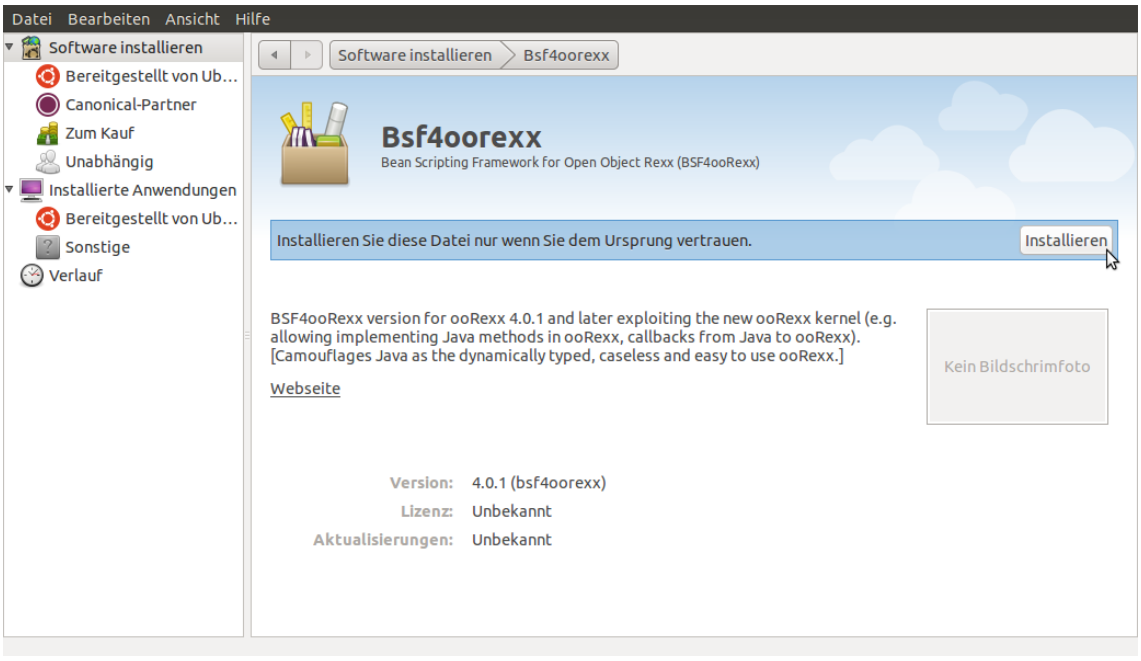


Figure 15: Installing the Debian package by using the Ubuntu Software Center.

When the user clicks on “Install”, he or she will be asked to authenticate as root user. Only if this step succeeds, the installation will start. It is necessary, because the system must

[8] At this point is has to be mentioned, that the resulting packages have been compiled and tested on Ubuntu 10.10 (Maverick Meerkat) only.

aprove that the user has sufficient privileges to make system wide changes by invoking the installer. This case is depicted in figure 16.

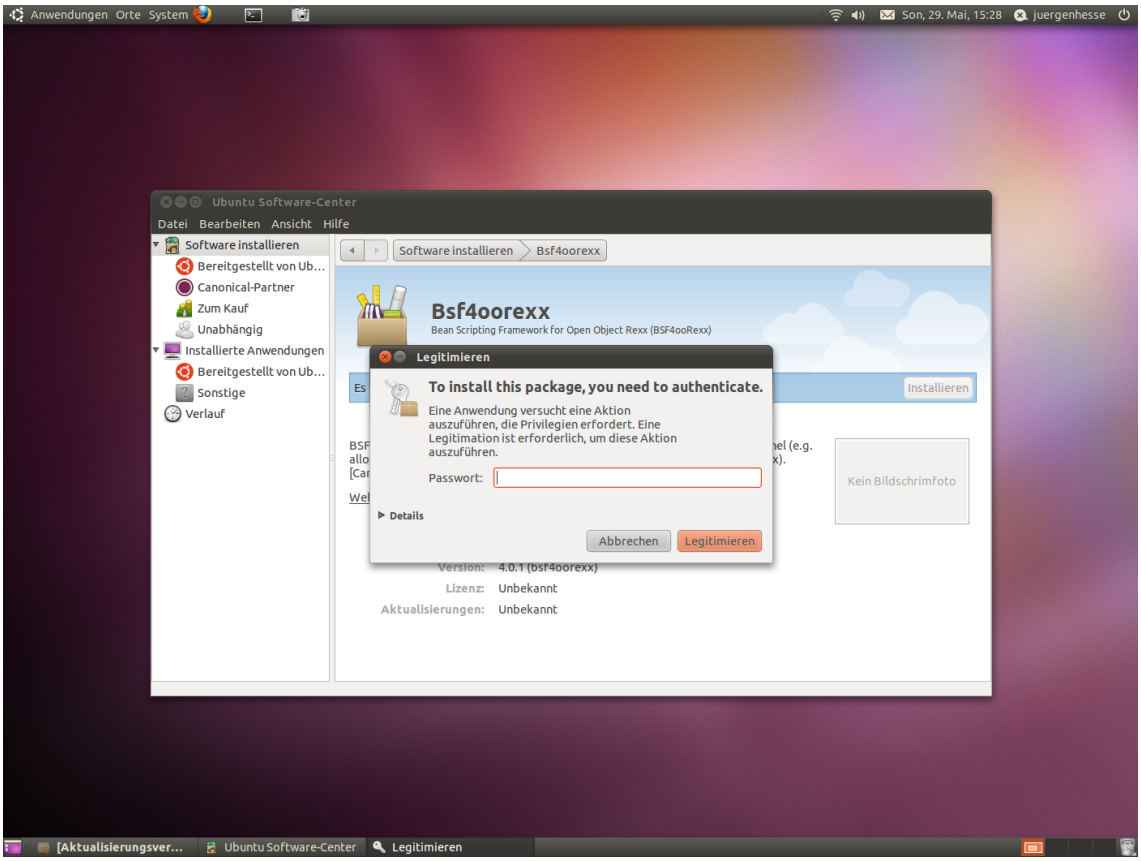


Figure 16: Authentication for the installation.

When the user has successfully been authenticated, the installation process starts, the required files are copied and the corresponding script are executed. Until this process is finished, the user is shown a nice progress bar, as depicted in figure 17.

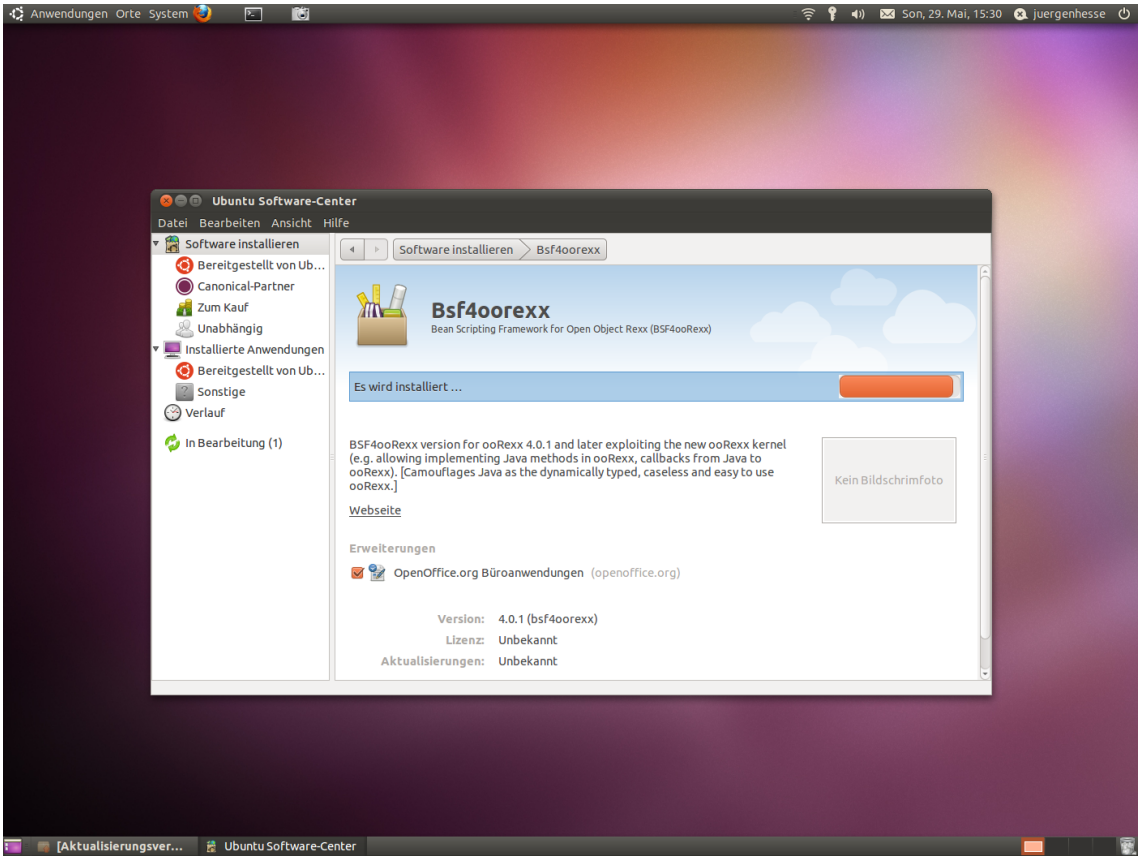


Figure 17: Installation process.

After the installation has been finished, the BSF4ooRexx package is shown in the installed-software-section of the Ubuntu Software Center, as depicted in figure 18.

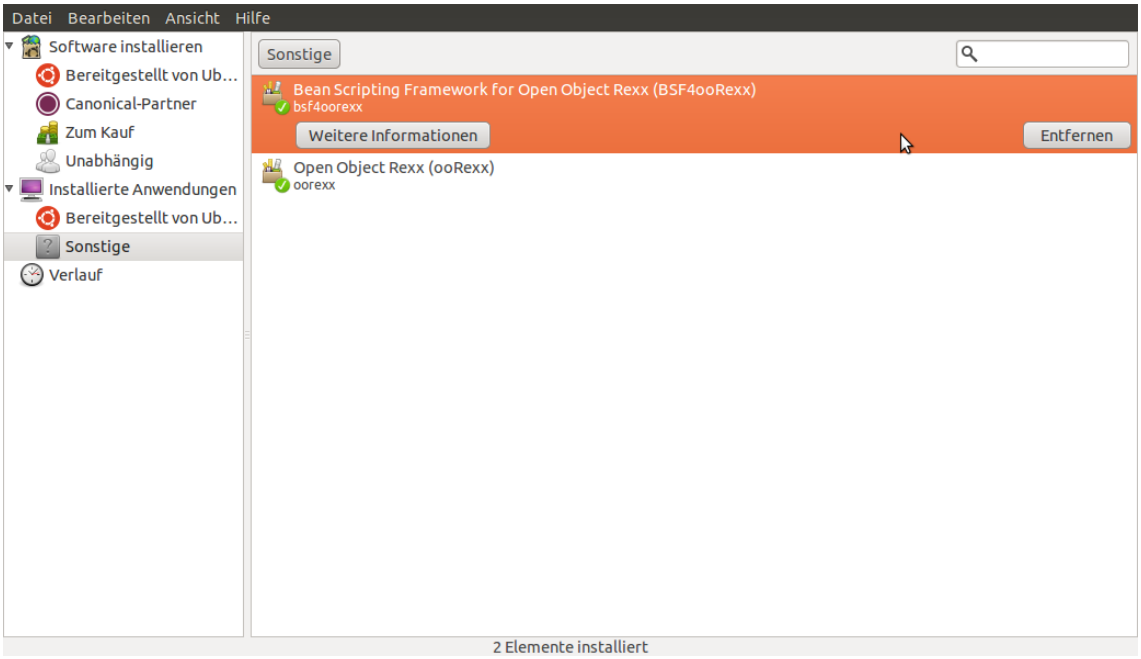


Figure 18: The Installed-Software-Section after installation.

Alternatively, the package can also be installed from the command line by invoking the command given in listing 27.

```
1 sudo dpkg -i bsf4oorexx-406.20110822.deb
```

Listing 27: Installing the package from the command line.

3.3.2 Reinstall

If OpenOffice was not installed before installation, its support can be enabled afterwards by reinstalling the package. Like above, there are two ways to carry out a reinstallation. The first possibility is to start the process by choosing the corresponding entry from the BSF4ooRexx Menu, as depicted in figure 19.

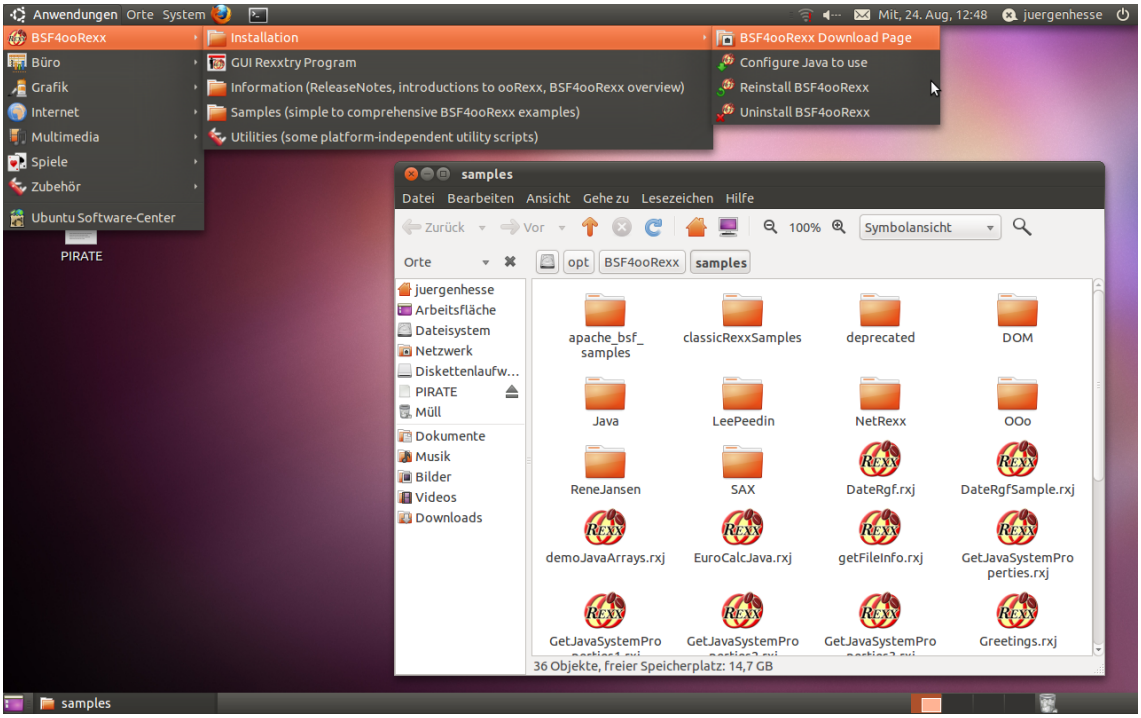


Figure 19: Reinstalling BSF4ooRexx from the menu.

After the reinstallation has successfully been carried out, the user gets a notification message.

Alternatively, the package can be reinstalled by double-clicking the Debian package file again. This will, just like in case of the initial installation, invoke the Ubuntu Software Center. The program will remember, that the package has already been installed and will show a reinstall- instead of an install button, as shown in figure 20.

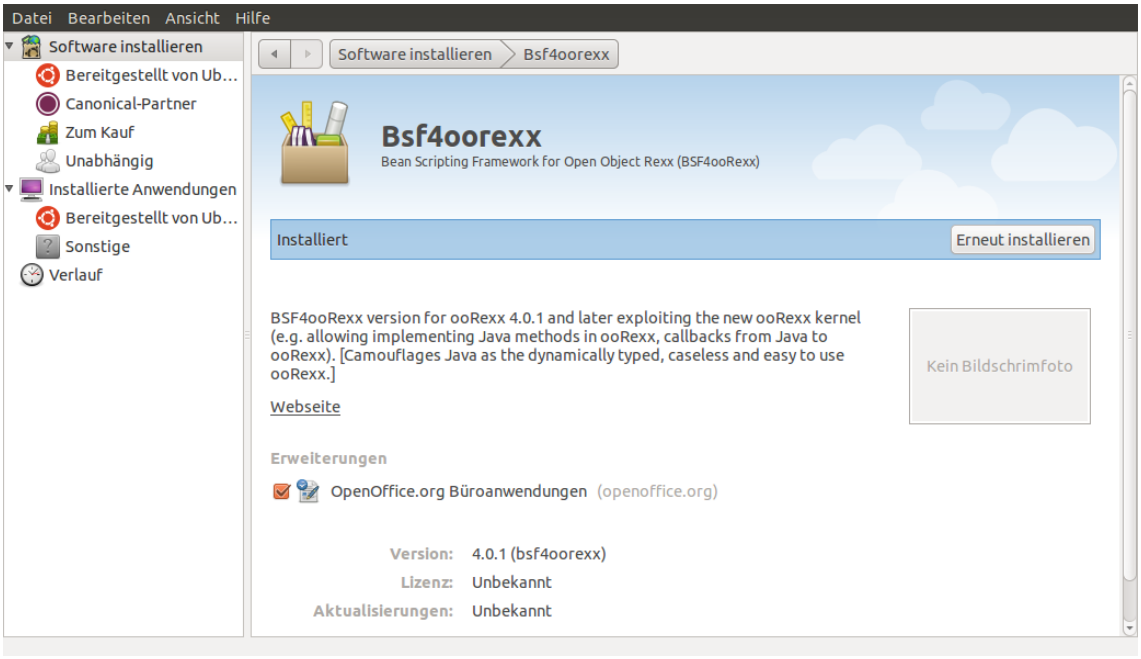


Figure 20: Reinstalling BSF4ooRexx using Ubuntu Software Center.

After clicking this button, the package will be reinstalled. During reinstallation, the user is shown a progress bar that looks like that shown during installation (shown in figure 17).

3.3.3 Uninstall

If the user wants to deinstall the package he or she has to open the Ubuntu Software Center. In the installed-software-section the BSF4ooRexx package appears. If a large number of other packages has been installed, the user can use the search field on the top left of the window to search for the package. The uninstall button appears right beside the package name, as shown in figure 21.

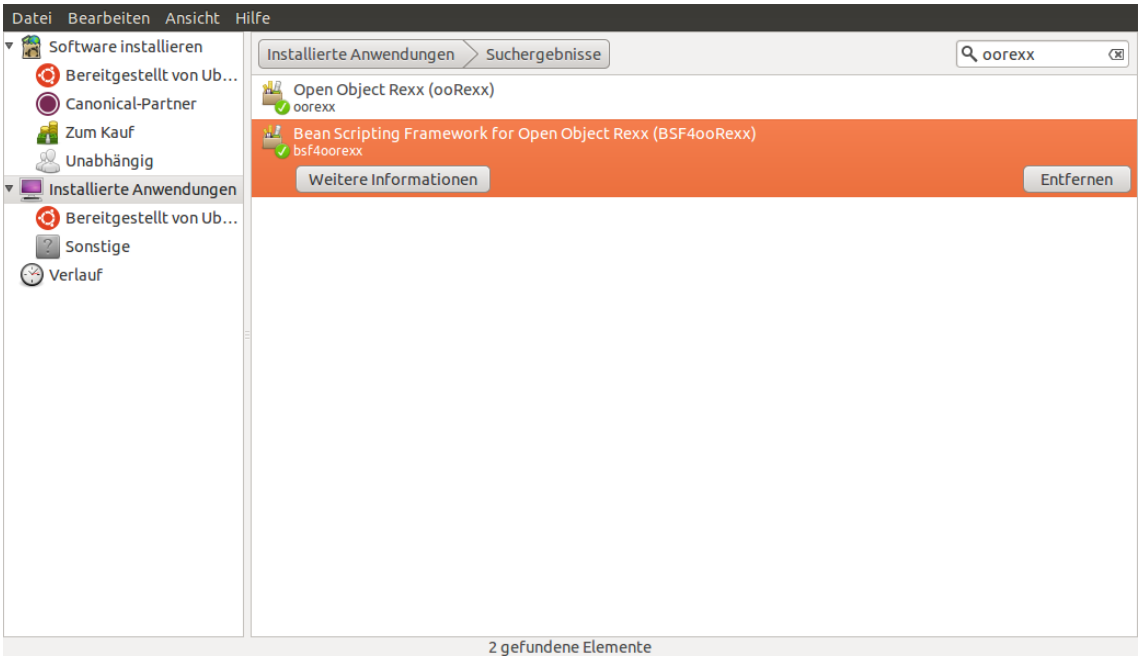


Figure 21: Uninstalling the package.

When the button has been clicked, the user must authenticate, just like before installation. After that the package is removed and all changes made during installation are reverted. During deinstallation, the user sees a progress bar, as depicted in figure 22.

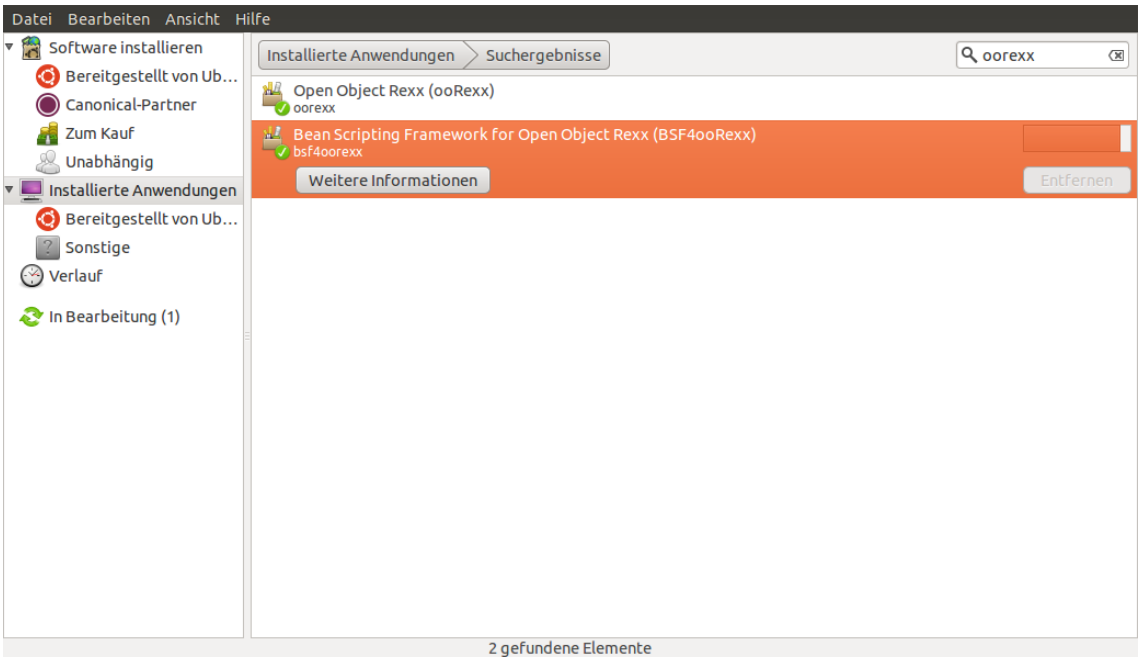


Figure 22: Deinstallation progress.

After the package has successfully been removed, is disappears from the installed-packages list, as shown in figure 23.

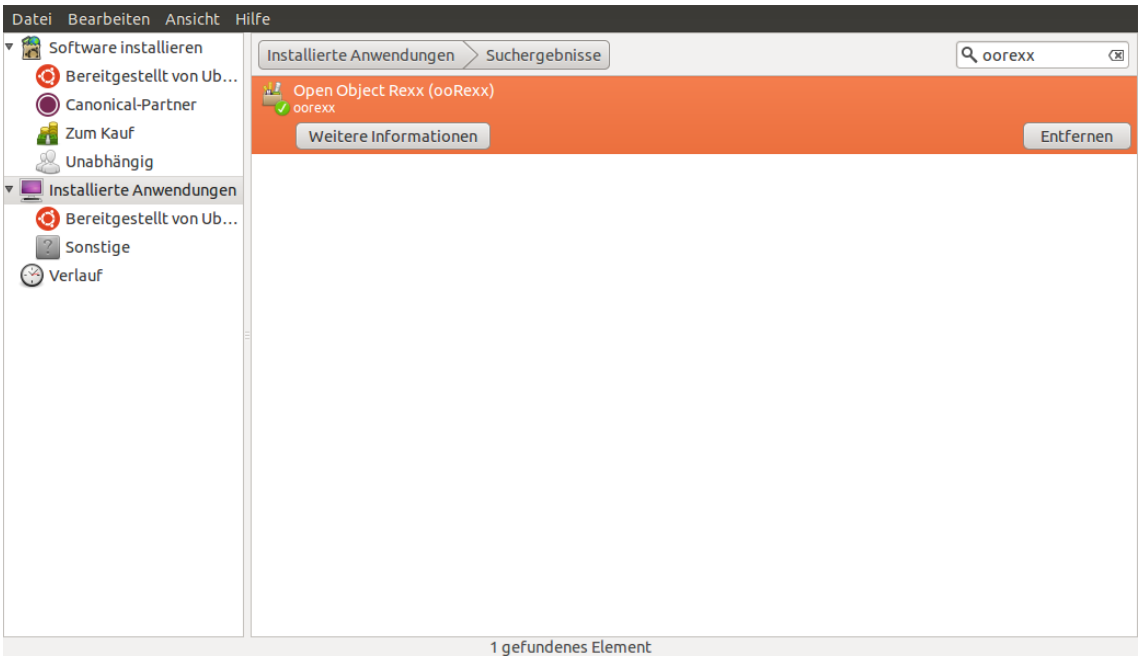


Figure 23: Deinstallation finished.

Alternatively, the package can be removed using the command line, by invoking the commands given in listing 28.

```
1 sudo dpkg -r bsf4oorexx
```

Listing 28: Uninstalling the package from the command line.

The effects are the same as they are when using the graphical uninstall mentioned above.

Additionally the user can uninstall BSF4ooRexx using the uninstall option in the BSF4ooRexx menu. In this case BSF4ooRexx will be removed completely, but the package entry will remain in the package manager and has to be removed by hand. This issue could only be solved by changing the install script contained in the BSF4ooRexx distribution, to automatically detect the current Linux version and the appropriate package removal command.

4 RPM Package for Red Hat-based Linux Distributions

Author: Jürgen Hesse

For the creation of the installer package for Red Hat-based Linux distributions, we used the RPM Package Management System, in short just RPM, which comes with every standard installation of a Red Hat-based linux distribution.
Chapter 4.1 will take a closer look on the features of RPM, followed by chapter 4.2, which shows how a package will be created. Last but not least, chapter 4.3 will show the package in action.

4.1 The RPM Package Management System

The RPM Package Manager (RPM) is a ... command line driven package management system capable of installing, uninstalling, ... and updating computer software packages. Each software package consists of files along with information about the package like its version, a description, and the like [20].

The basic functionality of RPM is very similar to the features of the Debian package management system, described in chapter 3. RPM uses a bunch of source files to create a package, based on the information defined in a control- or, what RPM calls it, package specification (in short “spec”) file. This file holds information like the package’s base name (i.e. the name without any version information), the version, a description, the requirements for installation and so on. The source-, specification- and resulting RPM-files are organized in a specific directory structure which can be built automatically using a helper program [21]. This step is desribed in more detail later in chapter 4.2.2.

Nevertheless, compared to the Debian system, RPM is more flexible (and therefore much more complex) regarding the features of its spec file. In contrast to a Debian package, a RPM

package is normally freshly compiled from its sources. Therefore, the package build process consists of serveral steps, or as RPM calls it, stages, that are executed one after another. Every stage is designed to read files from one directory, to carry out some operations on these files and to write them back to the same or another directory. Therefore, the system uses some predefined variables, which all start with a percent sign (%), to address these stages and directories. The available stages and their functions are described in chapter 4.2.3 in further detail. The following chapter will lead through the creation of such a RPM package.

4.2 Creating the RPM Package for Red Hat-based Linux Distributions

To create the desired RPM package, a number of tasks has to be carried out. This includes

1. installing the required programs, which is described in chapter 4.2.1,
2. creating the directory structure for building RPM packages, as described in chapter 4.2.2,
3. creating the specification file, that holds the packages meta data (e.g. name, version, installation prerequisites, etc.) and defines the steps of its build process (as shown in chapter 4.2.3), and finally
4. building the package, as shown in chapter 4.2.4.

4.2.1 Installing the required programs

In contrast to the preconditions for the Debian-based package, the RPM package needs some additional programs to be installed in order to gain the ability to create a new package [22]. The required components can be installed by executing the commands given in listing 29 as superuser on the command line.

```
1 yum install @development-tools
2 yum install rpm-build rpmdevtools rpmlint
```

Listing 29: Installing the required utilities.

It has to be mentioned, that some Red Hat-based distributions (e.g. Fedora 14 etc.) do not add the default user to the `sudoer`-file. This means that, in contrast to other systems like Ubuntu Linux, the default user is not able to execute commands in the name of a superuser by using the `sudo` command. There are two possible solutions for this issue. The first is, to add the corresponding entry to the `sudoer`-file according to the system's definitions. The second is the "`su -`" command to gain superuser (root) privileges. The latter one is more risky and should only be used if the user knows what he or she is doing. Nevertheless, this is the only situation (apart from the installation of the final package) where we will need to have root privileges.

The term `@development-tools` references a package of development tools, obviously, that allow, for example, to build the `rpm` package by using a Makefile. The programs `rpm-build`, `rpmdevtools`, and `rpmlint` are utilities that allow to create a basic package structure and to build the final package.

4.2.2 Creating the required directory structure

After installing the required utilities, the next step is to create the basic structure of the package [21, 22]. This is done by using the command given in listing 30.

```
1 rpmdev-setuptree
```

Listing 30: Setup of the basic package structure.

The command is executed with the privilieges of a standard user and creates a directory called `rpmbuild` in the user's home directory. The `rpmbuild` directory is the starting point

for all RPM-packages that will be built on this system by this user. According to the official documentation [21]: *"The "rpmdev-setuptree" program will create a ~/rpmbuild directory. Underneath "rpmbuild" are a set of subdirectories (such as SPECS and BUILD), which you will use for creating your packages."*

The resulting directory structure is shown in figure 24.

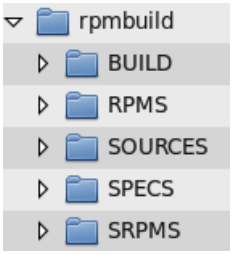


Figure 24: The structure of the rpmbuild directory.

These directories will be addressed from within the spec file (which will be created in chapter 4.2.3) during the package's build process by using predefined variables, sometimes referred to as macros. These variables, or macros, and their target directories' purposes are described in table 1.

Macro	Points To	Purpose
`%_specdir`	`~/rpmbuild/SPECS`	Contains the RPM specification (spec) files.
`%_sourcedir`	`~/rpmbuild/SOURCES`	Holds pristine source packages, e.g. a packed `.zip` file.
`%_builddir`	`~/rpmbuild/BUILD`	Source files are unpacked (and compiled) in a subdirectory of this folder.
`%_buildrootdir`	`~/rpmbuild/BUILDROOT`	Files are installed here during the `%install` stage.
`%_rpmdir`	`~/rpmbuild/RPMS`	Binary RPMs are stored here after the build process has finished.
`%_srcrpmdir`	`~/rpmbuild/SRPMS`	Source RPMs are stored here.

Table 1: Predefined variables (macros) for accessing the RPM build directory structure, as described in [21].

The `SOURCES` folder, as described above, will only contain the source files of the package, in our case the BSF4ooRexx distribution as a `.zip` file. During the stages of the build process

it will be unpacked and moved to the computer's filesystem. All necessary settings for this build process, are defined in the specs file, which is described in the following chapter.

4.2.3 Creating the Specifications (spec) File

Basically, the spec file is a simple text file, which is located in the `SPECS` directory of the `rpmbuild` folder. As already mentioned, it defines the packages meta data and the actions that will be taken during the build process. The user does not have to write the complete file on his or her own from the scratch. It is possible to create a basic spec file using the command shown in listing 31 [22].

```
1 cd ~/rpmbuild/SPECS
2 rpmdev-newspec bsf4oorexx
```

Listing 31: Creating a basic spec file.

The resulting specifications file for the BSF4ooRexx package is called `bsf4oorexx.spec` and is stored in the `SPECS` directory. After creation it is mostly empty and has to be filled according to the requirements. The content of the completely filled spec file is shown in listing 32.

```
1  Name:           bsf4oorexx
2  Version:        406
3  Release:        20110822
4  Summary:        Bean Scripting Framework for Open Object Rexx (BSF4ooRexx)
5
6  Group:          Development/Tools
7  License:        Apache Software License 2.0
8  URL:            http://sourceforge.net/projects/bsf4oorexx/
9  #Source0:
10 BuildRoot:      %{_tmppath}/%{name}-%{version}-%{release}-root-%(%{__id_u} -n)
11
12 Autoreq:        0
13 BuildRequires:  unzip
14 Requires:       oorexx >= 4.1.0
15 Requires:       java-1.6.0-openjdk >= 1.6.0
16
17 %description
```

```
18 BSF4Rexx version for ooRexx 4.1.0 and later, exploiting the new ooRexx kernel (e
       ↪ .g. allowing implementation of Java methods in ooRexx, callbacks from Java
       ↪  to ooRexx). [Camouflages Java as the dynamically typed, caseless and easy
       ↪  to use ooRexx.]
19
20 %prep
21 unzip -d %{_builddir}/BSF4ooRexxInstall %{_sourcedir}/BSF4ooRexx_install
22
23 %build
24
25 %install
26 rm -rf %{buildroot}
27 mkdir -p %{buildroot}/tmp/
28 cp -r BSF4ooRexxInstall %{buildroot}/tmp/
29
30 %post
31 rm -rf /opt/BSF4ooRexx
32 cp -R %{_sourcedir}/BSF4ooRexxInstall /tmp/
33 #cd /tmp/BSF4ooRexxInstall/bsf4oorexx/install/
34 rexx /tmp/BSF4ooRexxInstall/bsf4oorexx/install/setupAllAndRun.rex install /@q:
35
36 %preun
37 #cd /opt/BSF4ooRexx
38 rexx /opt/BSF4ooRexx/install/setupAllAndRun.rex uninstall /@q:
39
40 %clean
41
42 %files
43 %defattr(-,root,root,-)
44 %doc
45 /tmp/*
46
47 %changelog
```

Listing 32: The bsf4oorexx.spec file.

Most of the file's settings are quite self-explanatory. `Name` and `Version`, of course, define the resulting package's name and version. The version for BSF4ooRexx is defined in the file `BSF.CLS` and is 406 at the time of writing this document. `Release`, in this case, indicates the date of the distribution's current release. Here it was 20110822 which stands for August, 22nd 2011. `Summary` defines a short summary of the packages contents. The next setting, named `Group`, is very important for the package management system. It defines to which group of software this package belongs. We decided to use Development/Tools, because BSF4ooRexx can be seen as extension to ooRexx. `License` and `URL` define, under which license the package and its content can be used and where it can be found on the internet. A very useful setting is `BuildRoot`. It defines the name used for the build-

subdirectory in the `BUILDROOT` folder. Using placeholder variables for name (`%{name}`), version (`%{version}`) and release (`%{release}`) it allows to build multiple versions of the package, whithout overwriting already existing older releases or versions.

The next block of settings has to do with requirements. `Autoreq` defines, if the package management system should automatically try to determine required additional software during the package's installation. This can be a very useful feature, but in our case, we exactly know which requirements must be met and, therefore, this feature has been turned off, as the 0 (zero) indicates[9]. The setting `BuildRequires` allows to define packages that must be installed, in order for the build to succeed. The programs (packages) listed here, are used during the stages of the build process e.g. for compiling the source files. We don't want to compile any files. The only thing we do is unpack the BSF4ooRexx distribution file. Therefore, the only package listed here is `unzip`. In contrast to the build process requirements, `Requires` lists packages that have to be present, to execute the files within this package. Because BSF4ooRexx needs Java[10] and ooRexx to work properly, both have been added here with their required version. As one can see, the `Requires` keyword can be used multiple time for defining a larger number of required programs.

`%description` is a macro that allows the user to define a longer description for the package. In contrast to the `Summary` tag, it can be used as a place for important information the user should see before the package is installed.

After that, the stages definition starts. The most important stages (and their purpose), that every package uses, are described in table 2. The directory macros used here have already been defined in table 1 in chapter 4.2.2.

[9] During development, the automatic determination of additional software led to a large number of additional packages that had to be installed. Adding the `Autoreq` parameter solved this issue.

[10] For running BSF4ooRexx the Java Runtime Environment (JRE) is required. Nevertheless, there is no explicit package for the JRE availabe. It is included in the JDK package defined here.

Stage	Reads	Writes	Action
%prep	%_sourcedir	%_builddir	The preparation stage reads the sources from %_sourcedir and unpacks them to a subdirectory of the build directory (%_builddir).
%build	%_builddir	%_builddir	During the build stage, the unpacked sources are compiled within the build directory.
%check	%_builddir	%_builddir	This stage can be used to check, if the compiled programs work properly.
%install	%_builddir	%_buildrootdir	This is the most interesting stage for us. It reads the contents of the build directory and writes the files that will be installed to the users filesystem to the buildroot directory.

Table 2: Stages and their functions, as described in [21].

As one can see from this definition, the system primarily aims at creating packages directly from the source code. Normally, the source code in unpacked, configured, compiled, tested and installed inside the buildroot directory. Based upon that the package is built, containing the required files only. In our case it is not really necessary to compile any source code. Therefore, the %build and %check stages, for example, are not used. %check is not even listed in listing 32.

During the preparation stage, as already mentioned, the BSF4ooRexx distribution file is unpacked to the build directory using the unzip command. Because we don't need to compile any files, the %build stage does not contain any commands. Reaching the %install stage, the buildroot directory is cleaned and freshly populated with a tmp directory and the unpacked BSF4ooRexx files. This is necessary, because the contents of the BSF4ooRexx distribution will be placed in the systems /tmp directory during installation.

The last two important settings are %post and %preun. The first one, %post, allows the definition of commands that will be executed, after the package has been installed and the files have been moved to the filesystem. In our case, an existing (old) BSF4ooRexx directory under /opt will here be removed, the BSF4ooRexx install folder contained in the

package is moved to `/tmp` and the according setup script is started. After this step is complete, BSF4ooRexx is installed and fully functional. Just like in case of the Debian package, there are no separate RPM packages for 32 and 64 bit installations required, because the BSF4ooRexx distribution's install scripts are used here, which can handle both situations.

The second setting, `%preun`, is used, when before the package is uninstalled. It runs the uninstall script that comes with BSF4ooRexx, which removes all settings and files that have been made or created during installation.

After these settings have been configured successfully, the package can finally be built.

4.2.4 Building the Final RPM Package

To finally built the RPM package, the commands given in listing 33 have to be executed from the command line.

```
1 cd ~/rpmbuild/SPECS
2 rpmbuild -ba bsf4oorexx.spec
```

Listing 33: Building the RPM package.

This will cause `rpmbuild` to read the spec file and to go through all stages in order to build the final package. The `-ba` option tells `rpmbuild` to build all packages, also the source package which is stored in `~/rpmbuild/SRPMS`.

When the build process is finished, the resulting package is stored in `~/rpmbuild/RPMS`. From there it can be distributed and installed. The according use cases are covered in chapter 4.3.

4.3 Use Cases

The install, reinstall and uninstall features of the BSF4ooRexx RPM package are explained in the following chapters[11].

4.3.1 Install

The installation of a RPM package can either be carried out graphically, e.g. by using the Add/Remove Software option from Linux' Administration menu or manually by invoking the corresponding commands from the command line. To install the package graphically the user must simply double-click on the file created in the steps above. On Fedora 14, a message box will appear, allowing the user to install the file. This case is shown in figure 25.

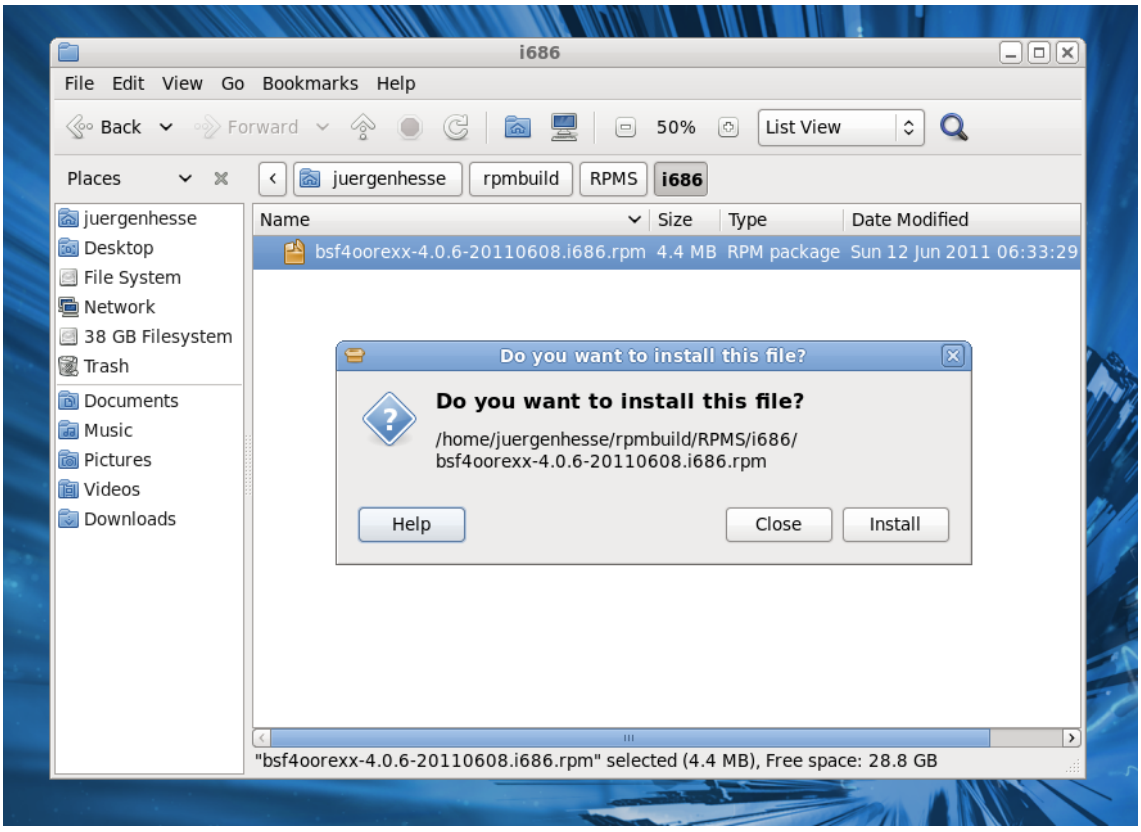


Figure 25: Installing the RPM package by double-clicking.

[11] At this point is has to be mentioned, that the resulting packages have been compiled and tested on Fedora 14 only.

After clicking on the “Install” button, the user has to proof, that he or the has sufficient privileges to install new software and that the package really should be installed, as depicted in figure 26.

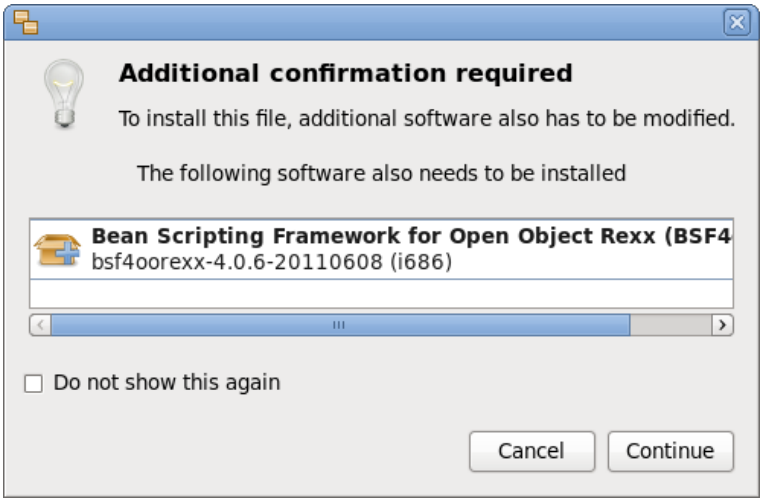


Figure 26: Authentication for the installation.

If the authentication and the confirmation have succeeded, the installation starts. The user sees a progress bar (shown in figure 27), indicating the state of the installation.

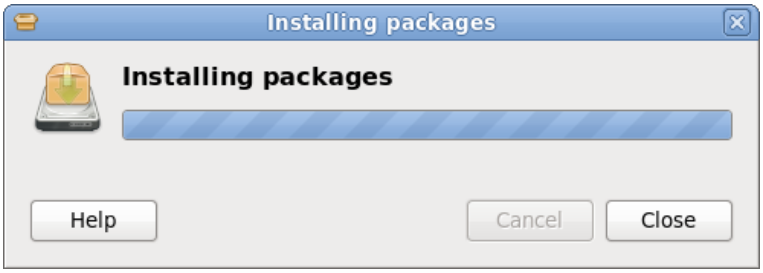


Figure 27: Installation progress.

When the installation has finished, the BSF4ooRexx package is shown in the Application’s menu and in the installed-software-section of the software management GUI. This case is shown in figure 28.

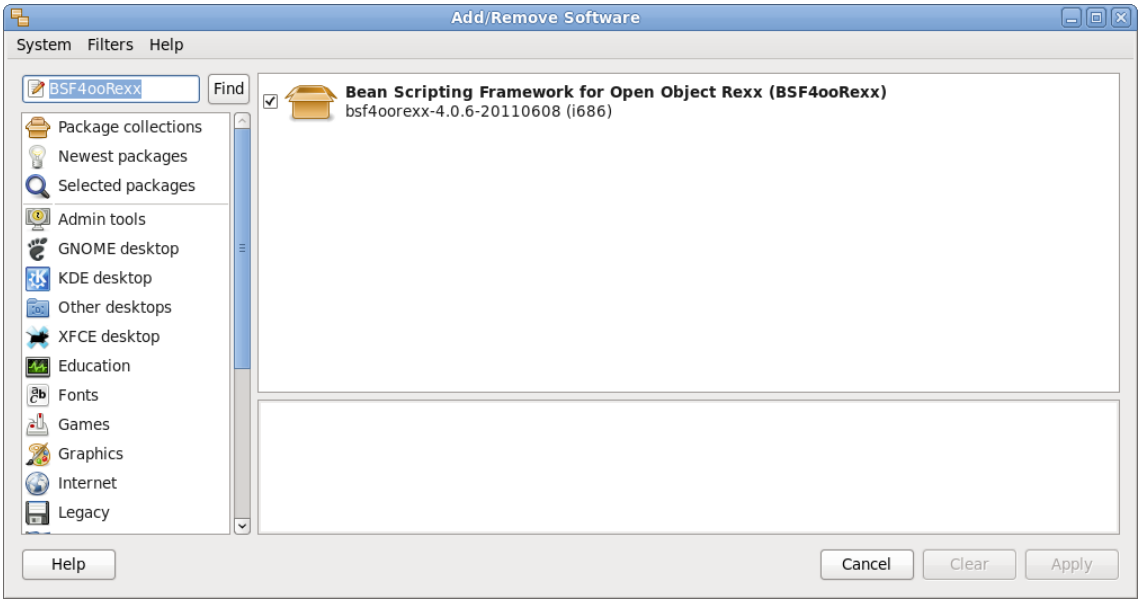


Figure 28: The Installed-Software-Section after installation.

Alternatively, the package can also be installed from the command line by invoking the command given in listing 34 as superuser.

```
1 rpm -i bsf4oorexx-406-20110822.i686.rpm
```

Listing 34: Installing the package from the command line.

4.3.2 Reinstall

If OpenOffice was not installed before installation, its support can be enabled afterwards by reinstalling the package. Unfortunately, the graphical RPM system does not support reinstallation of standalone RPM packages. Therefore, the easiest way is to start the process by choosing the corresponding entry from the BSF4ooRexx Menu, as depicted in figure 29.

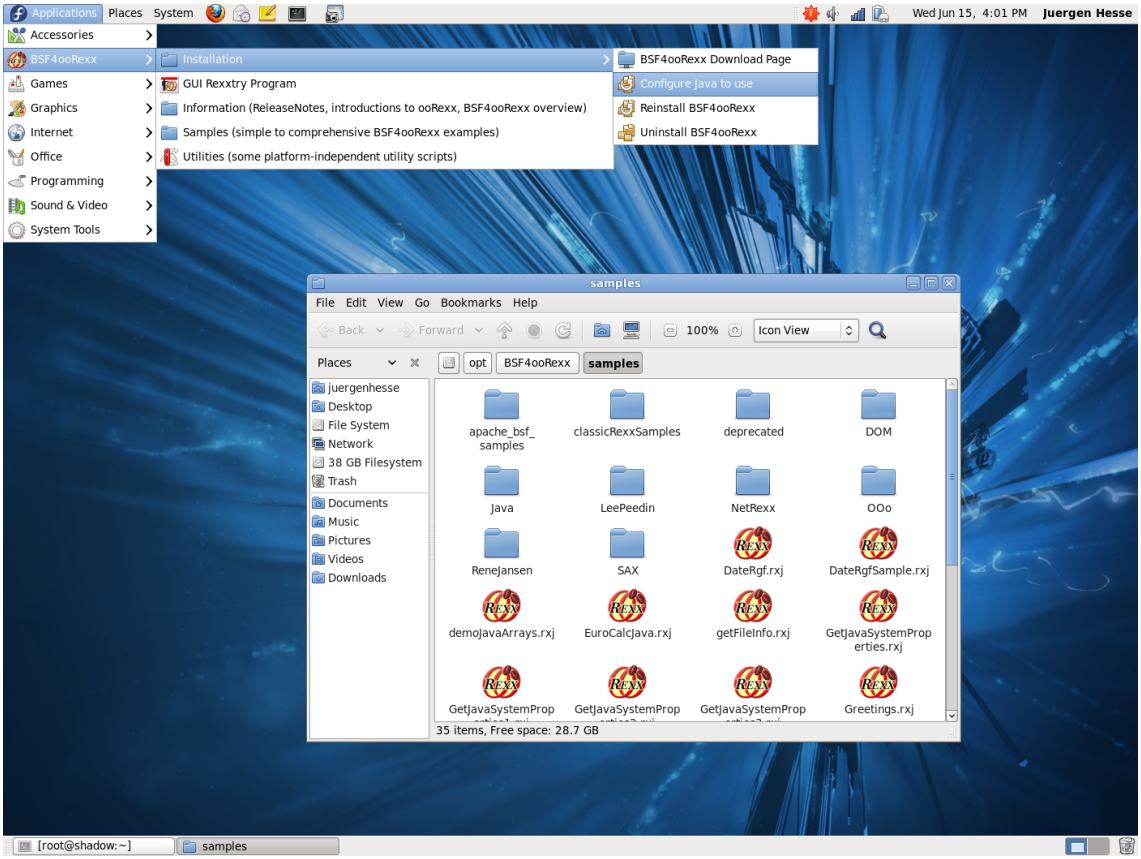


Figure 29: Reinstalling BSF4ooRexx from the menu.

After the reinstallation has successfully been carried out, the user gets a nice message.

4.3.3 Uninstall

If the user wants to deinstall the package he or she has to open the system's software management GUI. In the installed-software-section the BSF4ooRexx package appears. If a large number of other packages has been installed, the user can use the search field on the top left of the window to search for the package. To uninstall the package, the user has to remove the check in front of the package icon and to click on the "Apply" button, as shown in figure 30.

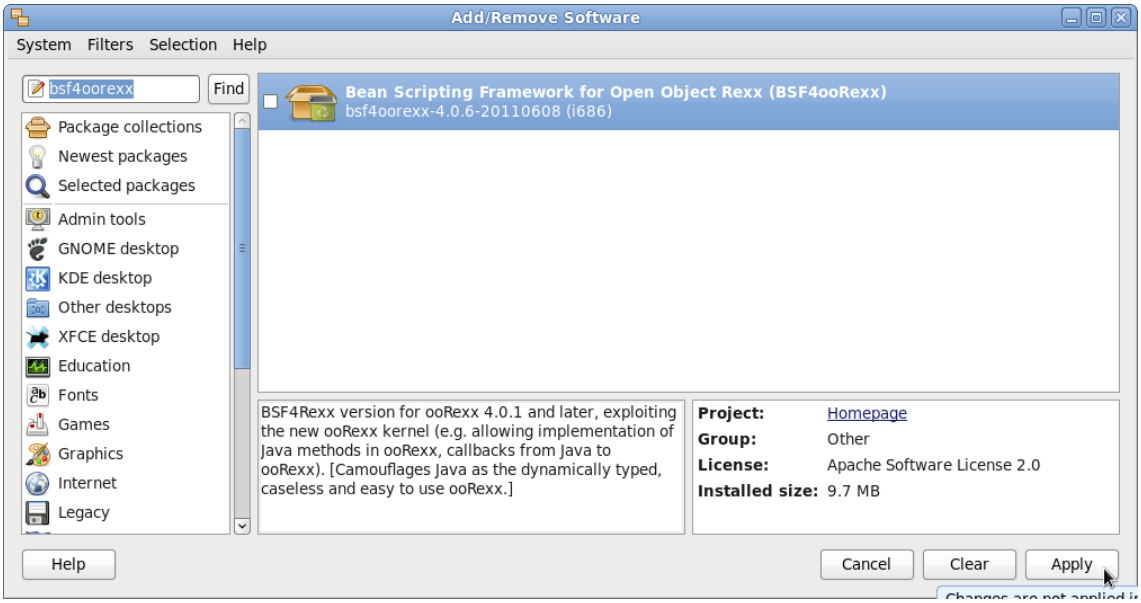


Figure 30: Uninstalling the package.

Before the uninstall process starts, the user must again proof, that he or she has sufficient privileges to remove software. This situation is depicted in figure 31.



Figure 31: Authentication for uninstalling the BSF4ooRexx package.

When the autentication has succeeded, the uninstall process starts. Until it is finished, the user sees a progress bar at the bottom left of the software management window, as shown in figure 32.

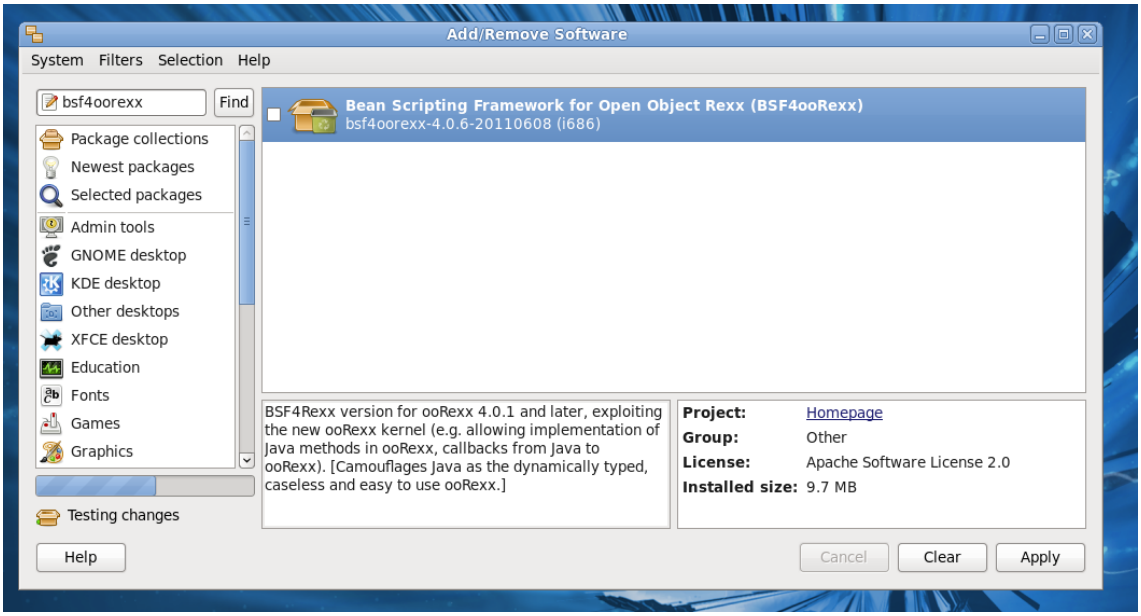


Figure 32: Deinstallation progress.

After the package has successfully been removed, is disappears from the installed-packages list.

Alternatively, the package can be removed using the command line, by invoking the command given in listing 35 as superuser.

```
1 rpm -e bsf4oorexx
```

Listing 35: Uninstalling the package from the command line.

The effects are the same as they are when using the graphical uninstall mentioned above.

Additionally the user can uninstall BSF4ooRexx using the uninstall option in the BSF4ooRexx menu. In this case BSF4ooRexx will be removed completely, but the package entry will remain in the package manager and has to be removed by hand. This issue could only be solved by changing the install script contained in the BSF4ooRexx distribution, to automatically detect the current Linux version and the appropriate package removal command.

5 Iceberg Installer for Mac OS X

Authors: Jürgen Hesse, Manuel Paar

For the creation of the installer package for Mac OS X, we used a program called Iceberg [23]. Beside the facts that it is free and easy to use, it fulfills our requirements for flexibility of configuration. It allows to define different types of scripts (as shown in chapter 5.3.3) as well as the software requirements (e.g. that Java is installed in the correct version) that have to be met before the installation process is started.
There are also alternatives to Iceberg e.g. PackageMaker that comes with XCode, Apple's Integrated Development Environment [24]. The disadvantages of these alternatives is, that they are either not free or are lacking important features, such as the possibility of defining postflight scripts that are executed after installation.
Chapter 5.1 will describe necessary preliminary works, chapter 5.2 will take a closer look on the features of Iceberg, followed by chapter 5.3, which shows how an installer will be created. Last but not least, chapter 5.4 will show the installer in action.

5.1 Preliminary Works

This chapter describes the necessary preliminary works, like gathering the source folders for BSF4ooRexx and ooRexx, as well as creating the install and uninstall scripts.

5.1.1 The Source Folder's Structure

For an easier handling, it is recommended to copy the installation files in a sources folder, which contains all required files for building the installer. Its structure is shown in figure 33.

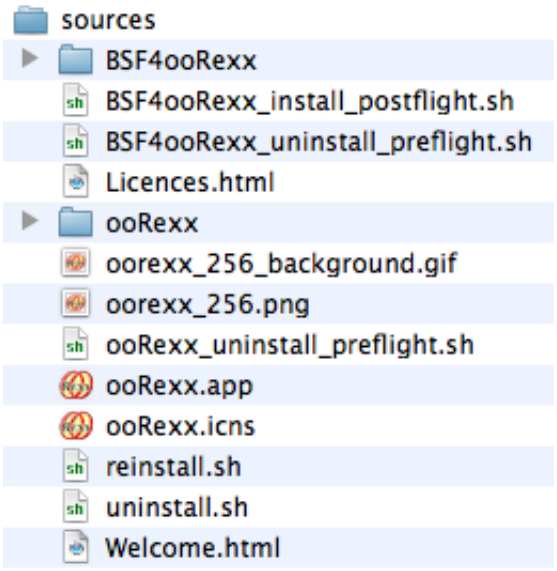


Figure 33: The sources folder.

The `BSF4ooRexx` folder contains all files and folders that are necessary to successfully use BSF4ooRexx. Its structure, depicted in figure 34, is based on the BSF4ooRexx framework defined in [1].

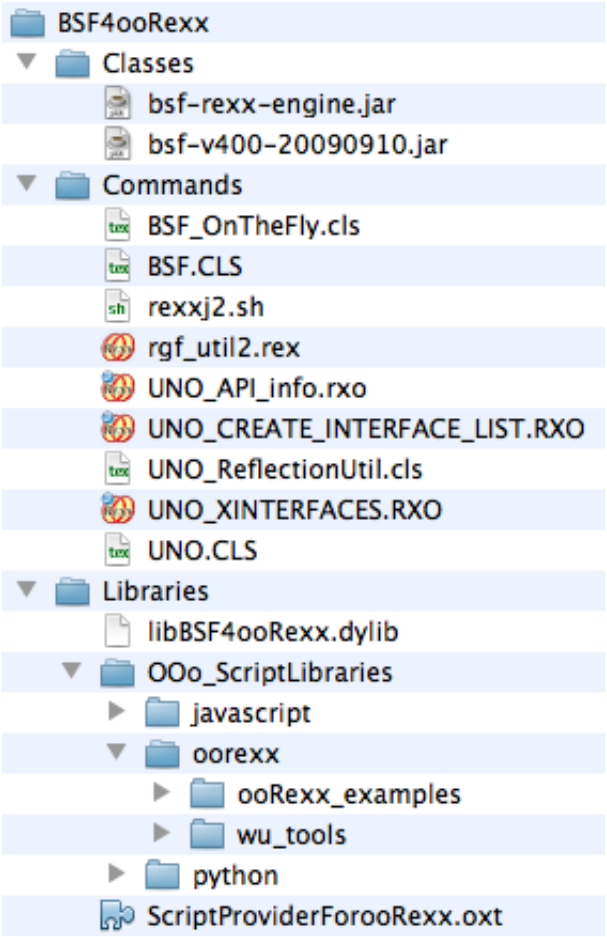


Figure 34: The `BSF4ooRexx` folder.

The second folder is the source folder for ooRexx. Because there was no official ooRexx package available at the time of writing this thesis, we were kindly allowed to use the unofficial ooRexx 4.1.1. Its contents is shown in figure 35.

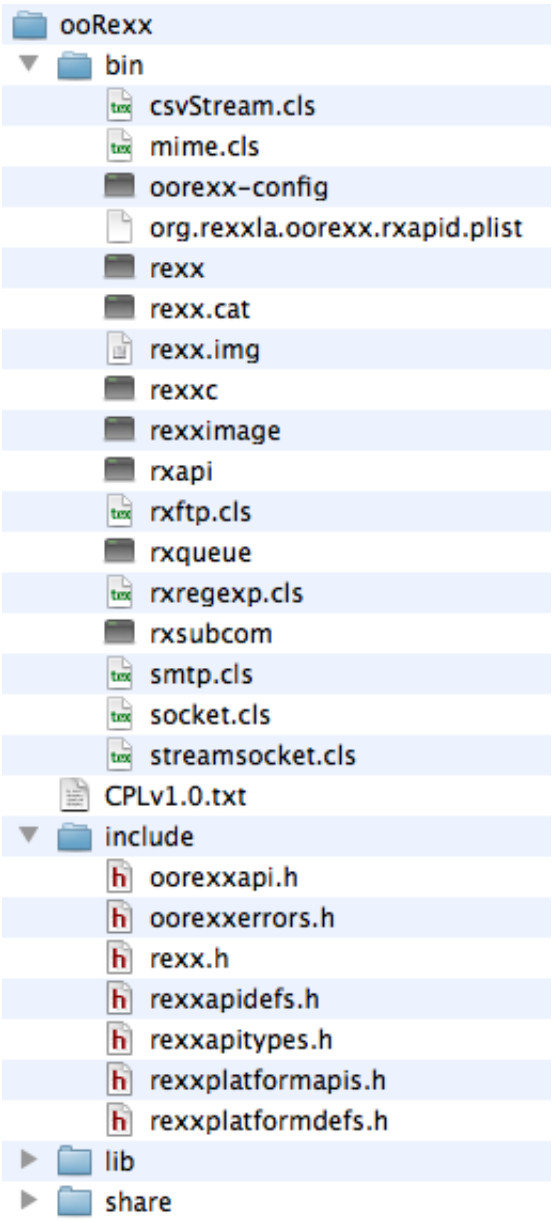


Figure 35: The `ooRexx` folder.

Like the `BSF4ooRexx` folder, the `ooRexx.app` folder is also based on the application bundle developed in [1]. Its name was changed to ooRexx because the bundle now contains ooRexx and BSF4ooRexx documentations and the samples. Furthermore the menu bar was modified to support both of them. Its complete and updated source code can be found in chapter 7.5.1 in the Appendix. The structure of the `ooRexx` folder is depicted in figure 36.

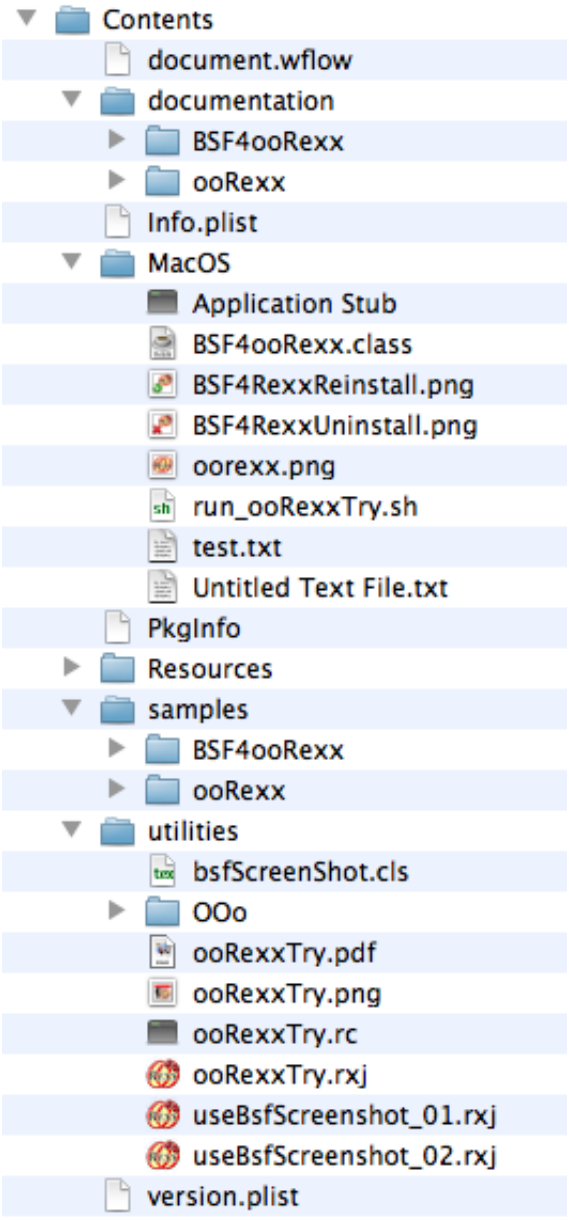


Figure 36: The contents of `ooRexx.app`.

5.1.2 The Install Script

The install script, which is shown in full length in chapter 7.5.2 in the Appendix, creates symbolic links to ooRexx and BSF4ooRexx in the `/usr/bin` and `/usr/lib` directories. Furthermore it links the ooRexx manual pages to `/usr/share/man/man1` and adds the C++ header files to `/usr/include`. After copying all ooRexx-related files, the ooRexx `rxapi` daemon is restarted.

Finally, the BSF4ooRexx support for OpenOffice/LibreOffice is installed by adding the corresponding `JAR` files to the `CLASSPATH` environment variable and copying the ooRexx macros to the OpenOffice/LibreOffice's script directory. This step will only be executed, if OpenOffice/LibreOffice is installed. If, not it is omitted.

Last but no least, the `lsregister` command is used to refresh all file association settings. As a result, all ooRexx and BSF4ooRexx script files are shown with their correct file type icons in the Mac OS X finder.

5.1.3 The Uninstall and Reinstall Scripts

The uninstall and reinstall process is primarily handled by four different scripts. All of which can be found in chapter 7.2 in the Appendix.

uninstall.sh: This script is executed if the user presses the uninstall or reinstall item in the ooRexx menu bar. It needs a parameter, indicating which process has to be started. If the parameter is "`reinstall`", the currently present ooRexx and BSF4ooRexx installation will be removed and newly installed, using the script described in chapter 7.5.4 in the Appendix. On the other hand, if the parameter is "`uninstall`", the installation will be removed.

ooRexx_uninstall_preflight.sh: This script removes all symbolic links that were created by the installer script from `/usr/bin`, `/usr/lib`, `/usr/include` and `/usr/share/man/man1`. In addition, it unloads the ooRexx `rxapi` daemon, removes it and starts `BSF4ooRexx_uninstall_preflight.sh`.

BSF4ooRexx_uninstall_preflight.sh: Like the previous script, it removes the created symbolic links from the `/usr/bin` and `/usr/lib` directories. Furthermore the ooRexx and BSF4ooRexx directories in `/opt` are removed and the `ooRexx.app` bundle in `/Applications` is deleted. Finally the `CLASSPATH` entries in `/etc/profile` are

removed.

If the parameter "`uninstall`" has previously been set, the uninstall process is finished here. Otherwise, the `reinstall.sh` script gets invoked.

reinstall.sh: This script installs a fresh copy of ooRexx and BSF4ooRexx in the `/opt` directory and `ooRexx.app` in the `/Applications` folder. Afterwards the install script, described in chapter 5.1.2, is executed.

5.2 The Installer

"Iceberg is an Integrated Packaging Environment (IPE) that allows you to create packages or metapackages conforming to the Mac OS X specifications" [23]. It has a graphical user interface for creating and editing installer projects. After installation, it is available in the `/Applications` folder of Mac OS X. Its features will be discussed in further detail in the following chapter.

5.3 Creating the Install Package for Mac OS X with Iceberg

To create a new project, the user has to start Iceberg and select `File` → `New Project...` from its menu bar. In the appearing assistant, the subitem `Package` below the item `Core Templates` has to be chosen. In the next step the project name and its root directory have to be set. After finishing the assistant, Iceberg creates an empty project and opens it in its main window, as shown in figure 37.

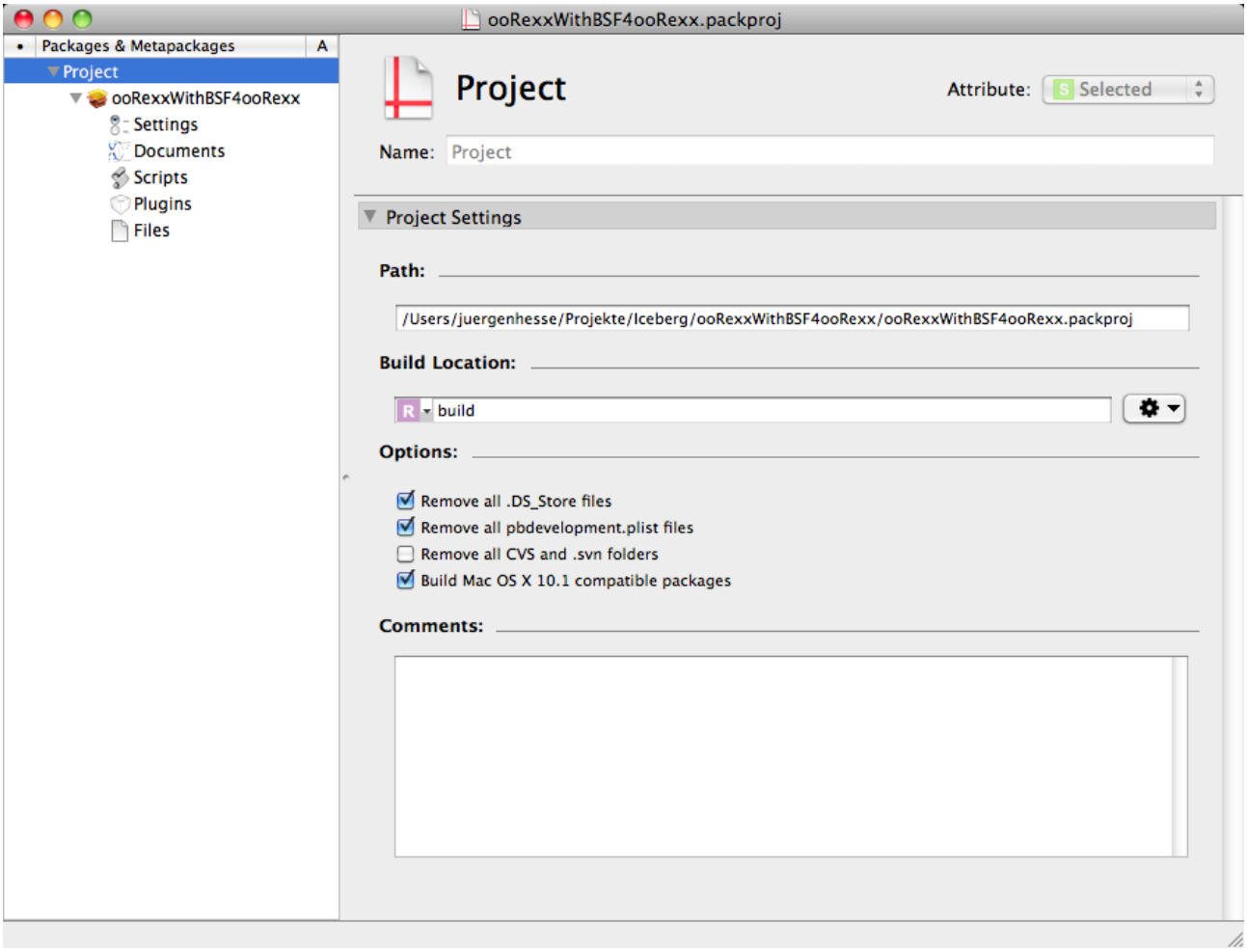


Figure 37: The Iceberg main window.

The left frame contains all open projects. The right frame shows the settings for the currently selected project. These settings are split into the following parts:

- `Settings`,
- `Documents`,
- `Scripts`,
- `Plugins` and
- `Files`.

The next chapters will take a closer look at these parts.

5.3.1 Settings

In the `Settings` part, depicted in figure 38, the user can specify general settings such as the package's title, the version and its icon file. The icon must be available in the Apple-specific `.icns` format. The procedure for creating such an icon file is described in chapter 3.3.1 "Creating the BSF4ooRexx Application Bundle" in [1].
Iceberg allows to select between the usage of the absolute or relative path to the icon file. The absolute path is workstation-specific and may be used if the project is developed only on a single computer. The relative path, on the other hand, is based on the project's root directory which has been set in chapter 5.3. It may be used if the project is shared with other developers.

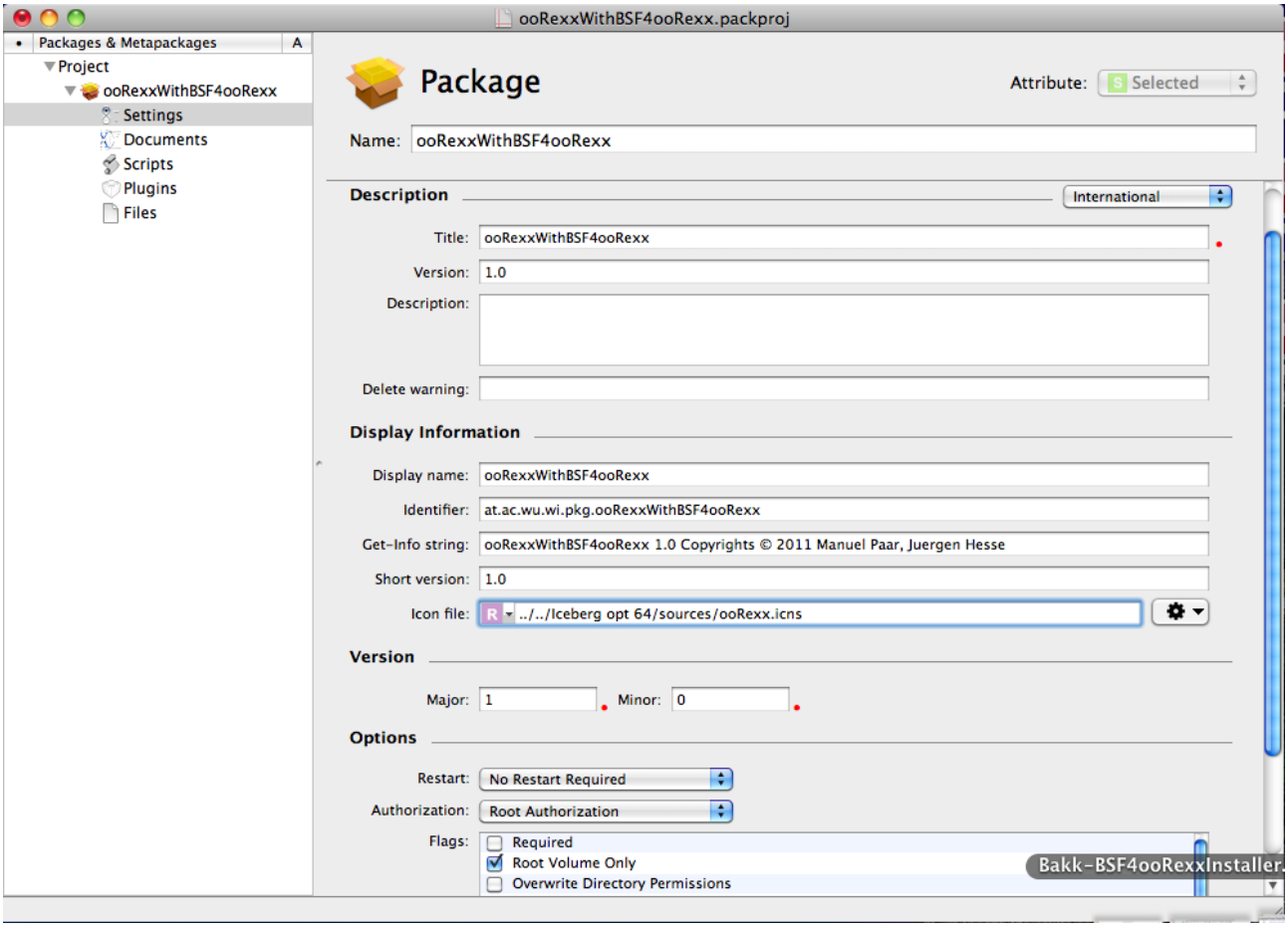


Figure 38: The Iceberg project settings.

A very important part in the `Settings` window is the `Options` section. It allows to define

- if a restart of the system is necessary after installation,
- if the user has to be authenticated as system administrator,
- if the program can be installed only onto the system's boot volume and
- further options regarding the file system.

In our case, we defined that `Root Authorization` is required, that the program can be installed to the `Root Volume Only` and that the installer should `Follow Symbolic Links`.

5.3.2 Documents

In this part, shown in figure 39, the user can specify the background image, the welcome text, the ReadMe file and the licence information which will be shown in the installer. Again, it is possible to choose between the absolute and the relative path for the given resources. The Welcome-, ReadMe- and Licence files can be defined in the `TXT`, `RTF`, `RTFd` and `HTML` format [25].

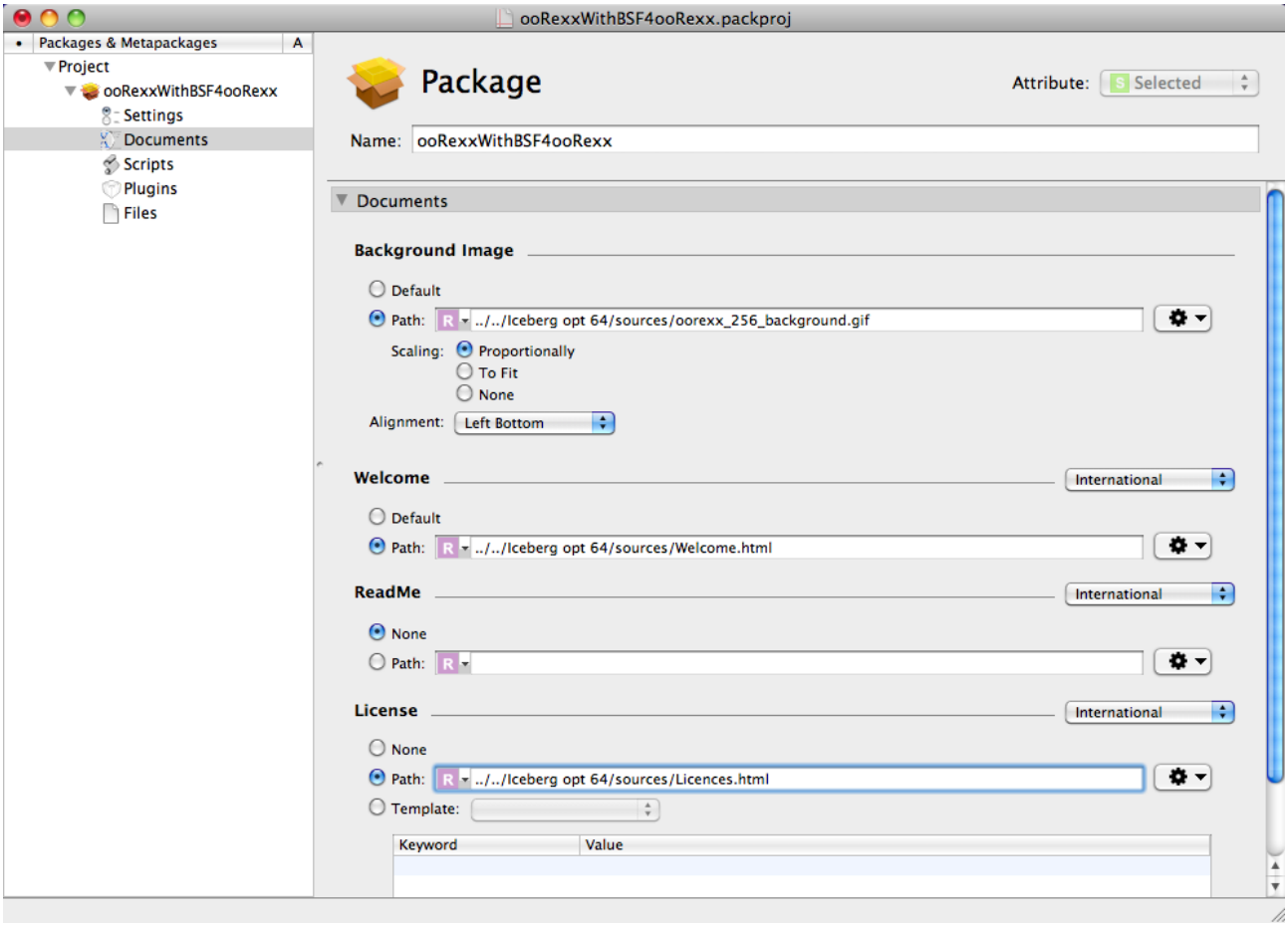


Figure 39: The Iceberg project's document settings.

5.3.3 Scripts

The `Installation Scripts` part of the `Scripts` settings window allows to define script files, that are executed before or after copying the program files to their destination. In our case, it was only necessary to execute a so-called postflight script, which runs after copying the files. This setting is shown in figure 40. The complete source code of the postflight script is shown in the Appendix in chapter 7.5.2.

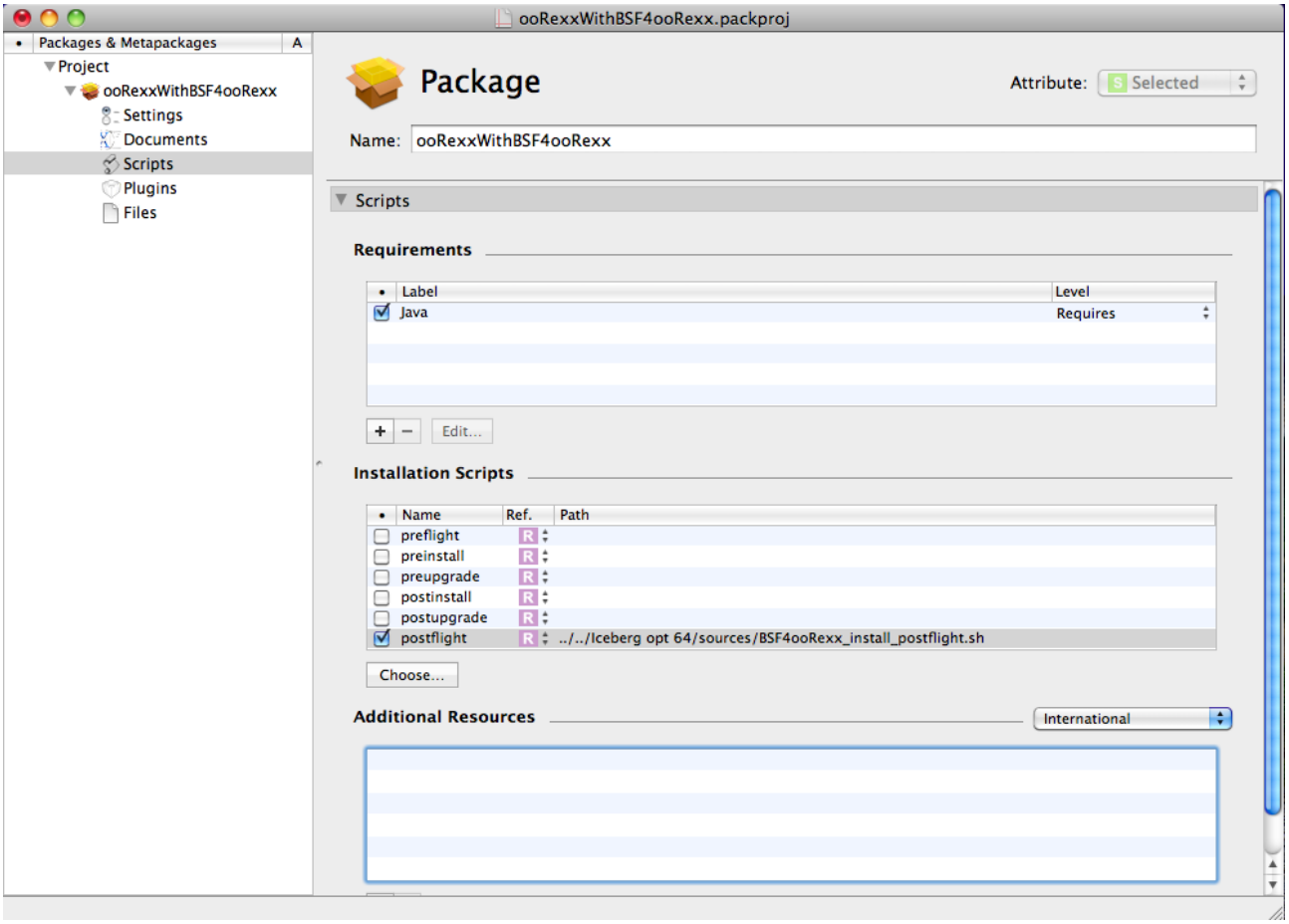


Figure 40: The Iceberg project's script settings.

The `Scripts` settings also allow to define the requirements the system has to fulfill before the program can be installed. In our case, the only requirement was, that Java 1.6 or higher is installed. To check if the correct Java version is present, the settings shown in figure 41 have been made. They check if the Java Bundle located at `/System/Library/Java/JavaVirtualMachines/1.6.0.jdk` has a `plist` value for the key `CFBundleIdentifier` that is equal to `com.apple.javajdk16`. If the requirement is not met, the message defined in the `Alert Dialog` section is shown and the user will not be able to start the installation[12].

[12] For running BSF4ooRexx the Java Runtime Environment (JRE) is required. Nevertheless, there is no explicit package for the JRE availabe on Mac OS X. It is included in the JDK package defined here.

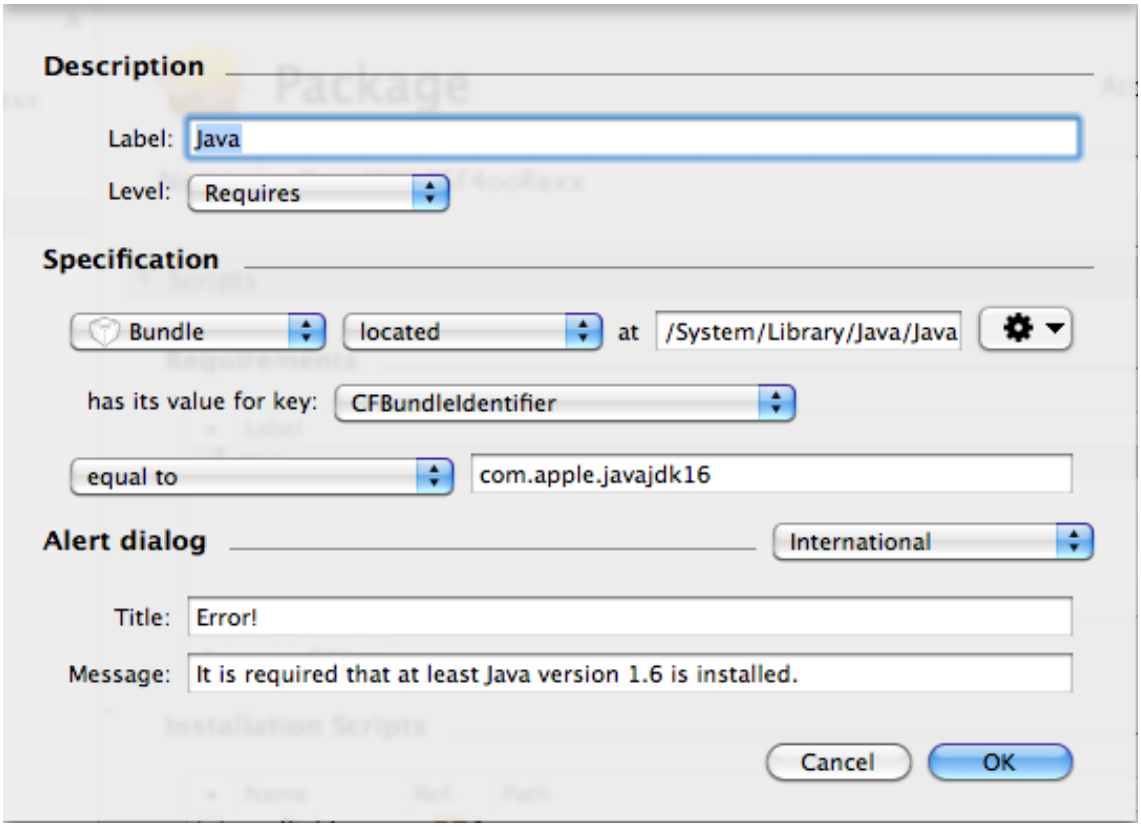


Figure 41: Defining the installer's requirements.

5.3.4 Plugins

The `Plugin` settings, shown in figure 42, allow to add additional panes or steps to the installation process. In our case it was not necessary to add such steps, therefore the default setting were used.

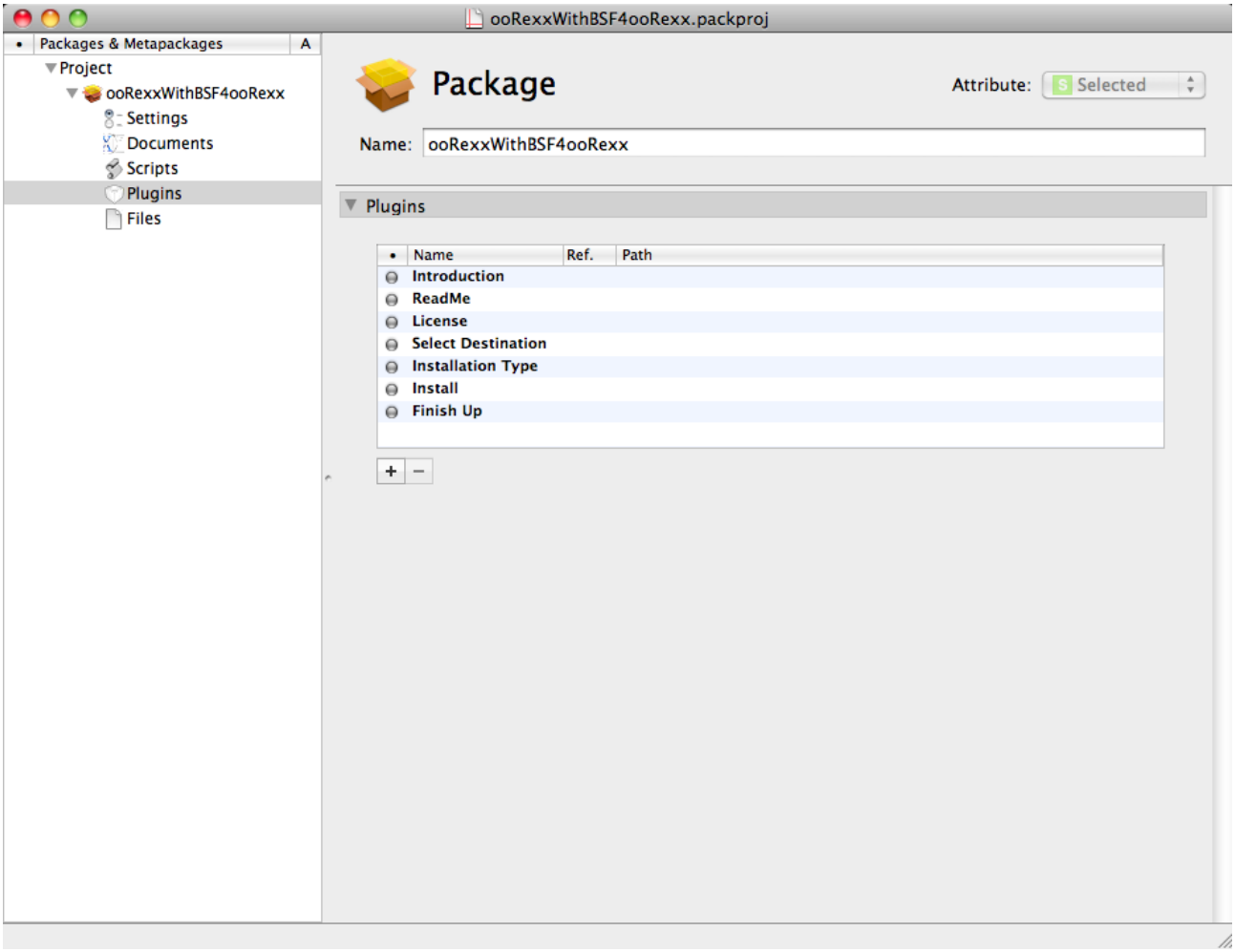


Figure 42: The Iceberg project's plugin settings.

5.3.5 Files

The `File` settings, depicted in figure 43, allow to define which files the installer will copy to which destination. Files and folders can be added either by using the context menu or by drag-and-drop.

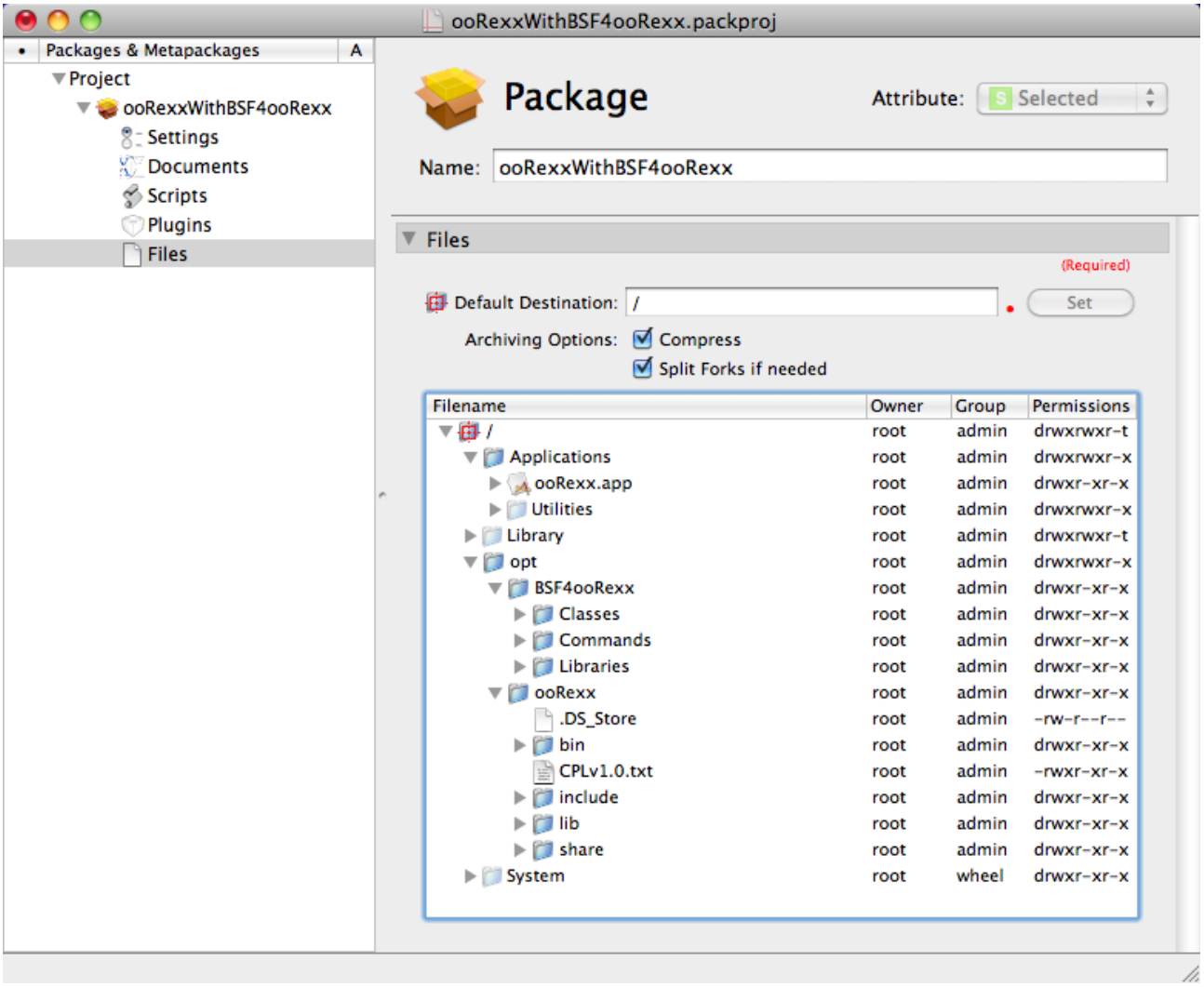


Figure 43: The Iceberg project's file settings.

First, the `/opt` folder has to be created. This is done by right-clicking the "/"-icon and choosing "New Folder" in the context menu. Then the `BSF4ooRexx` and `ooRexx` folders from the `sources` folder, created in chapter 5.1.1, have to be added by drag-and-drop.

The next step is to drag the `ooRexx.app` into the `Applications` folder. In order to allow the reinstallation of the package, the sources folder, which's contents are described in chapter 5.1.1, has to be added to the application bundle itself. Therefore, the bundle's structure has to be expanded by right-clicking on `ooRexx.app` and choosing "Expand All" from the context menu. This makes it possible to drag the `sources` folder into the `Contents` folder of the bundle.

Alternatively to using the GUI interface of Iceberg, the Iceberg project file (`.packproj`) can be edited manually. It is a XML file which contains all necessary information for creating an installer package. Due to the high complexity of this file, it was decided to use the graphical way instead.

After adding all files and folders, the installer package can be built.

5.3.6 Building the Package

There are three ways to build the installation package.

Using the Iceberg GUI: To build the package the user simply has to choose `Build` → `Build` from the menu bar.

Using the command line: The command `freeze`, which is installed together with Iceberg, can be used to build the installer from the command line [26].

Using the Makefile: Alternatively, it is also possible to use the Makefile shown in chapter 7.5.9 in the Appendix. This Makefile also creates up-to-date versions of the ooRexx and BSF4ooRexx folders in Iceberg's sources folder.

5.4 Use Cases

The usage of the installer, reinstaller and uninstaller for Mac OS X is explained in the following chapters.

5.4.1 Install

After double-clicking on the installer package, the installer starts and shows a dialog as depicted in figure 44.



Figure 44: Welcome dialog of the ooRexx and BSF4ooRexx installer for Mac OS X.

When the “`Next`” button will be clicked, the license agreement, as shown in figure 45, will be displayed.

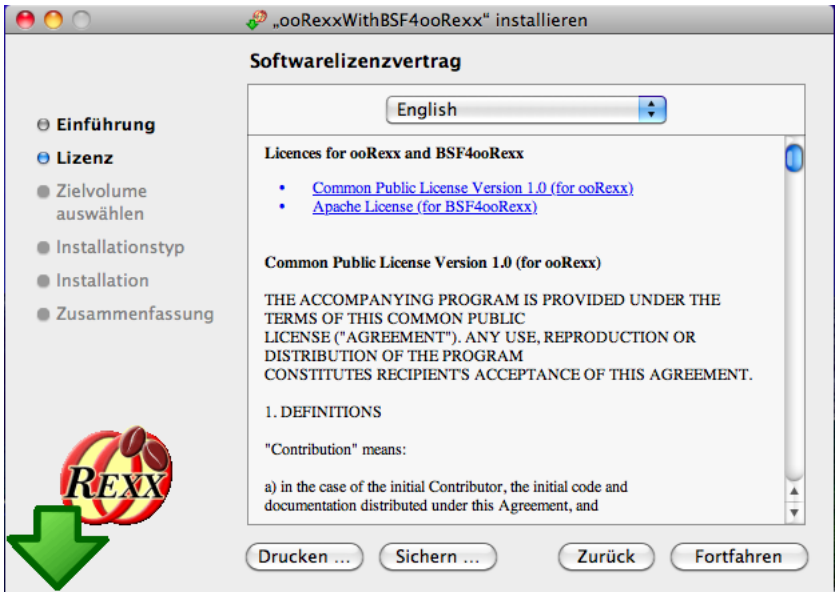


Figure 45: Licence agreement of the ooRexx and BSF4ooRexx installer for Mac OS X.

After accepting the license, the user has to choose the system drive, as shown in figure 46.



Figure 46: Installation target of the ooRexx and BSF4ooRexx installer for Mac OS X.

The next screen, depicted in figure 47, shows a summary of the installation process.



Figure 47: Installation summary of the ooRexx and BSF4ooRexx installer for Mac OS X.

After clicking on the “`Install`” button the installation, as shown in figure 48, begins.

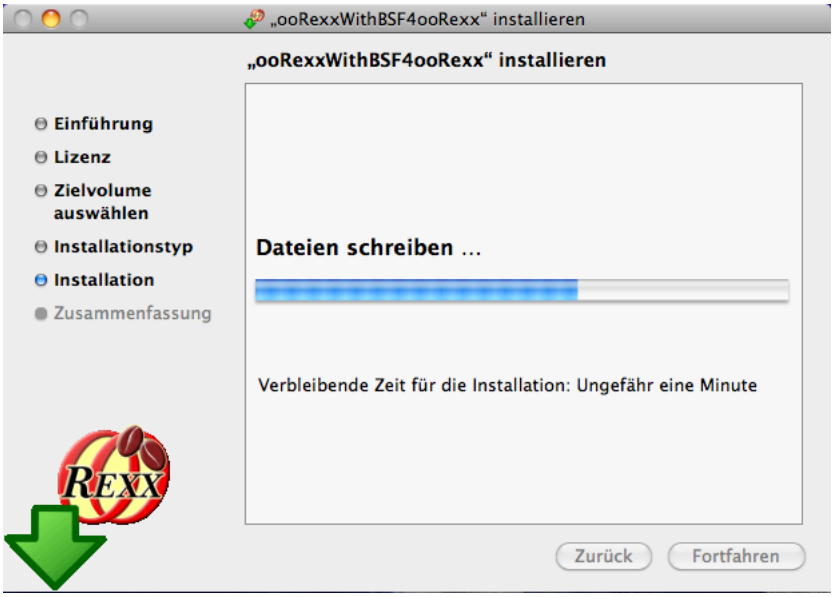


Figure 48: Installing window of the ooRexx and BSF4ooRexx installer for Mac OS X.

The installation was successful, when the dialog depicted in figure 49 is displayed.



Figure 49: Finish dialog of the ooRexx and BSF4ooRexx installer for Mac OS X.

5.4.2 Reinstall

The reinstaller can be started from the ooRexx menu bar by choosing `Installation` → `Reinstall ooRexx with BSF4ooRexx`. After clicking on the menu item, the message

shown in figure 50 will be displayed.

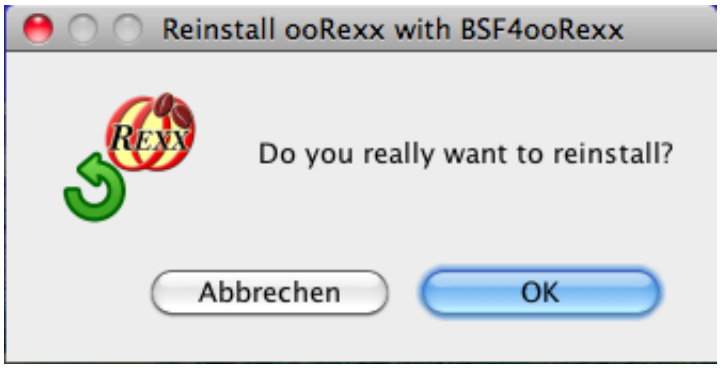


Figure 50: Reinstall message of the ooRexx and BSF4ooRexx reinstaller for Mac OS X.

The reinstallation was successful, when the dialog depicted in figure 51 is shown.

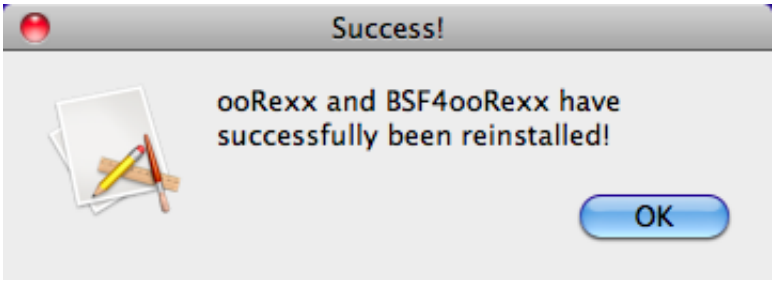


Figure 51: Finish dialog of the ooRexx and BSF4ooRexx reinstaller for Mac OS X.

5.4.3 Uninstall

The uninstaller can be started from the ooRexx menu bar by choosing `Installation` → `Uninstall ooRexx with BSF4ooRexx`. After clicking on the menu item, the message shown in figure 52 will be displayed.



Figure 52: Uninstall message of the ooRexx and BSF4ooRexx uninstaller for Mac OS X.

The uninstallation was successful, when the dialog depicted in figure 53 is shown.

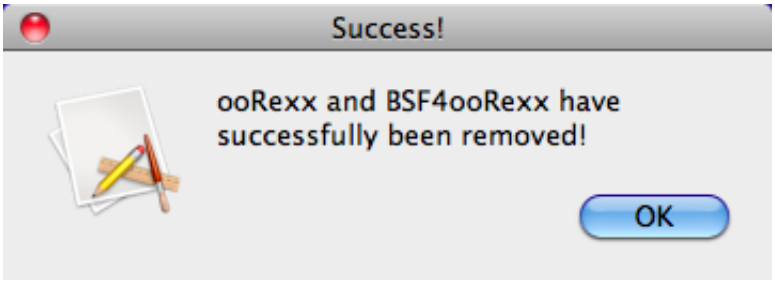


Figure 53: Finish dialog of the ooRexx and BSF4ooRexx uninstaller for Mac OS X.

6 Roundup and Outlook

Authors: Jürgen Hesse, Manuel Paar

After intensive work on the mentioned installers, it can be said that each of which has its own advantages and disadvantages. The NSIS installer for example, is a very mighty and flexible tool, which can be extensively customized. On the other hand, it does not have a graphical interface and needs an external editor for creating the installation scripts. Regarding usability, the creation of installation scripts using NSIS is comparable to the creation of Debian- or Red Hat-based packages. In contrast to these, Iceberg is primarily using a graphical interface, which makes it very comfortable to use on the cost of customisability.

Nevertheless, it can be critically noted, that the work on the installers created during this thesis will never be finished completely. The reason for this, are the continuous adjustments on the underlaying operating systems, e.g. changes on registry keys on Windows, which will require permanent maintainance of the installers, even if the hardest part of the work is finished. This fact is especially true for the Linux packages: On the one hand, there are differences between the particular versions of operating systems from the same distribution and, on the other hand, there are even larger differences between particular distributions that use the same package management system (e.g. Debian or Ubuntu).

Personally, we can say that the creation of the Mac OS X installer was the biggest challenge for us, due to its characteristics (e.g. its special Java implementation or the Application Bundle structure). Therefore, we decided to develop this installer together.

Finally, it can be stated, that the ease of installation of BSF4ooRexx on the mentioned operating systems using the created installer packages was largely improved. We hope this helps making BSF4ooRexx more appealing to a larger number of users.

7 Appendix

In the following chapters we have collected the used programs, the source codes which are necessary to create the installers for each operating system.

7.1 Used Programs

Authors: Jürgen Hesse, Manuel Paar

To carry out the tasks described in this work, the programs explained in the following chapters have been used.

7.1.1 Windows

HM NIS Editor: HM NIS Edit is an editor for NSIS. It is free and is available from [4].

NSIS (Nullsoft Scriptable Install System): Nullsoft Scriptable Install System (NSIS) is a script based Installer for Windows. It was developed by Nullsoft. NSIS is open source software and licensed under different free software licences [3].

7.1.2 Debian Linux

Debian Package Management System (deb): The Debian package management system is capable of installing, uninstalling and updating software programs organized in packages. The system comes with every Debian-based Linux distribution and doesn't have to be installed separately.

7.1.3 Red Hat Linux

RPM Package Management System (rpm): *The RPM Package Manager (RPM) is a ... command line driven package management system capable of installing, uninstalling, ... and updating computer software packages. Each software package consists of files along with information about the package like its version, a description, and the like* [20].

The RPM system comes with every Red Hat-based Linux distribution and does not have to be installed separately.

7.1.4 Mac OS X

Iceberg: *"Iceberg is an Integrated Packaging Environment (IPE) that allows you to create packages or metapackages conforming to the Mac OS X specifications"* [23]. It has a graphical user interface for creating and editing installer projects.

7.2 Windows Source Codes

Author: Manuel Paar

The following chapters contain the source codes for the Windows installer.

7.2.1 The NSIS Installation Script for 32 bit Windows

The following listing contains the installation script for 32 bit Windows. It is described in detail in chapter 2.2.

```
; Name: BSF4ooRexx_x32.nsi
; Purpose: NSIS script for BSF4ooRexx (32 bit)
; Author: Juergen Hesse, Manuel Paar
; Version: 1.0
;
; ------------------- Apache Version 2.0 license --------------------
;    Copyright (C) 2011 Juergen Hesse, Manuel Paar
;
;    Licensed under the Apache License, Version 2.0 (the "License");
;    you may not use this file except in compliance with the License.
;    You may obtain a copy of the License at
;
;          http://www.apache.org/licenses/LICENSE-2.0
;
;    Unless required by applicable law or agreed to in writing, software
;    distributed under the License is distributed on an "AS IS" BASIS,
;    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
;    implied.
;    See the License for the specific language governing permissions and
;    limitations under the License.
; -------------------------------------------------------------------

; load more functions
!include "MUI.nsh"
!include "EnvVarUpdate.nsh"

; set product name and uninstall registry-key
!define PRODUCT_NAME "BSF4ooRexx"
!define PRODUCT_UNINST_KEY "Software\Microsoft\Windows\CurrentVersion\Uninstall\
    ↪ ${PRODUCT_NAME}"
!define PRODUCT_UNINST_ROOT_KEY "HKLM"

; TO CUSTOMIZE
; Source-Folder
!define BSF_ROOT ".\BSF4ooRexx_install\bsf4oorexx"
; Script Location (getBSFVersion.rex, ...)
!define SCRIPT_PATH "."
```

```
37
38 SetCompressor lzma
39 ; MUI Settings
40 !define MUI_ABORTWARNING
41 !define MUI_ICON "${BSF_ROOT}\install\images\windows\bsf4rexx-install.ico"
42 !define MUI_UNICON "${BSF_ROOT}\install\images\windows\bsf4rexx-uninstall.ico"
43 ; Welcome page
44 !insertmacro MUI_PAGE_WELCOME
45 ; License page
46 !define MUI_LICENSEPAGE_CHECKBOX
47 !insertmacro MUI_PAGE_LICENSE "${BSF_ROOT}\ApacheLicense2.0.txt"
48 ; Instfiles page
49 !insertmacro MUI_PAGE_INSTFILES
50 ; Finish page
51 !insertmacro MUI_PAGE_FINISH
52 ; Uninstaller pages
53 !insertmacro MUI_UNPAGE_INSTFILES
54 ; Language files
55 !insertmacro MUI_LANGUAGE "English"
56 ; Reserve files
57 !insertmacro MUI_RESERVEFILE_INSTALLOPTIONS
58
59 Var JAVA_INSTALLATION_MSG
60 Var REXX_HOME
61 Var REXX_VER
62 Var REXX_INSTALLATION_MSG
63 Var OO_HOME
64 Var LO_HOME
65 Var WINDOWS_ARCHITECTURE
66 Var PRODUCT_VERSION
67
68 Name "${PRODUCT_NAME} $PRODUCT_VERSION"
69 OutFile "${PRODUCT_NAME}_Setup_x32.exe"
70 InstallDir "$PROGRAMFILES\${PRODUCT_NAME}"
71 XPStyle on
72 ShowInstDetails show
73 ShowUnInstDetails show
74
75 ; copy all files to the destination folder
76 ; create start menue entries
77 ; create file associations
78 Section "BSF4ooRexx" SEC01
79   SetShellVarContext all
80   SetAutoClose false
81   SetOverwrite ifnewer
82
83   ; copy all files to the destination folder
84   SetOutPath "$INSTDIR"
85   File "${BSF_ROOT}\addLinks2OOoSnippets.rxo"
86   File "${BSF_ROOT}\ApacheLicense2.0.txt"
87   File "${BSF_ROOT}\asm_LICENSE.TXT"
88   File "${BSF_ROOT}\bsf-rexx-engine.jar"
89   File "${BSF_ROOT}\bsf-v400-20090910.jar"
90   File "${BSF_ROOT}\BSF.CLS"
91   File "${BSF_ROOT}\install\32\BSF4ooRexx.dll"
92   File "${BSF_ROOT}\BSF_OnTheFly.cls"
```

```
93    File "${BSF_ROOT}\credits.txt"
94    File "${BSF_ROOT}\janino_LICENSE.TXT"
95    File "${BSF_ROOT}\refcardBSF4ooRexx.pdf"
96    File "${BSF_ROOT}\refcardOOo.pdf"
97    File "/oname=rexxj.cmd" "${BSF_ROOT}\rexxj2.cmd"
98    File "${BSF_ROOT}\rexxj2.cmd"
99    File "${BSF_ROOT}\rexxj2-32.sh"
100   File "${BSF_ROOT}\rexxj2-64.sh"
101   File "${BSF_ROOT}\rexxj2.sh"
102   File "${BSF_ROOT}\rgf_util2.rex"
103   File "${BSF_ROOT}\UNO.CLS"
104   File "${BSF_ROOT}\UNO_API_info.rxo"
105   File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
106   File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
107   File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
108   File ".\${PRODUCT_NAME}_Setup_x32_reinstall.exe"
109
110   File /r "${BSF_ROOT}\information"
111   File /r "${BSF_ROOT}\install"
112   File /r "${BSF_ROOT}\samples"
113   File /r "${BSF_ROOT}\Templates"
114   File /r "${BSF_ROOT}\utilities"
115
116   ; create start menue entries
117   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}"
118   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\GUI RexxTry Program (ooRexxTry.rxj
          ↪ ).lnk" "$INSTDIR\utilities\ooRexxTry.rxj" "" "$INSTDIR\install\images\
          ↪ windows\oorexxtry.ico"
119   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Information (ReleaseNotes,
          ↪ introductions to ooRexx, ${PRODUCT_NAME} overview).lnk" "$INSTDIR\
          ↪ information"
120
121   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}\Installation"
122   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\${PRODUCT_NAME}
          ↪ Download Page.lnk" "http://sourceforge.net/projects/bsf4oorexx/"
123
124   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Samples (simple to comprehensive $
          ↪ {PRODUCT_NAME} examples).lnk" "$INSTDIR\samples"
125   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Utilities (some platform-
          ↪ independent utility scripts).lnk" "$INSTDIR\utilities"
126
127   ; create file associations
128   WriteRegStr HKCR ".JREXX" "" "RexxJavaScript"
129   WriteRegStr HKCR ".JREXX" "Content Type" "text/x-rexx-java"
130   WriteRegStr HKCR ".JREXX" "PerceivedType" "text"
131
132   WriteRegStr HKCR ".ORX" "" "REXXScript"
133   WriteRegStr HKCR ".ORX" "Content Type" "text/x-rexx"
134   WriteRegStr HKCR ".ORX" "PerceivedType" "text"
135
136   WriteRegStr HKCR ".REXX" "" "REXXScript"
137   WriteRegStr HKCR ".REXX" "Content Type" "text/x-rexx"
138   WriteRegStr HKCR ".REXX" "PerceivedType" "text"
139
140   WriteRegStr HKCR ".REXXJ" "" "RexxJavaScript"
141   WriteRegStr HKCR ".REXXJ" "Content Type" "text/x-rexx-java"
```

```
142   WriteRegStr HKCR ".REXXJ" "PerceivedType" "text"
143
144   WriteRegStr HKCR ".RXJ" "" "RexxJavaScript"
145   WriteRegStr HKCR ".RXJ" "Content Type" "text/x-rexx-java"
146   WriteRegStr HKCR ".RXJ" "PerceivedType" "text"
147   WriteRegStr HKCR ".RXJ\ShellNew" "FileName" "$INSTDIR\Templates\RexxJavaScript
        ↪  (Rexx script using Java) executed by ooRexx.rxj"
148
149   WriteRegStr HKCR ".TESTGROUP" "" "REXXScript"
150   WriteRegStr HKCR ".TESTGROUP" "Content Type" "text/x-rexx"
151   WriteRegStr HKCR ".TESTGROUP" "PerceivedType" "text"
152
153   WriteRegStr HKCR ".TESTUNIT" "" "REXXScript"
154   WriteRegStr HKCR ".TESTUNIT" "Content Type" "text/x-rexx"
155   WriteRegStr HKCR ".TESTUNIT" "PerceivedType" "text"
156
157   WriteRegStr HKCR "RexxJavaScript" "" "RexxJavaScript (Rexx script using Java)
        ↪ executed by ooRexx"
158   WriteRegStr HKCR "RexxJavaScript\DefaultIcon" "" "$INSTDIR\install\images\
        ↪ windows\bsf4oorexx.ico,0"
159   WriteRegStr HKCR "RexxJavaScript\shell" "" "open"
160   WriteRegStr HKCR "RexxJavaScript\shell\edit" "" "Edit"
161   WriteRegStr HKCR "RexxJavaScript\shell\edit\command" "" 'notepad.exe "%1"'
162   WriteRegStr HKCR "RexxJavaScript\shell\open" "" "Run"
163   WriteRegStr HKCR "RexxJavaScript\shell\open\command" "" '"$REXX_HOME\rexx.exe"
        ↪  "%1" %*'
164
165   WriteRegStr HKCR ".CLS" "" "REXXScript"
166   WriteRegStr HKCR ".CLS" "Content Type" "text/x-rexx"
167   WriteRegStr HKCR ".CLS" "PerceivedType" "text"
168
169   DeleteRegKey HKCR ".rex"
170   WriteRegStr HKCR ".rex" "" "REXXScript"
171   WriteRegStr HKCR ".rex" "Content Type" "text/x-rexx"
172   WriteRegStr HKCR ".rex" "PerceivedType" "text"
173   WriteRegStr HKCR ".rex\ShellNew" "FileName" "$INSTDIR\Templates\RexxScript (
        ↪ Rexx script) executed by ooRexx.rex"
174
175   DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
        ↪ \.rxj"
176   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
        ↪ \.rxj\OpenWithList" "a" "rexx.exe"
177   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
        ↪ \.rxj\OpenWithList" "MRUList" "a"
178   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
        ↪ \.rxj\OpenWithProgids" "RexxJavaScript" ""
179
180   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx" "Extension" ".rex"
181   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java" "Extension" ".
        ↪ rxj"
182
183   WriteRegStr HKCR "REXXScript" "" "RexxScript (Rexx script) executed by ooRexx"
184   WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$INSTDIR\install\images\windows\
        ↪ oorexx.ico,0"
185   WriteRegStr HKCR "REXXScript\shell" "" "open"
186   WriteRegStr HKCR "REXXScript\shell\edit" "" "Edit"
```

```
187   WriteRegStr HKCR "REXXScript\shell\edit\command" "" 'notepad.exe "%1"'
188   WriteRegStr HKCR "REXXScript\shell\open" "" "Run"
189   WriteRegStr HKCR "REXXScript\shell\open\command" "" '"$REXX_HOME\rexx.exe" "%1
          ↪ " %*'
190
191   ; set CLASSPATH and PATH
192   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
193   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-v400-20090910.jar"
194
195   ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$INSTDIR"
196
197   StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
198
199   x64:
200       ReadRegStr $1 HKLM "SOFTWARE\Wow6432Node\JavaSoft\Java Runtime Environment
              ↪ \1.6" "RuntimeLib"
201       Goto AddJavaRuntimeLib
202
203   x32:
204       ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
              ↪ RuntimeLib"
205       Goto AddJavaRuntimeLib
206
207   AddJavaRuntimeLib:
208       ${StrRep} $2 "$1" "\jvm.dll" ""
209       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$2"
210
211   ; is OpenOffice installed?
212   StrCmp "" $OO_HOME SearchLibreOffice OOFound
213
214   ; yes, so copy necessary files
215   OOFound:
216      KillProcDLL::KillProc "soffice.exe"
217      KillProcDLL::KillProc "soffice.bin"
218      SetOutPath "$OO_HOMEprogram"
219      File "${BSF_ROOT}\install\32\BSF4ooRexx.dll"
220      File "${BSF_ROOT}\BSF.CLS"
221      File "${BSF_ROOT}\UNO.CLS"
222      File "${BSF_ROOT}\UNO_API_info.rxo"
223      File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
224      File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
225      File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
226
227      SetOutPath "$OO_HOMEBasis\share\Scripts\javascript"
228      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
             ↪ HowtoCreateApiInfo"
229
230      SetOutPath "$OO_HOMEBasis\share\Scripts"
231      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
232
233      SetOutPath "$OO_HOMEBasis\share\Scripts\python"
234      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
235
236      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEprogram\"
237      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEBasis\program\classes\
             ↪ unoil.jar"
```

```
238       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\ridl.jar"
239       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\jurt.jar"
240       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\juh.jar"
241
242       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$OO_HOMEprogram"
243
244       SetOutPath "$INSTDIR\install"
245       ExecWait '"$OO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
              ↪ .oxt'
246       StrCmp "" $LO_HOME AddOfficeReg LOFound
247
248    SearchLibreOffice:
249       ; is LibreOffice installed?
250       StrCmp "" $LO_HOME Done LOFound
251
252    ; yes, so copy necessary files
253    LOFound:
254       KillProcDLL::KillProc "soffice.exe"
255       KillProcDLL::KillProc "soffice.bin"
256       SetOutPath $LO_HOME"program"
257       File "${BSF_ROOT}\install\32\BSF4ooRexx.dll"
258       File "${BSF_ROOT}\BSF.CLS"
259       File "${BSF_ROOT}\UNO.CLS"
260       File "${BSF_ROOT}\UNO_API_info.rxo"
261       File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
262       File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
263       File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
264
265       SetOutPath "$LO_HOMEBasis\share\Scripts\javascript"
266       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
              ↪ HowtoCreateApiInfo"
267
268       SetOutPath "$LO_HOMEBasis\share\Scripts"
269       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
270
271       SetOutPath "$LO_HOMEBasis\share\Scripts\python"
272       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
273
274       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEprogram\"
275       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEBasis\program\classes\
              ↪ unoil.jar"
276       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\ridl.jar"
277       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\jurt.jar"
278       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\juh.jar"
279
280       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$LO_HOMEprogram"
281
282       SetOutPath "$INSTDIR\install"
283       ExecWait '"$LO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
              ↪ .oxt'
284       Goto AddOfficeReg
285
286    AddOfficeReg:
287       WriteRegStr HKCR ".RXO" "" "RexxOOoScript"
288       WriteRegStr HKCR ".RXO" "Content Type" "text/x-rexx-java-ooo"
289       WriteRegStr HKCR ".RXO" "PerceivedType" "text"
```

```
290       WriteRegStr HKCR ".RXO\ShellNew" "FileName" "$INSTDIR\Templates\
              ↪ RexxOOoScript (Rexx script using OpenOffice.org) executed by ooRexx.
              ↪ rxo"
291
292       WriteRegStr HKCR "RexxOOoScript" "" "RexxOOoScript (Rexx script using
              ↪ OpenOffice.org/LibreOffice) executed by ooRexx"
293       WriteRegStr HKCR "RexxOOoScript\DefaultIcon" "" "$INSTDIR\install\images\
              ↪ windows\oorexx4ooo.ico,0"
294       WriteRegStr HKCR "RexxOOoScript\shell" "" "open"
295       WriteRegStr HKCR "RexxOOoScript\shell\edit" "" "Edit"
296       WriteRegStr HKCR "RexxOOoScript\shell\edit\command" "" 'notepad.exe "%1"'
297       WriteRegStr HKCR "RexxOOoScript\shell\open" "" "Run"
298       WriteRegStr HKCR "RexxOOoScript\shell\open\command" "" '"$REXX_HOME\rexx.
              ↪ exe" "%1" %*'
299
300       WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo" "
              ↪ Extension" ".rxo"
301       Goto Done
302
303     Done:
304       ; reload file associations
305       ExecWait "$INSTDIR\install\associationChanged.exe"
306   SectionEnd
307
308   ; Add Uninstall and Reinstall Entry to the start menue
309   Section -AdditionalIcons
310     CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Uninstall ${
          ↪ PRODUCT_NAME}.lnk" "$INSTDIR\uninst.exe" "" "$INSTDIR\install\images\
          ↪ windows\bsf4rexx-uninstall.ico"
311     CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Reinstall ${
          ↪ PRODUCT_NAME}.lnk" "$INSTDIR\${PRODUCT_NAME}_Setup_x32_reinstall.exe" ""
          ↪  "$INSTDIR\install\images\windows\bsf4rexx-reinstall.ico"
312   SectionEnd
313
314   ; create Uninstaller and Uninstall registry entries
315   Section -Post
316     WriteUninstaller "$INSTDIR\uninst.exe"
317     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayName" "
          ↪ $(^Name)"
318     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
          ↪ UninstallString" "$INSTDIR\uninst.exe"
319     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayIcon" "
          ↪ $INSTDIR\install\images\windows\bsf4oorexx.ico"
320     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayVersion
          ↪ " "$PRODUCT_VERSION"
321     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoRepair" 0
322     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoModify" 0
323     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "ModifyPath" "
          ↪ $INSTDIR\${PRODUCT_NAME}_Setup_x32_reinstall.exe"
324   SectionEnd
325
326   ; if no silent mode, show success message
327   Function un.onUninstSuccess
328     IfSilent jump_if_silent jump_if_not_silent
329
330     jump_if_not_silent:
```

```
331       HideWindow
332       MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully uninstalled.
              ↪ " IDOK
333
334     jump_if_silent:
335
336   FunctionEnd
337
338   ; if no silent mode, ask user if he want to uninstall
339   Function un.onInit
340     Call un.getBSFVersion
341
342     IfSilent jump_if_silent jump_if_not_silent
343
344     jump_if_not_silent:
345       MessageBox MB_ICONQUESTION|MB_YESNO|MB_DEFBUTTON2 "Do you want to uninstall
              ↪  $(^Name) and all the components?" IDYES +2
346       Abort
347
348     jump_if_silent:
349
350   FunctionEnd
351
352   ; undo all changes
353   Section Uninstall
354     SetShellVarContext all
355
356     ; is OpenOffice/LibreOffice installed?
357     Call un.FindOpenOffice
358     Call un.FindLibreOffice
359     StrCmp "" $OO_HOME SearchLibreOffice OOFound
360
361     ; OpenOffice found: so delete files
362     OOFound:
363       KillProcDLL::KillProc "soffice.exe"
364       KillProcDLL::KillProc "soffice.bin"
365
366       RmDir /r "$OO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
367       RmDir /r "$OO_HOMEBasis\share\Scripts\oorexx"
368       RmDir /r "$OO_HOMEBasis\share\Scripts\python\pythonSamples"
369
370       ExecWait "$OO_HOMEprogram\unopkg.exe remove --shared
              ↪ ScriptProviderForooRexx.oxt"
371
372       Delete "$OO_HOMEprogram\BSF4ooRexx.dll"
373       Delete "$OO_HOMEprogram\BSF.CLS"
374       Delete "$OO_HOMEprogram\UNO.CLS"
375       Delete "$OO_HOMEprogram\UNO_API_info.rxo"
376       Delete "$OO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
377       Delete "$OO_HOMEprogram\UNO_ReflectionUtil.cls"
378       Delete "$OO_HOMEprogram\UNO_XINTERFACES.RXO"
379
380       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEprogram\"
381       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEBasis\program\classes
              ↪ \unoil.jar"
382       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\ridl.jar"
```

```
383       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\jurt.jar"
384       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\juh.jar"
385
386       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$OO_HOMEprogram"
387       StrCmp "" $LO_HOME RemOfficeReg LOFound
388
389    SearchLibreOffice:
390       ; is LibreOffice installed?
391       StrCmp "" $LO_HOME Done LOFound
392
393    ; LibreOffice found: so delete files
394    LOFound:
395       KillProcDLL::KillProc "soffice.exe"
396       KillProcDLL::KillProc "soffice.bin"
397
398       RmDir /r "$LO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
399       RmDir /r "$LO_HOMEBasis\share\Scripts\oorexx"
400       RmDir /r "$LO_HOMEBasis\share\Scripts\python\pythonSamples"
401
402       ExecWait "$LO_HOMEprogram\unopkg.exe remove --shared
              ↪ ScriptProviderForooRexx.oxt"
403
404       Delete "$LO_HOMEprogram\BSF4ooRexx.dll"
405       Delete "$LO_HOMEprogram\BSF.CLS"
406       Delete "$LO_HOMEprogram\UNO.CLS"
407       Delete "$LO_HOMEprogram\UNO_API_info.rxo"
408       Delete "$LO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
409       Delete "$LO_HOMEprogram\UNO_ReflectionUtil.cls"
410       Delete "$LO_HOMEprogram\UNO_XINTERFACES.RXO"
411
412       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEprogram\"
413       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEBasis\program\classes
              ↪ \unoil.jar"
414       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\ridl.jar"
415       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\jurt.jar"
416       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\juh.jar"
417
418       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$LO_HOMEprogram"
419       Goto RemOfficeReg
420
421    RemOfficeReg:
422       DeleteRegKey HKCR ".RXO"
423       DeleteRegKey HKCR "RexxOOoScript"
424       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo"
425       Goto Done
426
427    Done:
428       StrCmp $WINDOWS_ARCHITECTURE "64" OS64 OS32
429
430    OS64:
431       ; change registry mode to 64 bit
432       SetRegView 64
433
434    OS32:
435       DeleteRegKey ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}"
436       SetRegView 32 ; change registry mode to 32 bit
```

```
437       DeleteRegKey HKCR ".JREXX"
438       DeleteRegKey HKCR ".ORX"
439       DeleteRegKey HKCR ".REXX"
440       DeleteRegKey HKCR ".REXXJ"
441       DeleteRegKey HKCR ".RXJ"
442       DeleteRegKey HKCR ".TESTGROUP"
443       DeleteRegKey HKCR ".TESTUNIT"
444       DeleteRegKey HKCR "RexxJavaScript"
445       DeleteRegKey HKCR ".CLS"
446       DeleteRegKey HKCR ".rex"
447       DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\
              ↪ FileExts\.rxj"
448       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx"
449       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java"
450
451       WriteRegStr HKCR "REXXScript" "" "ooRexx Rexx Progam"
452       WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$REXX_HOME\rexx.exe,0"
453
454       ; reload file associations
455       ExecWait "$INSTDIR\install\associationChanged.exe"
456
457       RMDir /r "$INSTDIR"
458       RMDir /r "$SMPROGRAMS\${PRODUCT_NAME}"
459       RMDir "$INSTDIR"
460       RMDir "$SMPROGRAMS\${PRODUCT_NAME}"
461
462       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
463       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-v400-20090910.
              ↪ jar"
464
465       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$INSTDIR"
466
467       StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
468
469       x64:
470           ReadRegStr $1 HKLM "SOFTWARE\Wow6432Node\JavaSoft\Java Runtime
                  ↪ Environment\1.6" "RuntimeLib"
471           Goto RemJavaRuntimeLib
472
473       x32:
474           ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
                  ↪ RuntimeLib"
475           Goto RemJavaRuntimeLib
476
477       RemJavaRuntimeLib:
478           ${un.StrRep} $2 "$1" "\jvm.dll" ""
479           ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$2"
480
481       SetAutoClose false
482   SectionEnd
483
484   ; check if Java and ooRexx is installed
485   ; find OpenOffice/LibreOffice folder if is installed
486   Function .onInit
487       Push $0
488       Push $1
```

```
489     Push $2
490     InitPluginsDir
491
492     GetVersion::WindowsPlatformArchitecture
493     Pop $WINDOWS_ARCHITECTURE ;32 or 64
494
495     Call getBSFVersion
496     Call FindooRexx
497     ; is ooRexx installed?
498     StrCmp "" $REXX_INSTALLATION_MSG RexxSuccess RexxNoSuccess
499
500     ; yes, it is
501     RexxSuccess:
502         Call FindJava
503         ; is Java installed?
504         StrCmp "" $JAVA_INSTALLATION_MSG JavaSuccess JavaNoSuccess
505
506     ; yes, it is
507     JavaSuccess:
508         Call FindOpenOffice
509         Call FindLibreOffice
510         ; check if a version of BSF4ooRexx is already installed
511         ReadRegStr $0 ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
                ↪ DisplayName"
512         ReadRegStr $1 ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
                ↪ DisplayVersion"
513         StrCmp $0 "$(^Name)" FoundBSF
514         Goto Done
515
516     JavaNoSuccess:
517         MessageBox MB_OK|MB_ICONEXCLAMATION $JAVA_INSTALLATION_MSG
518         Goto Error
519
520     RexxNoSuccess:
521         MessageBox MB_OK|MB_ICONEXCLAMATION $REXX_INSTALLATION_MSG
522         Goto Error
523
524     ; BSF4ooRexx is already installed
525     ; if no silent mode, ask user what to do
526     ; if silent mode, uninstall the existing version
527     FoundBSF:
528         IfSilent jump_if_silent jump_if_not_silent
529
530         jump_if_not_silent:
531                 MessageBox MB_ICONQUESTION|MB_YESNO "Version $1 of ${
                        ↪ PRODUCT_NAME} is installed. You have to uninstall it
                        ↪ before installing this version. Would you like to
                        ↪ uninstall it?" IDNO err
532
533         jump_if_silent:
534
535         ExecWait '"$INSTDIR\uninst.exe" /S _?=$INSTDIR'
536         Goto Done
537     err:
538         MessageBox MB_OK|MB_ICONEXCLAMATION "The installation was aborted."
539         Abort
```

```
540
541     Error:
542         RMDir /r "$INSTDIR"
543         Abort
544
545     Done:
546         Pop $2
547         Pop $1
548         Pop $0
549 FunctionEnd
550
551 ; read the BSF4ooRexx version by using getBSFVersion.rex
552 Function getBSFVersion
553     Push $0
554
555     SetOutPath "$INSTDIR"
556     File "${SCRIPT_PATH}\getBSFVersion.rex"
557     File "${BSF_ROOT}\BSF.CLS"
558     ExecWait "rexx getBSFVersion.rex"
559
560     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
561     FileRead $0 $PRODUCT_VERSION
562     FileClose $0
563
564     Delete "$INSTDIR\getBSFVersion.rex"
565     Delete "$INSTDIR\BSF.CLS"
566     Delete "$INSTDIR\BSFVersion.txt"
567
568     Pop $0
569 FunctionEnd
570
571 ; read the BSF4ooRexx version by using getBSFVersion.rex
572 ; Uninstall function
573 Function un.getBSFVersion
574     Push $0
575
576     SetOutPath "$INSTDIR"
577     File "${SCRIPT_PATH}\getBSFVersion.rex"
578     File "${BSF_ROOT}\BSF.CLS"
579     ExecWait "rexx getBSFVersion.rex"
580
581     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
582     FileRead $0 $PRODUCT_VERSION
583     FileClose $0
584
585     Delete "$INSTDIR\getBSFVersion.rex"
586     Delete "$INSTDIR\BSF.CLS"
587     Delete "$INSTDIR\BSFVersion.txt"
588
589     Pop $0
590 FunctionEnd
591
592 ; read Java version by using java_version.rex
593 Function FindJava
594     Push $0
595     Push $1
```

```
596
597     SetOutPath "$INSTDIR"
598     File "${SCRIPT_PATH}\java_version.rex"
599     ExecWait "rexx java_version.rex bitness" $0
600     ExecWait "rexx java_version.rex version" $1
601     Delete "$INSTDIR\java_version.rex"
602     StrCmp $0 "64" JavaVerNotCorrect CheckVersion
603
604     CheckVersion:
605         StrCmp $1 "1" Done JavaVerNotCorrect
606
607     JavaVerNotCorrect:
608         StrCpy $JAVA_INSTALLATION_MSG "You need Java version 1.4 (32 bit) or
                ↪ newer to run this program."
609
610     Done:
611         Pop $1
612         Pop $0
613 FunctionEnd
614
615 ; read ooRexx version from registry
616 ; and bitness by using rexx_version.rex
617 Function FindooRexx
618     Push $0
619     Push $1
620     Push $2
621     Push $3
622
623     ReadRegStr $REXX_VER HKLM "Software\Microsoft\Windows\CurrentVersion\
            ↪ Uninstall\ooRexx" "DisplayVersion"
624     StrCmp "" "$REXX_VER" RexxNotPresent
625     ReadRegStr $0 HKLM "SYSTEM\CurrentControlSet\Control\Session Manager\
            ↪ Environment" "REXX_HOME"
626     StrCmp "" $0 RexxNotPresent CheckRexxVer
627
628     RexxNotPresent:
629         StrCpy $REXX_INSTALLATION_MSG "ooRexx is not installed on your computer.
                ↪  You need version 4.1.0 (32 bit) or newer to run this program."
630         Goto Done
631
632     CheckRexxVer:
633         GetFullPathName /SHORT $REXX_HOME "$0"
634         StrCpy $0 $REXX_VER 1 0
635         StrCpy $1 $REXX_VER 1 2
636         StrCpy $2 $REXX_VER 1 4
637         StrCpy $REXX_VER "$0$1$2"
638         IntCmp 410 $REXX_VER FoundCorrectRexxVer FoundCorrectRexxVer
                ↪ RexxVerNotCorrect
639
640     FoundCorrectRexxVer:
641         IfFileExists "$REXX_HOME\rexx.exe" CheckRexxBitness RexxNotPresent
642
643     CheckRexxBitness:
644         SetOutPath "$INSTDIR"
645         File "${SCRIPT_PATH}\rexx_version.rex"
646         ExecWait "rexx rexx_version.rex" $3
```

```
647         Delete "$INSTDIR\rexx_version.rex"
648         StrCmp $3 "32" Done RexxVerNotCorrect
649
650     RexxVerNotCorrect:
651         StrCpy $REXX_INSTALLATION_MSG "The version of ooRexx installed on your
                ↪ computer is $0.$1.$2 ($3 bit). Version 4.1.0 (32 bit) or newer is
                ↪ required to run this program."
652
653     Done:
654         Pop $3
655         Pop $2
656         Pop $1
657         Pop $0
658 FunctionEnd
659
660 ; find OpenOffice folder
661 Function FindOpenOffice
662     Push $0
663
664     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
665
666     x64:
667         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\
                ↪ OpenOffice.org\3" "OFFICEINSTALLLOCATION"
668         StrCmp "" $0 Done OOFound
669
670     x32:
671         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
                ↪ OFFICEINSTALLLOCATION"
672         StrCmp "" $0 Done OOFound
673
674     OOFound:
675         GetFullPathName $OO_HOME "$0"
676         Goto Done
677
678     Done:
679         Pop $0
680 FunctionEnd
681
682 ; find OpenOffice folder (uninstall function)
683 Function un.FindOpenOffice
684     Push $0
685
686     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
687
688     x64:
689         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\
                ↪ OpenOffice.org\3" "OFFICEINSTALLLOCATION"
690         StrCmp "" $0 Done OOFound
691
692     x32:
693         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
                ↪ OFFICEINSTALLLOCATION"
694         StrCmp "" $0 Done OOFound
695
696     OOFound:
```

```
697         GetFullPathName $OO_HOME "$0"
698         Goto Done
699
700     Done:
701         Pop $0
702 FunctionEnd
703
704 ; find LibreOffice folder
705 Function FindLibreOffice
706     Push $0
707
708     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
709
710     x64:
711         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice
                ↪ \3" "OFFICEINSTALLLOCATION"
712         StrCmp "" $0 Done LOFound
713
714     x32:
715         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
716         StrCmp "" $0 Done LOFound
717
718     LOFound:
719         GetFullPathName $LO_HOME "$0"
720         Goto Done
721
722     Done:
723         Pop $0
724 FunctionEnd
725
726 ; find LibreOffice folder (uninstall function)
727 Function un.FindLibreOffice
728     Push $0
729
730     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
731
732     x64:
733         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice
                ↪ \3" "OFFICEINSTALLLOCATION"
734         StrCmp "" $0 Done LOFound
735
736     x32:
737         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
738         StrCmp "" $0 Done LOFound
739
740     LOFound:
741         GetFullPathName $LO_HOME "$0"
742         Goto Done
743
744     Done:
745         Pop $0
746 FunctionEnd
```

Listing 36: The NSIS Installation Script for 32bit Windows.

7.2.2 The NSIS Installation Script for 64 bit Windows

The following listing contains the installation script for 64 bit Windows. It is described in detail in chapter 2.2.

```
; Name: BSF4ooRexx_x64.nsi
; Purpose: NSIS script for BSF4ooRexx (64 bit)
; Author: Juergen Hesse, Manuel Paar
; Version: 1.0
;
; ------------------- Apache Version 2.0 license --------------------
;    Copyright (C) 2011 Juergen Hesse, Manuel Paar
;
;    Licensed under the Apache License, Version 2.0 (the "License");
;    you may not use this file except in compliance with the License.
;    You may obtain a copy of the License at
;
;          http://www.apache.org/licenses/LICENSE-2.0
;
;    Unless required by applicable law or agreed to in writing, software
;    distributed under the License is distributed on an "AS IS" BASIS,
;    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
;    implied.
;    See the License for the specific language governing permissions and
;    limitations under the License.
; -------------------------------------------------------------------

; load more functions
!include "MUI.nsh"
!include "EnvVarUpdate.nsh"

; set product name and uninstall registry-key
!define PRODUCT_NAME "BSF4ooRexx"
!define PRODUCT_UNINST_KEY "Software\Microsoft\Windows\CurrentVersion\Uninstall\
    ↪ ${PRODUCT_NAME}"
!define PRODUCT_UNINST_ROOT_KEY "HKLM"

; TO CUSTOMIZE
; Source-Folder
!define BSF_ROOT ".\BSF4ooRexx_install\bsf4oorexx"
; Script Location (getBSFVersion.rex, ...)
!define SCRIPT_PATH "."

SetCompressor lzma

; MUI Settings
!define MUI_ABORTWARNING
!define MUI_ICON "${BSF_ROOT}\install\images\windows\bsf4rexx-install.ico"
!define MUI_UNICON "${BSF_ROOT}\install\images\windows\bsf4rexx-uninstall.ico"
; Welcome page
!insertmacro MUI_PAGE_WELCOME
; License page
!define MUI_LICENSEPAGE_CHECKBOX
!insertmacro MUI_PAGE_LICENSE "${BSF_ROOT}\ApacheLicense2.0.txt"
```

```
49  ; Instfiles page
50  !insertmacro MUI_PAGE_INSTFILES
51  ; Finish page
52  !insertmacro MUI_PAGE_FINISH
53  ; Uninstaller pages
54  !insertmacro MUI_UNPAGE_INSTFILES
55  ; Language files
56  !insertmacro MUI_LANGUAGE "English"
57  ; Reserve files
58  !insertmacro MUI_RESERVEFILE_INSTALLOPTIONS
59
60  Var JAVA_INSTALLATION_MSG
61  Var REXX_HOME
62  Var REXX_VER
63  Var REXX_INSTALLATION_MSG
64  Var OO_HOME
65  Var LO_HOME
66  Var WINDOWS_ARCHITECTURE
67  Var PRODUCT_VERSION
68
69  Name "${PRODUCT_NAME} $PRODUCT_VERSION"
70  OutFile "${PRODUCT_NAME}_Setup_x64.exe"
71  InstallDir "$PROGRAMFILES64\${PRODUCT_NAME}"
72  XPStyle on       ; wenn XP, dann XP-Style verwenden
73  ShowInstDetails show
74  ShowUnInstDetails show
75
76  ; copy all files to the destination folder
77  ; create start menue entries
78  ; create file associations
79  Section "BSF4ooRexx" SEC01
80    SetShellVarContext all
81    SetAutoClose true
82    SetOverwrite ifnewer
83
84    ; copy all files to the destination folder
85    SetOutPath "$INSTDIR"
86    File "${BSF_ROOT}\addLinks2OOoSnippets.rxo"
87    File "${BSF_ROOT}\ApacheLicense2.0.txt"
88    File "${BSF_ROOT}\asm_LICENSE.TXT"
89    File "${BSF_ROOT}\bsf-rexx-engine.jar"
90    File "${BSF_ROOT}\bsf-v400-20090910.jar"
91    File "${BSF_ROOT}\BSF.CLS"
92    File "${BSF_ROOT}\install\64\BSF4ooRexx.dll"
93    File "${BSF_ROOT}\BSF_OnTheFly.cls"
94    File "${BSF_ROOT}\credits.txt"
95    File "${BSF_ROOT}\janino_LICENSE.TXT"
96    File "${BSF_ROOT}\refcardBSF4ooRexx.pdf"
97    File "${BSF_ROOT}\refcardOOo.pdf"
98    File "/oname=rexxj.cmd" "${BSF_ROOT}\rexxj2.cmd"
99    File "${BSF_ROOT}\rexxj2.cmd"
100   File "${BSF_ROOT}\rexxj2-32.sh"
101   File "${BSF_ROOT}\rexxj2-64.sh"
102   File "${BSF_ROOT}\rexxj2.sh"
103   File "${BSF_ROOT}\rgf_util2.rex"
104   File "${BSF_ROOT}\UNO.CLS"
```

```
105   File "${BSF_ROOT}\UNO_API_info.rxo"
106   File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
107   File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
108   File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
109   File ".\${PRODUCT_NAME}_Setup_x64_reinstall.exe"
110
111   File /r "${BSF_ROOT}\information"
112   File /r "${BSF_ROOT}\install"
113   File /r "${BSF_ROOT}\samples"
114   File /r "${BSF_ROOT}\Templates"
115   File /r "${BSF_ROOT}\utilities"
116
117   ; create start menue entries
118   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}"
119   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\GUI RexxTry Program (ooRexxTry.rxj
          ↪ ).lnk" "$INSTDIR\utilities\ooRexxTry.rxj" "" "$INSTDIR\install\images\
          ↪ windows\oorexxtry.ico"
120   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Information (ReleaseNotes,
          ↪ introductions to ooRexx, ${PRODUCT_NAME} overview).lnk" "$INSTDIR\
          ↪ information"
121
122   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}\Installation"
123   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\${PRODUCT_NAME}
          ↪ Download Page.lnk" "http://sourceforge.net/projects/bsf4oorexx/"
124
125   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Samples (simple to comprehensive $
          ↪ {PRODUCT_NAME} examples).lnk" "$INSTDIR\samples"
126   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Utilities (some platform-
          ↪ independent utility scripts).lnk" "$INSTDIR\utilities"
127
128   ; create file associations
129   WriteRegStr HKCR ".JREXX" "" "RexxJavaScript"
130   WriteRegStr HKCR ".JREXX" "Content Type" "text/x-rexx-java"
131   WriteRegStr HKCR ".JREXX" "PerceivedType" "text"
132
133   WriteRegStr HKCR ".ORX" "" "REXXScript"
134   WriteRegStr HKCR ".ORX" "Content Type" "text/x-rexx"
135   WriteRegStr HKCR ".ORX" "PerceivedType" "text"
136
137   WriteRegStr HKCR ".REXX" "" "REXXScript"
138   WriteRegStr HKCR ".REXX" "Content Type" "text/x-rexx"
139   WriteRegStr HKCR ".REXX" "PerceivedType" "text"
140
141   WriteRegStr HKCR ".REXXJ" "" "RexxJavaScript"
142   WriteRegStr HKCR ".REXXJ" "Content Type" "text/x-rexx-java"
143   WriteRegStr HKCR ".REXXJ" "PerceivedType" "text"
144
145   WriteRegStr HKCR ".rxj" "" "RexxJavaScript"
146   WriteRegStr HKCR ".rxj" "Content Type" "text/x-rexx-java"
147   WriteRegStr HKCR ".rxj" "PerceivedType" "text"
148   WriteRegStr HKCR ".rxj\ShellNew" "FileName" "$INSTDIR\Templates\RexxJavaScript
          ↪  (Rexx script using Java) executed by ooRexx.rxj"
149
150   WriteRegStr HKCR ".TESTGROUP" "" "REXXScript"
151   WriteRegStr HKCR ".TESTGROUP" "Content Type" "text/x-rexx"
152   WriteRegStr HKCR ".TESTGROUP" "PerceivedType" "text"
```

```
153
154   WriteRegStr HKCR ".TESTUNIT" "" "REXXScript"
155   WriteRegStr HKCR ".TESTUNIT" "Content Type" "text/x-rexx"
156   WriteRegStr HKCR ".TESTUNIT" "PerceivedType" "text"
157
158   WriteRegStr HKCR "RexxJavaScript" "" "RexxJavaScript (Rexx script using Java)
         ↪ executed by ooRexx"
159   WriteRegStr HKCR "RexxJavaScript\DefaultIcon" "" "$INSTDIR\install\images\
         ↪ windows\bsf4oorexx.ico,0"
160   WriteRegStr HKCR "RexxJavaScript\shell" "" "open"
161   WriteRegStr HKCR "RexxJavaScript\shell\edit" "" "Edit"
162   WriteRegStr HKCR "RexxJavaScript\shell\edit\command" "" 'notepad.exe "%1"'
163   WriteRegStr HKCR "RexxJavaScript\shell\open" "" "Run"
164   WriteRegStr HKCR "RexxJavaScript\shell\open\command" "" '"$REXX_HOME\rexx.exe"
         ↪  "%1" %*'
165
166   WriteRegStr HKCR ".CLS" "" "REXXScript"
167   WriteRegStr HKCR ".CLS" "Content Type" "text/x-rexx"
168   WriteRegStr HKCR ".CLS" "PerceivedType" "text"
169
170   DeleteRegKey HKCR ".rex"
171   WriteRegStr HKCR ".rex" "" "REXXScript"
172   WriteRegStr HKCR ".rex" "Content Type" "text/x-rexx"
173   WriteRegStr HKCR ".rex" "PerceivedType" "text"
174   WriteRegStr HKCR ".rex\ShellNew" "FileName" "$INSTDIR\Templates\RexxScript (
         ↪ Rexx script) executed by ooRexx.rex"
175
176   DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
         ↪ \.rxj"
177   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
         ↪ \.rxj\OpenWithList" "a" "rexx.exe"
178   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
         ↪ \.rxj\OpenWithList" "MRUList" "a"
179   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
         ↪ \.rxj\OpenWithProgids" "RexxJavaScript" ""
180
181   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx" "Extension" ".rex"
182   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java" "Extension" ".
         ↪ rxj"
183
184   WriteRegStr HKCR "REXXScript" "" "RexxScript (Rexx script) executed by ooRexx"
185   WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$INSTDIR\install\images\windows\
         ↪ oorexx.ico,0"
186   WriteRegStr HKCR "REXXScript\shell" "" "open"
187   WriteRegStr HKCR "REXXScript\shell\edit" "" "Edit"
188   WriteRegStr HKCR "REXXScript\shell\edit\command" "" 'notepad.exe "%1"'
189   WriteRegStr HKCR "REXXScript\shell\open" "" "Run"
190   WriteRegStr HKCR "REXXScript\shell\open\command" "" '"$REXX_HOME\rexx.exe" "%1
         ↪ " %*'
191
192   ; set CLASSPATH and PATH
193   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
194   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-v400-20090910.jar"
195
196   ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$INSTDIR"
197
```

```
198   ; change registry mode to 64 bit
199   SetRegView 64
200   ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
          ↪ RuntimeLib"
201   ${StrRep} $2 "$1" "\jvm.dll" ""
202   ${StrRep} $2 "$2" "client" "server"
203   ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$2"
204
205   ; is OpenOffice installed?
206   StrCmp "" $OO_HOME SearchLibreOffice OOFound
207
208   ; yes, so copy necessary files
209   OOFound:
210      KillProcDLL::KillProc "soffice.exe"
211      KillProcDLL::KillProc "soffice.bin"
212      SetOutPath "$OO_HOMEprogram"
213      File "${BSF_ROOT}\install\64\BSF4ooRexx.dll"
214      File "${BSF_ROOT}\BSF.CLS"
215      File "${BSF_ROOT}\UNO.CLS"
216      File "${BSF_ROOT}\UNO_API_info.rxo"
217      File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
218      File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
219      File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
220
221      SetOutPath "$OO_HOMEBasis\share\Scripts\javascript"
222      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
          ↪ HowtoCreateApiInfo"
223
224      SetOutPath "$OO_HOMEBasis\share\Scripts"
225      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
226
227      SetOutPath "$OO_HOMEBasis\share\Scripts\python"
228      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
229
230      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEprogram\"
231      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEBasis\program\classes\
          ↪ unoil.jar"
232      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\ridl.jar"
233      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\jurt.jar"
234      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\juh.jar"
235
236      ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$OO_HOMEprogram"
237
238      SetOutPath "$INSTDIR\install"
239      ExecWait '"$OO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
          ↪ .oxt'
240      StrCmp "" $LO_HOME AddOfficeReg LOFound
241
242   SearchLibreOffice:
243      ; is LibreOffice installed?
244      StrCmp "" $LO_HOME Done LOFound
245
246   ; yes, so copy necessary files
247   LOFound:
248      KillProcDLL::KillProc "soffice.exe"
249      KillProcDLL::KillProc "soffice.bin"
```

```
250       SetOutPath $LO_HOME"program"
251       File "${BSF_ROOT}\install\64\BSF4ooRexx.dll"
252       File "${BSF_ROOT}\BSF.CLS"
253       File "${BSF_ROOT}\UNO.CLS"
254       File "${BSF_ROOT}\UNO_API_info.rxo"
255       File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
256       File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
257       File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
258
259       SetOutPath "$LO_HOMEBasis\share\Scripts\javascript"
260       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
              ↪ HowtoCreateApiInfo"
261
262       SetOutPath "$LO_HOMEBasis\share\Scripts"
263       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
264
265       SetOutPath "$LO_HOMEBasis\share\Scripts\python"
266       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
267
268       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEprogram\"
269       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEBasis\program\classes\
              ↪ unoil.jar"
270       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\ridl.jar"
271       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\jurt.jar"
272       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\juh.jar"
273
274       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$LO_HOMEprogram"
275
276       SetOutPath "$INSTDIR\install"
277       ExecWait '"$LO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
              ↪ .oxt'
278       Goto AddOfficeReg
279
280    AddOfficeReg:
281       WriteRegStr HKCR ".RXO" "" "RexxOOoScript"
282       WriteRegStr HKCR ".RXO" "Content Type" "text/x-rexx-java-ooo"
283       WriteRegStr HKCR ".RXO" "PerceivedType" "text"
284       WriteRegStr HKCR ".RXO\ShellNew" "FileName" "$INSTDIR\Templates\
              ↪ RexxOOoScript (Rexx script using OpenOffice.org) executed by ooRexx.
              ↪ rxo"
285
286       WriteRegStr HKCR "RexxOOoScript" "" "RexxOOoScript (Rexx script using
              ↪ OpenOffice.org/LibreOffice) executed by ooRexx"
287       WriteRegStr HKCR "RexxOOoScript\DefaultIcon" "" "$INSTDIR\install\images\
              ↪ windows\oorexx4ooo.ico,0"
288       WriteRegStr HKCR "RexxOOoScript\shell" "" "open"
289       WriteRegStr HKCR "RexxOOoScript\shell\edit" "" "Edit"
290       WriteRegStr HKCR "RexxOOoScript\shell\edit\command" "" 'notepad.exe "%1"'
291       WriteRegStr HKCR "RexxOOoScript\shell\open" "" "Run"
292       WriteRegStr HKCR "RexxOOoScript\shell\open\command" "" '"$REXX_HOME\rexx.
              ↪ exe" "%1" %*'
293
294       WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo" "
              ↪ Extension" ".rxo"
295       Goto Done
296
```

```
297   Done:
298      ; reload file associations
299      ExecWait "$INSTDIR\install\associationChanged.exe"
300 SectionEnd
301
302 ; Add Uninstall and Reinstall Entry to the start menue
303 Section -AdditionalIcons
304   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Uninstall ${
          ↪ PRODUCT_NAME}.lnk" "$INSTDIR\uninst.exe" "" "$INSTDIR\install\images\
          ↪ windows\bsf4rexx-uninstall.ico"
305   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Reinstall ${
          ↪ PRODUCT_NAME}.lnk" "$INSTDIR\${PRODUCT_NAME}_Setup_x64_reinstall.exe" ""
          ↪  "$INSTDIR\install\images\windows\bsf4rexx-reinstall.ico"
306 SectionEnd
307
308 ; create Uninstaller and Uninstall registry entries
309 Section -Post
310   WriteUninstaller "$INSTDIR\uninst.exe"
311   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayName" "
          ↪ $(^Name)"
312   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
          ↪ UninstallString" "$INSTDIR\uninst.exe"
313   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayIcon" "
          ↪ $INSTDIR\install\images\windows\bsf4oorexx.ico"
314   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayVersion
          ↪ " "$PRODUCT_VERSION"
315   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoRepair" 0
316   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoModify" 0
317   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "ModifyPath" "
          ↪ $INSTDIR\${PRODUCT_NAME}_Setup_x64_reinstall.exe"
318 SectionEnd
319
320 ; if no silent mode, show success message
321 Function un.onUninstSuccess
322   IfSilent jump_if_silent jump_if_not_silent
323
324   jump_if_not_silent:
325      HideWindow
326      MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully uninstalled.
          ↪ " IDOK
327
328   jump_if_silent:
329
330 FunctionEnd
331
332 ; if no silent mode, ask user if he want to uninstall
333 Function un.onInit
334   Call un.getBSFVersion
335   IfSilent jump_if_silent jump_if_not_silent
336
337   jump_if_not_silent:
338      MessageBox MB_ICONQUESTION|MB_YESNO|MB_DEFBUTTON2 "Do you want to uninstall
          ↪  $(^Name) and all the components?" IDYES +2
339      Abort
340
341   jump_if_silent:
```

```
342
343 FunctionEnd
344
345 ; undo all changes
346 Section Uninstall
347   SetShellVarContext all
348
349   ; is OpenOffice/LibreOffice installed?
350   Call un.FindOpenOffice
351   Call un.FindLibreOffice
352   StrCmp "" $OO_HOME SearchLibreOffice OOFound
353
354   ; OpenOffice found: so delete files
355   OOFound:
356      KillProcDLL::KillProc "soffice.exe"
357      KillProcDLL::KillProc "soffice.bin"
358
359      RmDir /r "$OO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
360      RmDir /r "$OO_HOMEBasis\share\Scripts\oorexx"
361      RmDir /r "$OO_HOMEBasis\share\Scripts\python\pythonSamples"
362
363      ExecWait "$OO_HOMEprogram\unopkg.exe remove --shared
         ↪ ScriptProviderForooRexx.oxt"
364
365      Delete "$OO_HOMEprogram\BSF4ooRexx.dll"
366      Delete "$OO_HOMEprogram\BSF.CLS"
367      Delete "$OO_HOMEprogram\UNO.CLS"
368      Delete "$OO_HOMEprogram\UNO_API_info.rxo"
369      Delete "$OO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
370      Delete "$OO_HOMEprogram\UNO_ReflectionUtil.cls"
371      Delete "$OO_HOMEprogram\UNO_XINTERFACES.RXO"
372
373      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEprogram\"
374      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEBasis\program\classes
         ↪ \unoil.jar"
375      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\ridl.jar"
376      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\jurt.jar"
377      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\juh.jar"
378
379      ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$OO_HOMEprogram"
380      StrCmp "" $LO_HOME RemOfficeReg LOFound
381
382   SearchLibreOffice:
383      ; is LibreOffice installed?
384      StrCmp "" $LO_HOME Done LOFound
385
386   ; LibreOffice found: so delete files
387   LOFound:
388      KillProcDLL::KillProc "soffice.exe"
389      KillProcDLL::KillProc "soffice.bin"
390
391      RmDir /r "$LO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
392      RmDir /r "$LO_HOMEBasis\share\Scripts\oorexx"
393      RmDir /r "$LO_HOMEBasis\share\Scripts\python\pythonSamples"
394
395      ExecWait "$LO_HOMEprogram\unopkg.exe remove --shared
```

```
396
397       Delete "$LO_HOMEprogram\BSF4ooRexx.dll"
398       Delete "$LO_HOMEprogram\BSF.CLS"
399       Delete "$LO_HOMEprogram\UNO.CLS"
400       Delete "$LO_HOMEprogram\UNO_API_info.rxo"
401       Delete "$LO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
402       Delete "$LO_HOMEprogram\UNO_ReflectionUtil.cls"
403       Delete "$LO_HOMEprogram\UNO_XINTERFACES.RXO"
404
405       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEprogram\"
406       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEBasis\program\classes
              ↪ \unoil.jar"
407       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\ridl.jar"
408       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\jurt.jar"
409       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\juh.jar"
410
411       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$LO_HOMEprogram"
412       Goto RemOfficeReg
413
414    RemOfficeReg:
415       DeleteRegKey HKCR ".RXO"
416       DeleteRegKey HKCR "RexxOOoScript"
417       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo"
418       Goto Done
419
420    Done:
421       SetRegView 64
422       DeleteRegKey ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}"
423       SetRegView 32
424       DeleteRegKey HKCR ".JREXX"
425       DeleteRegKey HKCR ".ORX"
426       DeleteRegKey HKCR ".REXX"
427       DeleteRegKey HKCR ".REXXJ"
428       DeleteRegKey HKCR ".RXJ"
429       DeleteRegKey HKCR ".TESTGROUP"
430       DeleteRegKey HKCR ".TESTUNIT"
431       DeleteRegKey HKCR "RexxJavaScript"
432       DeleteRegKey HKCR ".CLS"
433       DeleteRegKey HKCR ".rex"
434       DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\
              ↪ FileExts\.rxj"
435       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx"
436       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java"
437
438       WriteRegStr HKCR "REXXScript" "" "ooRexx Rexx Progam"
439       WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$REXX_HOME\rexx.exe,0"
440
441       ; reload file associations
442       ExecWait "$INSTDIR\install\associationChanged.exe"
443
444       RMDir /r "$INSTDIR"
445       RMDir /r "$SMPROGRAMS\${PRODUCT_NAME}"
446       RMDir "$INSTDIR"
447       RMDir "$SMPROGRAMS\${PRODUCT_NAME}"
448
```

```
449       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
450       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-v400-20090910.
              ↪ jar"
451
452       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$INSTDIR"
453
454       SetRegView 64
455       ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
              ↪ RuntimeLib"
456       ${un.StrRep} $2 "$1" "\jvm.dll" ""
457       ${un.StrRep} $2 "$2" "client" "server"
458       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$2"
459
460       SetAutoClose true
461   SectionEnd
462
463   ; check if Java and ooRexx is installed
464   ; find OpenOffice/LibreOffice folder if is installed
465   Function .onInit
466       Push $0
467       Push $1
468       Push $2
469       InitPluginsDir
470
471       Call getBSFVersion
472       GetVersion::WindowsPlatformArchitecture
473       Pop $WINDOWS_ARCHITECTURE ;32 or 64
474       StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
475
476       x64:
477           Call FindooRexx
478           ; is ooRexx installed?
479           StrCmp "" $REXX_INSTALLATION_MSG RexxSuccess RexxNoSuccess
480           Goto FindBSF
481       x32:
482           MessageBox MB_OK|MB_ICONEXCLAMATION "You need a 64 bit Windows to start
                  ↪ this installer."
483           Abort
484
485       ; yes, it is
486       RexxSuccess:
487           Call FindJava
488           ; is Java installed?
489           StrCmp "" $JAVA_INSTALLATION_MSG JavaSuccess JavaNoSuccess
490
491       ; yes, it is
492       JavaSuccess:
493           Call FindOpenOffice
494           Call FindLibreOffice
495           Goto FindBSF
496
497       JavaNoSuccess:
498           MessageBox MB_OK|MB_ICONEXCLAMATION $JAVA_INSTALLATION_MSG
499           Goto Error
500
501       RexxNoSuccess:
```

```
502         MessageBox MB_OK|MB_ICONEXCLAMATION $REXX_INSTALLATION_MSG
503         Goto Error
504
505     FindBSF:
506         ; check if a version of BSF4ooRexx is already installed
507         SetRegView 64
508         ReadRegStr $0 ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
                ↪ DisplayName"
509         ReadRegStr $1 ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
                ↪ DisplayVersion"
510         SetRegView 32
511         StrCmp $0 "$(^Name)" FoundBSF
512         Goto Done
513
514     ; BSF4ooRexx is already installed
515     ; if no silent mode, ask user what to do
516     ; if silent mode, uninstall the existing version
517     FoundBSF:
518         IfSilent jump_if_silent jump_if_not_silent
519
520         jump_if_not_silent:
521                 MessageBox MB_ICONQUESTION|MB_YESNO "Version $1 of ${
                        ↪ PRODUCT_NAME} is installed. You have to uninstall it
                        ↪ before installing this version. Would you like to
                        ↪ uninstall it?" IDNO err
522
523         jump_if_silent:
524
525         ExecWait '"$INSTDIR\uninst.exe" /S _?=$INSTDIR'
526         Goto Done
527     err:
528         MessageBox MB_OK|MB_ICONEXCLAMATION "The installation was aborted."
529         Abort
530
531     Error:
532         RMDir /r "$INSTDIR"
533         Abort
534
535     Done:
536         Pop $2
537         Pop $1
538         Pop $0
539 FunctionEnd
540
541 ; read the BSF4ooRexx version by using getBSFVersion.rex
542 Function getBSFVersion
543     Push $0
544
545     SetOutPath "$INSTDIR"
546     File "${SCRIPT_PATH}\getBSFVersion.rex"
547     File "${BSF_ROOT}\BSF.CLS"
548     ExecWait "rexx getBSFVersion.rex"
549
550     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
551     FileRead $0 $PRODUCT_VERSION
552     FileClose $0
```

```
553
554     Delete "$INSTDIR\getBSFVersion.rex"
555     Delete "$INSTDIR\BSF.CLS"
556     Delete "$INSTDIR\BSFVersion.txt"
557
558     Pop $0
559 FunctionEnd
560
561 ; read the BSF4ooRexx version by using getBSFVersion.rex
562 ; uninstall function
563 Function un.getBSFVersion
564     Push $0
565
566     SetOutPath "$INSTDIR"
567     File "${SCRIPT_PATH}\getBSFVersion.rex"
568     File "${BSF_ROOT}\BSF.CLS"
569     ExecWait "rexx getBSFVersion.rex"
570
571     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
572     FileRead $0 $PRODUCT_VERSION
573     FileClose $0
574
575     Delete "$INSTDIR\getBSFVersion.rex"
576     Delete "$INSTDIR\BSF.CLS"
577     Delete "$INSTDIR\BSFVersion.txt"
578
579     Pop $0
580 FunctionEnd
581
582 ; read Java version by using java_version.rex
583 Function FindJava
584     Push $0
585     Push $1
586
587     SetOutPath "$INSTDIR"
588     File "${SCRIPT_PATH}\java_version.rex"
589     ExecWait "rexx java_version.rex bitness" $0
590     ExecWait "rexx java_version.rex version" $1
591     Delete "$INSTDIR\java_version.rex"
592     StrCmp $0 "64" CheckVersion JavaVerNotCorrect
593
594     CheckVersion:
595         StrCmp $1 "1" Done JavaVerNotCorrect
596
597     JavaVerNotCorrect:
598         StrCpy $JAVA_INSTALLATION_MSG "You need Java version 1.4 (64 bit) or
            ↪ newer to run this program."
599
600     Done:
601         Pop $1
602         Pop $0
603 FunctionEnd
604
605 ; read ooRexx version from registry
606 ; and bitness by using rexx_version.rex
607 Function FindooRexx
```

```
608     Push $0
609     Push $1
610     Push $2
611     Push $3
612
613     ReadRegStr $REXX_VER HKLM "Software\Microsoft\Windows\CurrentVersion\
            ↪ Uninstall\ooRexx" "DisplayVersion"
614     StrCmp "" "$REXX_VER" RexxNotPresent
615     ReadRegStr $0 HKLM "SYSTEM\CurrentControlSet\Control\Session Manager\
            ↪ Environment" "REXX_HOME"
616     StrCmp "" $0 RexxNotPresent CheckRexxVer
617
618     RexxNotPresent:
619         StrCpy $REXX_INSTALLATION_MSG "ooRexx is not installed on your computer.
                ↪  You need version 4.1.0 (64 bit) or newer to run this program."
620         Goto Done
621
622     CheckRexxVer:
623         GetFullPathName /SHORT $REXX_HOME "$0"
624         StrCpy $0 $REXX_VER 1 0
625         StrCpy $1 $REXX_VER 1 2
626         StrCpy $2 $REXX_VER 1 4
627         StrCpy $REXX_VER "$0$1$2"
628         IntCmp 410 $REXX_VER FoundCorrectRexxVer FoundCorrectRexxVer
                ↪ RexxVerNotCorrect
629
630     FoundCorrectRexxVer:
631         IfFileExists "$REXX_HOME\rexx.exe" CheckRexxBitness RexxNotPresent
632
633     CheckRexxBitness:
634         SetOutPath "$INSTDIR"
635         File "${SCRIPT_PATH}\rexx_version.rex"
636         ExecWait "rexx rexx_version.rex" $3
637         Delete "$INSTDIR\rexx_version.rex"
638         StrCmp $3 "64" Done RexxVerNotCorrect
639
640     RexxVerNotCorrect:
641         StrCpy $REXX_INSTALLATION_MSG "The version of ooRexx installed on your
                ↪ computer is $0.$1.$2 ($3 bit). Version 4.1.0 (64 bit) or newer is
                ↪ required to run this program."
642
643     Done:
644         Pop $3
645         Pop $2
646         Pop $1
647         Pop $0
648 FunctionEnd
649
650 ; find OpenOffice folder
651 Function FindOpenOffice
652     Push $0
653
654     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\OpenOffice.
            ↪ org\3" "OFFICEINSTALLLOCATION"
655     StrCmp "" $0 OO_2 Done
656
```

```
657     OO_2:
658         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
                ↪ OFFICEINSTALLLOCATION"
659         StrCmp "" $0 Done OOFound
660 
661     OOFound:
662         GetFullPathName $OO_HOME "$0"
663         Goto Done
664 
665     Done:
666         Pop $0
667 FunctionEnd
668 
669 ; find OpenOffice folder (uninstall function)
670 Function un.FindOpenOffice
671     Push $0
672 
673     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\OpenOffice.
            ↪ org\3" "OFFICEINSTALLLOCATION"
674     StrCmp "" $0 OO_2 Done
675 
676     OO_2:
677         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
                ↪ OFFICEINSTALLLOCATION"
678         StrCmp "" $0 Done OOFound
679 
680     OOFound:
681         GetFullPathName $OO_HOME "$0"
682         Goto Done
683 
684     Done:
685         Pop $0
686 FunctionEnd
687 
688 ; find LibreOffice folder
689 Function FindLibreOffice
690     Push $0
691 
692     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice\3"
            ↪ "OFFICEINSTALLLOCATION"
693     StrCmp "" $0 LO_2 Done
694 
695     LO_2:
696         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
697         StrCmp "" $0 Done LOFound
698 
699     LOFound:
700         GetFullPathName $LO_HOME "$0"
701         Goto Done
702 
703     Done:
704         Pop $0
705 FunctionEnd
706 
707 ; find LibreOffice folder (uninstall function)
```

```
708 Function un.FindLibreOffice
709     Push $0
710
711     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice\3"
        ↪ "OFFICEINSTALLLOCATION"
712     StrCmp "" $0 LO_2 Done
713
714     LO_2:
715         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
            ↪ OFFICEINSTALLLOCATION"
716         StrCmp "" $0 Done LOFound
717
718     LOFound:
719         GetFullPathName $LO_HOME "$0"
720         Goto Done
721
722     Done:
723         Pop $0
724 FunctionEnd
```

Listing 37: The NSIS Installation Script for 64bit Windows.

7.2.3 The NSIS Reinstall Script for 32 bit Windows

The following listing contains the reinstall script for 32 bit Windows. It is described in detail in chapter 2.2.3.

```
; Name: BSF4ooRexx_x64_reinstall.nsi
; Purpose: NSIS reinstall script for BSF4ooRexx (32 bit)
; Author: Juergen Hesse, Manuel Paar
; Version: 1.0
;
; ------------------- Apache Version 2.0 license --------------------
;    Copyright (C) 2011 Juergen Hesse, Manuel Paar
;
;    Licensed under the Apache License, Version 2.0 (the "License");
;    you may not use this file except in compliance with the License.
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;
;          http://www.apache.org/licenses/LICENSE-2.0
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;    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
;    implied.
;    See the License for the specific language governing permissions and
;    limitations under the License.
; -------------------------------------------------------------------


; load more functions
!include "MUI.nsh"
!include "EnvVarUpdate.nsh"

; set product name and uninstall registry-key
!define PRODUCT_NAME "BSF4ooRexx"
!define PRODUCT_UNINST_KEY "Software\Microsoft\Windows\CurrentVersion\Uninstall\
    ↪ ${PRODUCT_NAME}"
!define PRODUCT_UNINST_ROOT_KEY "HKLM"

; TO CUSTOMIZE
; Source-Folder
!define BSF_ROOT ".\BSF4ooRexx_install\bsf4oorexx"
; Script Location (getBSFVersion.rex, ...)
!define SCRIPT_PATH "."

SetCompressor lzma

; MUI Settings
!define MUI_ABORTWARNING
!define MUI_ICON "${BSF_ROOT}\install\images\windows\bsf4rexx-install.ico"
!define MUI_UNICON "${BSF_ROOT}\install\images\windows\bsf4rexx-uninstall.ico"
; Uninstaller pages
!insertmacro MUI_UNPAGE_INSTFILES
; Language files
!insertmacro MUI_LANGUAGE "English"
```

```
49  ; Reserve files
50  !insertmacro MUI_RESERVEFILE_INSTALLOPTIONS
51
52  Var JAVA_INSTALLATION_MSG
53  Var REXX_HOME
54  Var REXX_VER
55  Var REXX_INSTALLATION_MSG
56  Var OO_HOME
57  Var LO_HOME
58  Var WINDOWS_ARCHITECTURE
59  Var SYSDRIVE
60  Var PRODUCT_VERSION
61
62  Name "${PRODUCT_NAME} $PRODUCT_VERSION"
63  OutFile "${PRODUCT_NAME}_Setup_x32_reinstall.exe"
64  InstallDir "$PROGRAMFILES\${PRODUCT_NAME}"
65  XPStyle on
66  ShowInstDetails show
67  ShowUnInstDetails show
68
69  ; uninstall old version
70  ; copy all files to the destination folder
71  ; create start menue entries
72  ; create file associations
73  Section "BSF4ooRexx" SEC01
74    SetShellVarContext all
75    SetAutoClose true
76    SetOverwrite ifnewer
77
78    ; uninstall old version
79    ExecWait '"$INSTDIR\uninst.exe" /S _?=$INSTDIR'
80
81    ; copy all files to the destination folder
82    SetOutPath "$INSTDIR"
83    File "${BSF_ROOT}\addLinks2OOoSnippets.rxo"
84    File "${BSF_ROOT}\ApacheLicense2.0.txt"
85    File "${BSF_ROOT}\asm_LICENSE.TXT"
86    File "${BSF_ROOT}\bsf-rexx-engine.jar"
87    File "${BSF_ROOT}\bsf-v400-20090910.jar"
88    File "${BSF_ROOT}\BSF.CLS"
89    File "${BSF_ROOT}\install\32\BSF4ooRexx.dll"
90    File "${BSF_ROOT}\BSF_OnTheFly.cls"
91    File "${BSF_ROOT}\credits.txt"
92    File "${BSF_ROOT}\janino_LICENSE.TXT"
93    File "${BSF_ROOT}\refcardBSF4ooRexx.pdf"
94    File "${BSF_ROOT}\refcardOOo.pdf"
95    File "/oname=rexxj.cmd" "${BSF_ROOT}\rexxj2.cmd"
96    File "${BSF_ROOT}\rexxj2.cmd"
97    File "${BSF_ROOT}\rexxj2-32.sh"
98    File "${BSF_ROOT}\rexxj2-64.sh"
99    File "${BSF_ROOT}\rexxj2.sh"
100   File "${BSF_ROOT}\rgf_util2.rex"
101   File "${BSF_ROOT}\UNO.CLS"
102   File "${BSF_ROOT}\UNO_API_info.rxo"
103   File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
104   File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
```

```
105   File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
106   CopyFiles "$SYSDRIVE\${PRODUCT_NAME}_Setup_x32_reinstall.exe" "$INSTDIR"
107   Delete "$SYSDRIVE\${PRODUCT_NAME}_Setup_x32_reinstall.exe"
108
109   File /r "${BSF_ROOT}\information"
110   File /r "${BSF_ROOT}\install"
111   File /r "${BSF_ROOT}\samples"
112   File /r "${BSF_ROOT}\Templates"
113   File /r "${BSF_ROOT}\utilities"
114
115   ; create start menue entries
116   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}"
117   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\GUI RexxTry Program (ooRexxTry.rxj
          ↪ ).lnk" "$INSTDIR\utilities\ooRexxTry.rxj" "" "$INSTDIR\install\images\
          ↪ windows\oorexxtry.ico"
118   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Information (ReleaseNotes,
          ↪ introductions to ooRexx, ${PRODUCT_NAME} overview).lnk" "$INSTDIR\
          ↪ information"
119
120   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}\Installation"
121   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\${PRODUCT_NAME}
          ↪ Download Page.lnk" "http://sourceforge.net/projects/bsf4oorexx/"
122
123   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Samples (simple to comprehensive $
          ↪ {PRODUCT_NAME} examples).lnk" "$INSTDIR\samples"
124   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Utilities (some platform-
          ↪ independent utility scripts).lnk" "$INSTDIR\utilities"
125
126   ; create file associations
127   WriteRegStr HKCR ".JREXX" "" "RexxJavaScript"
128   WriteRegStr HKCR ".JREXX" "Content Type" "text/x-rexx-java"
129   WriteRegStr HKCR ".JREXX" "PerceivedType" "text"
130
131   WriteRegStr HKCR ".ORX" "" "REXXScript"
132   WriteRegStr HKCR ".ORX" "Content Type" "text/x-rexx"
133   WriteRegStr HKCR ".ORX" "PerceivedType" "text"
134
135   WriteRegStr HKCR ".REXX" "" "REXXScript"
136   WriteRegStr HKCR ".REXX" "Content Type" "text/x-rexx"
137   WriteRegStr HKCR ".REXX" "PerceivedType" "text"
138
139   WriteRegStr HKCR ".REXXJ" "" "RexxJavaScript"
140   WriteRegStr HKCR ".REXXJ" "Content Type" "text/x-rexx-java"
141   WriteRegStr HKCR ".REXXJ" "PerceivedType" "text"
142
143   WriteRegStr HKCR ".RXJ" "" "RexxJavaScript"
144   WriteRegStr HKCR ".RXJ" "Content Type" "text/x-rexx-java"
145   WriteRegStr HKCR ".RXJ" "PerceivedType" "text"
146   WriteRegStr HKCR ".RXJ\ShellNew" "FileName" "$INSTDIR\Templates\RexxJavaScript
          ↪  (Rexx script using Java) executed by ooRexx.rxj"
147
148   WriteRegStr HKCR ".TESTGROUP" "" "REXXScript"
149   WriteRegStr HKCR ".TESTGROUP" "Content Type" "text/x-rexx"
150   WriteRegStr HKCR ".TESTGROUP" "PerceivedType" "text"
151
152   WriteRegStr HKCR ".TESTUNIT" "" "REXXScript"
```

```
153   WriteRegStr HKCR ".TESTUNIT" "Content Type" "text/x-rexx"
154   WriteRegStr HKCR ".TESTUNIT" "PerceivedType" "text"
155
156   WriteRegStr HKCR "RexxJavaScript" "" "RexxJavaScript (Rexx script using Java)
          ↪ executed by ooRexx"
157   WriteRegStr HKCR "RexxJavaScript\DefaultIcon" "" "$INSTDIR\install\images\
          ↪ windows\bsf4oorexx.ico,0"
158   WriteRegStr HKCR "RexxJavaScript\shell" "" "open"
159   WriteRegStr HKCR "RexxJavaScript\shell\edit" "" "Edit"
160   WriteRegStr HKCR "RexxJavaScript\shell\edit\command" "" 'notepad.exe "%1"'
161   WriteRegStr HKCR "RexxJavaScript\shell\open" "" "Run"
162   WriteRegStr HKCR "RexxJavaScript\shell\open\command" "" '"$REXX_HOME\rexx.exe"
          ↪  "%1" %*'
163
164   WriteRegStr HKCR ".CLS" "" "REXXScript"
165   WriteRegStr HKCR ".CLS" "Content Type" "text/x-rexx"
166   WriteRegStr HKCR ".CLS" "PerceivedType" "text"
167
168   DeleteRegKey HKCR ".rex"
169   WriteRegStr HKCR ".rex" "" "REXXScript"
170   WriteRegStr HKCR ".rex" "Content Type" "text/x-rexx"
171   WriteRegStr HKCR ".rex" "PerceivedType" "text"
172   WriteRegStr HKCR ".rex\ShellNew" "FileName" "$INSTDIR\Templates\RexxScript (
          ↪ Rexx script) executed by ooRexx.rex"
173
174   DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj"
175   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj\OpenWithList" "a" "rexx.exe"
176   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj\OpenWithList" "MRUList" "a"
177   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj\OpenWithProgids" "RexxJavaScript" ""
178
179   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx" "Extension" ".rex"
180   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java" "Extension" ".
          ↪ rxj"
181
182   WriteRegStr HKCR "REXXScript" "" "RexxScript (Rexx script) executed by ooRexx"
183   WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$INSTDIR\install\images\windows\
          ↪ oorexx.ico,0"
184   WriteRegStr HKCR "REXXScript\shell" "" "open"
185   WriteRegStr HKCR "REXXScript\shell\edit" "" "Edit"
186   WriteRegStr HKCR "REXXScript\shell\edit\command" "" 'notepad.exe "%1"'
187   WriteRegStr HKCR "REXXScript\shell\open" "" "Run"
188   WriteRegStr HKCR "REXXScript\shell\open\command" "" '"$REXX_HOME\rexx.exe" "%1
          ↪ " %*'
189
190   ; set CLASSPATH and PATH
191   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
192   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-v400-20090910.jar"
193
194   ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$INSTDIR"
195
196   StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
197
```

```
198   x64:
199       ReadRegStr $1 HKLM "SOFTWARE\Wow6432Node\JavaSoft\Java Runtime Environment
              ↪ \1.6" "RuntimeLib"
200       Goto AddJavaRuntimeLib
201
202   x32:
203       ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
              ↪ RuntimeLib"
204       Goto AddJavaRuntimeLib
205
206   AddJavaRuntimeLib:
207       ${StrRep} $2 "$1" "\jvm.dll" ""
208       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$2"
209
210   ; is OpenOffice installed?
211   StrCmp "" $OO_HOME SearchLibreOffice OOFound
212
213   ; yes, so copy necessary files
214   OOFound:
215      KillProcDLL::KillProc "soffice.exe"
216      KillProcDLL::KillProc "soffice.bin"
217      SetOutPath "$OO_HOMEprogram"
218      File "${BSF_ROOT}\install\32\BSF4ooRexx.dll"
219      File "${BSF_ROOT}\BSF.CLS"
220      File "${BSF_ROOT}\UNO.CLS"
221      File "${BSF_ROOT}\UNO_API_info.rxo"
222      File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
223      File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
224      File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
225
226      SetOutPath "$OO_HOMEBasis\share\Scripts\javascript"
227      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
              ↪ HowtoCreateApiInfo"
228
229      SetOutPath "$OO_HOMEBasis\share\Scripts"
230      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
231
232      SetOutPath "$OO_HOMEBasis\share\Scripts\python"
233      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
234
235      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEprogram\"
236      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEBasis\program\classes\
              ↪ unoil.jar"
237      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\ridl.jar"
238      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\jurt.jar"
239      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\juh.jar"
240
241      ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$OO_HOMEprogram"
242
243      SetOutPath "$INSTDIR\install"
244      ExecWait '"$OO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
              ↪ .oxt'
245      StrCmp "" $LO_HOME AddOfficeReg LOFound
246
247   SearchLibreOffice:
248      ; is LibreOffice installed?
```

```
249       StrCmp "" $LO_HOME Done LOFound
250
251    ; yes, so copy necessary files
252    LOFound:
253       KillProcDLL::KillProc "soffice.exe"
254       KillProcDLL::KillProc "soffice.bin"
255       SetOutPath $LO_HOME"program"
256       File "${BSF_ROOT}\install\32\BSF4ooRexx.dll"
257       File "${BSF_ROOT}\BSF.CLS"
258       File "${BSF_ROOT}\UNO.CLS"
259       File "${BSF_ROOT}\UNO_API_info.rxo"
260       File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
261       File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
262       File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
263
264       SetOutPath "$LO_HOMEBasis\share\Scripts\javascript"
265       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
             ↪ HowtoCreateApiInfo"
266
267       SetOutPath "$LO_HOMEBasis\share\Scripts"
268       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
269
270       SetOutPath "$LO_HOMEBasis\share\Scripts\python"
271       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
272
273       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEprogram\"
274       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEBasis\program\classes\
             ↪ unoil.jar"
275       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\ridl.jar"
276       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\jurt.jar"
277       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\juh.jar"
278
279       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$LO_HOMEprogram"
280
281       SetOutPath "$INSTDIR\install"
282       ExecWait '"$LO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
             ↪ .oxt'
283       Goto AddOfficeReg
284
285    AddOfficeReg:
286       WriteRegStr HKCR ".RXO" "" "RexxOOoScript"
287       WriteRegStr HKCR ".RXO" "Content Type" "text/x-rexx-java-ooo"
288       WriteRegStr HKCR ".RXO" "PerceivedType" "text"
289       WriteRegStr HKCR ".RXO\ShellNew" "FileName" "$INSTDIR\Templates\
             ↪ RexxOOoScript (Rexx script using OpenOffice.org) executed by ooRexx.
             ↪ rxo"
290
291       WriteRegStr HKCR "RexxOOoScript" "" "RexxOOoScript (Rexx script using
             ↪ OpenOffice.org/LibreOffice) executed by ooRexx"
292       WriteRegStr HKCR "RexxOOoScript\DefaultIcon" "" "$INSTDIR\install\images\
             ↪ windows\oorexx4ooo.ico,0"
293       WriteRegStr HKCR "RexxOOoScript\shell" "" "open"
294       WriteRegStr HKCR "RexxOOoScript\shell\edit" "" "Edit"
295       WriteRegStr HKCR "RexxOOoScript\shell\edit\command" "" 'notepad.exe "%1"'
296       WriteRegStr HKCR "RexxOOoScript\shell\open" "" "Run"
297       WriteRegStr HKCR "RexxOOoScript\shell\open\command" "" '"$REXX_HOME\rexx.
```

```
        ↪ exe" "%1" %*'
298
299     WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo" "
        ↪ Extension" ".rxo"
300     Goto Done
301   Done:
302     ; reload file associations
303     ExecWait "$INSTDIR\install\associationChanged.exe"
304     MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully reinstalled.
        ↪ "
305 SectionEnd
306
307 ; Add Uninstall and Reinstall Entry to the start menue
308 Section -AdditionalIcons
309   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Uninstall ${
      ↪ PRODUCT_NAME}.lnk" "$INSTDIR\uninst.exe" "" "$INSTDIR\install\images\
      ↪ windows\bsf4rexx-uninstall.ico"
310   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Reinstall ${
      ↪ PRODUCT_NAME}.lnk" "$INSTDIR\${PRODUCT_NAME}_Setup_x32_reinstall.exe" ""
      ↪  "$INSTDIR\install\images\windows\bsf4rexx-reinstall.ico"
311 SectionEnd
312
313 ; create Uninstaller and Uninstall registry entries
314 Section -Post
315   WriteUninstaller "$INSTDIR\uninst.exe"
316   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayName" "
      ↪ $(^Name)"
317   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
      ↪ UninstallString" "$INSTDIR\uninst.exe"
318   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayIcon" "
      ↪ $INSTDIR\install\images\windows\bsf4oorexx.ico"
319   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayVersion
      ↪ " "$PRODUCT_VERSION"
320   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoRepair" 0
321   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoModify" 0
322   WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "ModifyPath" "
      ↪ $INSTDIR\${PRODUCT_NAME}_Setup_x32_reinstall.exe"
323 SectionEnd
324
325 ; if no silent mode, show success message
326 Function un.onUninstSuccess
327   IfSilent jump_if_silent jump_if_not_silent
328
329   jump_if_not_silent:
330     HideWindow
331     MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully uninstalled.
        ↪ " IDOK
332
333   jump_if_silent:
334
335 FunctionEnd
336
337 ; if no silent mode, ask user if he want to uninstall
338 Function un.onInit
339   Call un.getBSFVersion
340   IfSilent jump_if_silent jump_if_not_silent
```

```
341
342   jump_if_not_silent:
343      MessageBox MB_ICONQUESTION|MB_YESNO|MB_DEFBUTTON2 "Do you want to uninstall
         ↪  $(^Name) and all the components?" IDYES +2
344      Abort
345
346   jump_if_silent:
347
348 FunctionEnd
349
350 ; undo all changes
351 Section Uninstall
352   SetShellVarContext all
353
354   ; is OpenOffice/LibreOffice installed?
355   Call un.FindOpenOffice
356   Call un.FindLibreOffice
357   StrCmp "" $OO_HOME SearchLibreOffice OOFound
358
359   ; OpenOffice found: so delete files
360   OOFound:
361      KillProcDLL::KillProc "soffice.exe"
362      KillProcDLL::KillProc "soffice.bin"
363
364      RMDir /r "$OO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
365      RMDir /r "$OO_HOMEBasis\share\Scripts\oorexx"
366      RMDir /r "$OO_HOMEBasis\share\Scripts\python\pythonSamples"
367
368      ExecWait "$OO_HOMEprogram\unopkg.exe remove --shared
         ↪ ScriptProviderForooRexx.oxt"
369
370      Delete "$OO_HOMEprogram\BSF4ooRexx.dll"
371      Delete "$OO_HOMEprogram\BSF.CLS"
372      Delete "$OO_HOMEprogram\UNO.CLS"
373      Delete "$OO_HOMEprogram\UNO_API_info.rxo"
374      Delete "$OO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
375      Delete "$OO_HOMEprogram\UNO_ReflectionUtil.cls"
376      Delete "$OO_HOMEprogram\UNO_XINTERFACES.RXO"
377
378      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEprogram\"
379      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEBasis\program\classes
         ↪ \unoil.jar"
380      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\ridl.jar"
381      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\jurt.jar"
382      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\juh.jar"
383
384      ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$OO_HOMEprogram"
385      StrCmp "" $LO_HOME RemOfficeReg LOFound
386
387   SearchLibreOffice:
388      ; is LibreOffice installed?
389      StrCmp "" $LO_HOME Done LOFound
390
391   ; LibreOffice found: so delete files
392   LOFound:
393      KillProcDLL::KillProc "soffice.exe"
```

```
394       KillProcDLL::KillProc "soffice.bin"
395
396       RMDir /r "$LO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
397       RMDir /r "$LO_HOMEBasis\share\Scripts\oorexx"
398       RMDir /r "$LO_HOMEBasis\share\Scripts\python\pythonSamples"
399
400       ExecWait "$LO_HOMEprogram\unopkg.exe remove --shared
              ↪ ScriptProviderForooRexx.oxt"
401
402       Delete "$LO_HOMEprogram\BSF4ooRexx.dll"
403       Delete "$LO_HOMEprogram\BSF.CLS"
404       Delete "$LO_HOMEprogram\UNO.CLS"
405       Delete "$LO_HOMEprogram\UNO_API_info.rxo"
406       Delete "$LO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
407       Delete "$LO_HOMEprogram\UNO_ReflectionUtil.cls"
408       Delete "$LO_HOMEprogram\UNO_XINTERFACES.RXO"
409
410       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEprogram\"
411       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEBasis\program\classes
              ↪ \unoil.jar"
412       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\ridl.jar"
413       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\jurt.jar"
414       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\juh.jar"
415
416       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$LO_HOMEprogram"
417       Goto RemOfficeReg
418
419    RemOfficeReg:
420       DeleteRegKey HKCR ".RXO"
421       DeleteRegKey HKCR "RexxOOoScript"
422       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo"
423       Goto Done
424
425    Done:
426       StrCmp $WINDOWS_ARCHITECTURE "64" OS64 OS32
427
428    OS64:
429       ; change registry mode to 64 bit
430       SetRegView 64
431
432    OS32:
433       DeleteRegKey ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}"
434       SetRegView 32 ; change registry mode to 32 bit
435       DeleteRegKey HKCR ".JREXX"
436       DeleteRegKey HKCR ".ORX"
437       DeleteRegKey HKCR ".REXX"
438       DeleteRegKey HKCR ".REXXJ"
439       DeleteRegKey HKCR ".RXJ"
440       DeleteRegKey HKCR ".TESTGROUP"
441       DeleteRegKey HKCR ".TESTUNIT"
442       DeleteRegKey HKCR "RexxJavaScript"
443       DeleteRegKey HKCR ".CLS"
444       DeleteRegKey HKCR ".rex"
445       DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\
              ↪ FileExts\.rxj"
446       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx"
```

```
447       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java"
448
449       WriteRegStr HKCR "REXXScript" "" "ooRexx Rexx Progam"
450       WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$REXX_HOME\rexx.exe,0"
451
452       ; reload file associations
453       ExecWait "$INSTDIR\install\associationChanged.exe"
454
455       RMDir /r "$INSTDIR"
456       RMDir /r "$SMPROGRAMS\${PRODUCT_NAME}"
457       RMDir "$INSTDIR"
458       RMDir "$SMPROGRAMS\${PRODUCT_NAME}"
459
460       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
461       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-v400-20090910.
              ↪ jar"
462
463       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$INSTDIR"
464
465       StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
466
467       x64:
468           ReadRegStr $1 HKLM "SOFTWARE\Wow6432Node\JavaSoft\Java Runtime
                  ↪ Environment\1.6" "RuntimeLib"
469           Goto RemJavaRuntimeLib
470
471       x32:
472           ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
                  ↪ RuntimeLib"
473           Goto RemJavaRuntimeLib
474
475       RemJavaRuntimeLib:
476           ${un.StrRep} $2 "$1" "\jvm.dll" ""
477           ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$2"
478
479       SetAutoClose false
480   SectionEnd
481
482   ; check if Java and ooRexx is installed
483   ; find OpenOffice/LibreOffice folder if is installed
484   Function .onInit
485       ReadEnvStr $SYSDRIVE SYSTEMDRIVE
486       InitPluginsDir
487
488       GetVersion::WindowsPlatformArchitecture
489       Pop $WINDOWS_ARCHITECTURE ;32 or 64
490
491       Call getBSFVersion
492       Call FindooRexx
493       ; is ooRexx installed?
494       StrCmp "" $REXX_INSTALLATION_MSG RexxSuccess RexxNoSuccess
495
496       ; yes, it is
497       RexxSuccess:
498           Call FindJava
499           ; is Java installed?
```

```
500         StrCmp "" $JAVA_INSTALLATION_MSG JavaSuccess JavaNoSuccess
501
502     ; yes, it is
503     JavaSuccess:
504         ; save reinstaller
505         CopyFiles "$INSTDIR\${PRODUCT_NAME}_Setup_x32_reinstall.exe" "$SYSDRIVE\
                ↪ "
506         Call FindOpenOffice
507         Call FindLibreOffice
508         Goto Done
509
510     JavaNoSuccess:
511         MessageBox MB_OK|MB_ICONEXCLAMATION $JAVA_INSTALLATION_MSG
512         Goto Error
513
514     RexxNoSuccess:
515         MessageBox MB_OK|MB_ICONEXCLAMATION $REXX_INSTALLATION_MSG
516         Goto Error
517
518     Error:
519         RMDir /r "$INSTDIR"
520         Abort
521
522     Done:
523 FunctionEnd
524
525 ; read the BSF4ooRexx version by using getBSFVersion.rex
526 Function getBSFVersion
527     Push $0
528
529     SetOutPath "$INSTDIR"
530     File "${SCRIPT_PATH}\getBSFVersion.rex"
531     File "${BSF_ROOT}\BSF.CLS"
532     ExecWait "rexx getBSFVersion.rex"
533
534     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
535     FileRead $0 $PRODUCT_VERSION
536     FileClose $0
537
538     Delete "$INSTDIR\getBSFVersion.rex"
539     Delete "$INSTDIR\BSF.CLS"
540     Delete "$INSTDIR\BSFVersion.txt"
541
542     Pop $0
543 FunctionEnd
544
545 ; read the BSF4ooRexx version by using getBSFVersion.rex
546 ; Uninstall function
547 Function un.getBSFVersion
548     Push $0
549
550     SetOutPath "$INSTDIR"
551     File "${SCRIPT_PATH}\getBSFVersion.rex"
552     File "${BSF_ROOT}\BSF.CLS"
553     ExecWait "rexx getBSFVersion.rex"
554
```

```
555     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
556     FileRead $0 $PRODUCT_VERSION
557     FileClose $0
558
559     Delete "$INSTDIR\getBSFVersion.rex"
560     Delete "$INSTDIR\BSF.CLS"
561     Delete "$INSTDIR\BSFVersion.txt"
562
563     Pop $0
564 FunctionEnd
565
566 ; read Java version by using java_version.rex
567 Function FindJava
568     Push $0
569     Push $1
570
571     SetOutPath "$INSTDIR"
572     File "${SCRIPT_PATH}\java_version.rex"
573     ExecWait "rexx java_version.rex bitness" $0
574     ExecWait "rexx java_version.rex version" $1
575     Delete "$INSTDIR\java_version.rex"
576     StrCmp $0 "64" JavaVerNotCorrect CheckVersion
577
578     CheckVersion:
579         StrCmp $1 "1" Done JavaVerNotCorrect
580
581     JavaVerNotCorrect:
582         StrCpy $JAVA_INSTALLATION_MSG "You need Java version 1.4 (32 bit) or
            ↪ newer to run this program."
583
584     Done:
585         Pop $1
586         Pop $0
587 FunctionEnd
588
589 ; read ooRexx version from registry
590 ; and bitness by using rexx_version.rex
591 Function FindooRexx
592     Push $0
593     Push $1
594     Push $2
595     Push $3
596
597     ReadRegStr $REXX_VER HKLM "Software\Microsoft\Windows\CurrentVersion\
        ↪ Uninstall\ooRexx" "DisplayVersion"
598     StrCmp "" "$REXX_VER" RexxNotPresent
599     ReadRegStr $0 HKLM "SYSTEM\CurrentControlSet\Control\Session Manager\
        ↪ Environment" "REXX_HOME"
600     StrCmp "" $0 RexxNotPresent CheckRexxVer
601
602     RexxNotPresent:
603         StrCpy $REXX_INSTALLATION_MSG "ooRexx is not installed on your computer.
            ↪  You need version 4.1.0 (32 bit) or newer to run this program."
604         Goto Done
605
606     CheckRexxVer:
```

```
607         GetFullPathName /SHORT $REXX_HOME "$0"
608         StrCpy $0 $REXX_VER 1 0
609         StrCpy $1 $REXX_VER 1 2
610         StrCpy $2 $REXX_VER 1 4
611         StrCpy $REXX_VER "$0$1$2"
612         IntCmp 410 $REXX_VER FoundCorrectRexxVer FoundCorrectRexxVer
                ↪ RexxVerNotCorrect
613
614     FoundCorrectRexxVer:
615         IfFileExists "$REXX_HOME\rexx.exe" CheckRexxBitness RexxNotPresent
616
617     CheckRexxBitness:
618         SetOutPath "$INSTDIR"
619         File "${SCRIPT_PATH}\rexx_version.rex"
620         ExecWait "rexx rexx_version.rex" $3
621         Delete "$INSTDIR\rexx_version.rex"
622         StrCmp $3 "32" Done RexxVerNotCorrect
623
624     RexxVerNotCorrect:
625         StrCpy $REXX_INSTALLATION_MSG "The version of ooRexx installed on your
                ↪ computer is $0.$1.$2 ($3 bit). Version 4.1.0 (32 bit) or newer is
                ↪ required to run this program."
626
627     Done:
628         Pop $3
629         Pop $2
630         Pop $1
631         Pop $0
632 FunctionEnd
633
634 ; find OpenOffice folder
635 Function FindOpenOffice
636     Push $0
637
638     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
639
640     x64:
641         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\
                ↪ OpenOffice.org\3" "OFFICEINSTALLLOCATION"
642         StrCmp "" $0 Done OOFound
643
644     x32:
645         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
                ↪ OFFICEINSTALLLOCATION"
646         StrCmp "" $0 Done OOFound
647
648     OOFound:
649         GetFullPathName $OO_HOME "$0"
650         Goto Done
651
652     Done:
653         Pop $0
654 FunctionEnd
655
656 ; find OpenOffice folder (uninstall function)
657 Function un.FindOpenOffice
```

```
658     Push $0
659
660     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
661
662     x64:
663         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\
                ↪ OpenOffice.org\3" "OFFICEINSTALLLOCATION"
664         StrCmp "" $0 Done OOFound
665
666     x32:
667         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
                ↪ OFFICEINSTALLLOCATION"
668         StrCmp "" $0 Done OOFound
669
670     OOFound:
671         GetFullPathName $OO_HOME "$0"
672         Goto Done
673
674     Done:
675         Pop $0
676 FunctionEnd
677
678 ; find LibreOffice folder
679 Function FindLibreOffice
680     Push $0
681
682     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
683
684     x64:
685         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice
                ↪ \3" "OFFICEINSTALLLOCATION"
686         StrCmp "" $0 Done LOFound
687
688     x32:
689         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
690         StrCmp "" $0 Done LOFound
691
692     LOFound:
693         GetFullPathName $LO_HOME "$0"
694         Goto Done
695
696     Done:
697         Pop $0
698 FunctionEnd
699
700 ; find LibreOffice folder (uninstall function)
701 Function un.FindLibreOffice
702     Push $0
703
704     StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
705
706     x64:
707         ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice
                ↪ \3" "OFFICEINSTALLLOCATION"
708         StrCmp "" $0 Done LOFound
```

```
709
710     x32:
711         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
712         StrCmp "" $0 Done LOFound
713
714     LOFound:
715         GetFullPathName $LO_HOME "$0"
716         Goto Done
717
718     Done:
719         Pop $0
720 FunctionEnd
```

Listing 38: The NSIS Reinstall Script for 32bit Windows.

7.2.4 The NSIS Reinstall Script for 64 bit Windows

The following listing contains the reinstall script for 64 bit Windows. It is described in detail in chapter 2.2.3.

```
; Name: BSF4ooRexx_x64_reinstall.nsi
; Purpose: NSIS reinstall script for BSF4ooRexx (64 bit)
; Author: Juergen Hesse, Manuel Paar
; Version: 1.0
;
; ------------------- Apache Version 2.0 license --------------------
;    Copyright (C) 2011 Juergen Hesse, Manuel Paar
;
;    Licensed under the Apache License, Version 2.0 (the "License");
;    you may not use this file except in compliance with the License.
;    You may obtain a copy of the License at
;
;          http://www.apache.org/licenses/LICENSE-2.0
;
;    Unless required by applicable law or agreed to in writing, software
;    distributed under the License is distributed on an "AS IS" BASIS,
;    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
;    implied.
;    See the License for the specific language governing permissions and
;    limitations under the License.
; -------------------------------------------------------------------

; load more functions
!include "MUI.nsh"
!include "EnvVarUpdate.nsh"

; set product name and uninstall registry-key
!define PRODUCT_NAME "BSF4ooRexx"
!define PRODUCT_UNINST_KEY "Software\Microsoft\Windows\CurrentVersion\Uninstall\
    ↪ ${PRODUCT_NAME}"
!define PRODUCT_UNINST_ROOT_KEY "HKLM"

; TO CUSTOMIZE
; Source-Folder
!define BSF_ROOT ".\BSF4ooRexx_install\bsf4oorexx"
; Script Location (getBSFVersion.rex, ...)
!define SCRIPT_PATH "."


SetCompressor lzma

; MUI Settings
!define MUI_ABORTWARNING
!define MUI_ICON "${BSF_ROOT}\install\images\windows\bsf4rexx-install.ico"
!define MUI_UNICON "${BSF_ROOT}\install\images\windows\bsf4rexx-uninstall.ico"
; Uninstaller pages
!insertmacro MUI_UNPAGE_INSTFILES
; Language files
!insertmacro MUI_LANGUAGE "English"
```

```
49  ; Reserve files
50  !insertmacro MUI_RESERVEFILE_INSTALLOPTIONS
51
52  Var JAVA_INSTALLATION_MSG
53  Var REXX_HOME
54  Var REXX_VER
55  Var REXX_INSTALLATION_MSG
56  Var OO_HOME
57  Var LO_HOME
58  Var WINDOWS_ARCHITECTURE
59  Var SYSDRIVE
60  Var PRODUCT_VERSION
61
62  Name "${PRODUCT_NAME} $PRODUCT_VERSION"
63  OutFile "${PRODUCT_NAME}_Setup_x64_reinstall.exe"
64  InstallDir "$PROGRAMFILES64\${PRODUCT_NAME}"
65  XPStyle on       ; wenn XP, dann XP-Style verwenden
66  ShowInstDetails show
67  ShowUnInstDetails show
68
69  ; uninstall old version
70  ; copy all files to the destination folder
71  ; create start menue entries
72  ; create file associations
73  Section "BSF4ooRexx" SEC01
74    SetShellVarContext all
75    SetAutoClose true
76    SetOverwrite ifnewer
77
78    ; uninstall old version
79    ExecWait '"$INSTDIR\uninst.exe" /S _?=$INSTDIR'
80
81    ; copy all files to the destination folder
82    SetOutPath "$INSTDIR"
83    File "${BSF_ROOT}\addLinks2OOoSnippets.rxo"
84    File "${BSF_ROOT}\ApacheLicense2.0.txt"
85    File "${BSF_ROOT}\asm_LICENSE.TXT"
86    File "${BSF_ROOT}\bsf-rexx-engine.jar"
87    File "${BSF_ROOT}\bsf-v400-20090910.jar"
88    File "${BSF_ROOT}\BSF.CLS"
89    File "${BSF_ROOT}\install\64\BSF4ooRexx.dll"
90    File "${BSF_ROOT}\BSF_OnTheFly.cls"
91    File "${BSF_ROOT}\credits.txt"
92    File "${BSF_ROOT}\janino_LICENSE.TXT"
93    File "${BSF_ROOT}\refcardBSF4ooRexx.pdf"
94    File "${BSF_ROOT}\refcardOOo.pdf"
95    File "/oname=rexxj.cmd" "${BSF_ROOT}\rexxj2.cmd"
96    File "${BSF_ROOT}\rexxj2.cmd"
97    File "${BSF_ROOT}\rexxj2-32.sh"
98    File "${BSF_ROOT}\rexxj2-64.sh"
99    File "${BSF_ROOT}\rexxj2.sh"
100   File "${BSF_ROOT}\rgf_util2.rex"
101   File "${BSF_ROOT}\UNO.CLS"
102   File "${BSF_ROOT}\UNO_API_info.rxo"
103   File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
104   File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
```

```
105   File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
106   CopyFiles "$SYSDRIVE\${PRODUCT_NAME}_Setup_x64_reinstall.exe" "$INSTDIR"
107   Delete "$SYSDRIVE\${PRODUCT_NAME}_Setup_x64_reinstall.exe"
108
109   File /r "${BSF_ROOT}\information"
110   File /r "${BSF_ROOT}\install"
111   File /r "${BSF_ROOT}\samples"
112   File /r "${BSF_ROOT}\Templates"
113   File /r "${BSF_ROOT}\utilities"
114
115   ; create start menue entries
116   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}"
117   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\GUI RexxTry Program (ooRexxTry.rxj
          ↪ ).lnk" "$INSTDIR\utilities\ooRexxTry.rxj" "" "$INSTDIR\install\images\
          ↪ windows\oorexxtry.ico"
118   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Information (ReleaseNotes,
          ↪ introductions to ooRexx, ${PRODUCT_NAME} overview).lnk" "$INSTDIR\
          ↪ information"
119
120   CreateDirectory "$SMPROGRAMS\${PRODUCT_NAME}\Installation"
121   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\${PRODUCT_NAME}
          ↪ Download Page.lnk" "http://sourceforge.net/projects/bsf4oorexx/"
122
123   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Samples (simple to comprehensive $
          ↪ {PRODUCT_NAME} examples).lnk" "$INSTDIR\samples"
124   CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Utilities (some platform-
          ↪ independent utility scripts).lnk" "$INSTDIR\utilities"
125
126   ; create file associations
127   WriteRegStr HKCR ".JREXX" "" "RexxJavaScript"
128   WriteRegStr HKCR ".JREXX" "Content Type" "text/x-rexx-java"
129   WriteRegStr HKCR ".JREXX" "PerceivedType" "text"
130
131   WriteRegStr HKCR ".ORX" "" "REXXScript"
132   WriteRegStr HKCR ".ORX" "Content Type" "text/x-rexx"
133   WriteRegStr HKCR ".ORX" "PerceivedType" "text"
134
135   WriteRegStr HKCR ".REXX" "" "REXXScript"
136   WriteRegStr HKCR ".REXX" "Content Type" "text/x-rexx"
137   WriteRegStr HKCR ".REXX" "PerceivedType" "text"
138
139   WriteRegStr HKCR ".REXXJ" "" "RexxJavaScript"
140   WriteRegStr HKCR ".REXXJ" "Content Type" "text/x-rexx-java"
141   WriteRegStr HKCR ".REXXJ" "PerceivedType" "text"
142
143   WriteRegStr HKCR ".rxj" "" "RexxJavaScript"
144   WriteRegStr HKCR ".rxj" "Content Type" "text/x-rexx-java"
145   WriteRegStr HKCR ".rxj" "PerceivedType" "text"
146   WriteRegStr HKCR ".rxj\ShellNew" "FileName" "$INSTDIR\Templates\RexxJavaScript
          ↪  (Rexx script using Java) executed by ooRexx.rxj"
147
148   WriteRegStr HKCR ".TESTGROUP" "" "REXXScript"
149   WriteRegStr HKCR ".TESTGROUP" "Content Type" "text/x-rexx"
150   WriteRegStr HKCR ".TESTGROUP" "PerceivedType" "text"
151
152   WriteRegStr HKCR ".TESTUNIT" "" "REXXScript"
```

```
153   WriteRegStr HKCR ".TESTUNIT" "Content Type" "text/x-rexx"
154   WriteRegStr HKCR ".TESTUNIT" "PerceivedType" "text"
155
156   WriteRegStr HKCR "RexxJavaScript" "" "RexxJavaScript (Rexx script using Java)
          ↪ executed by ooRexx"
157   WriteRegStr HKCR "RexxJavaScript\DefaultIcon" "" "$INSTDIR\install\images\
          ↪ windows\bsf4oorexx.ico,0"
158   WriteRegStr HKCR "RexxJavaScript\shell" "" "open"
159   WriteRegStr HKCR "RexxJavaScript\shell\edit" "" "Edit"
160   WriteRegStr HKCR "RexxJavaScript\shell\edit\command" "" 'notepad.exe "%1"'
161   WriteRegStr HKCR "RexxJavaScript\shell\open" "" "Run"
162   WriteRegStr HKCR "RexxJavaScript\shell\open\command" "" '"$REXX_HOME\rexx.exe"
          ↪  "%1" %*'
163
164   WriteRegStr HKCR ".CLS" "" "REXXScript"
165   WriteRegStr HKCR ".CLS" "Content Type" "text/x-rexx"
166   WriteRegStr HKCR ".CLS" "PerceivedType" "text"
167
168   DeleteRegKey HKCR ".rex"
169   WriteRegStr HKCR ".rex" "" "REXXScript"
170   WriteRegStr HKCR ".rex" "Content Type" "text/x-rexx"
171   WriteRegStr HKCR ".rex" "PerceivedType" "text"
172   WriteRegStr HKCR ".rex\ShellNew" "FileName" "$INSTDIR\Templates\RexxScript (
          ↪ Rexx script) executed by ooRexx.rex"
173
174   DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj"
175   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj\OpenWithList" "a" "rexx.exe"
176   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj\OpenWithList" "MRUList" "a"
177   WriteRegStr HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts
          ↪ \.rxj\OpenWithProgids" "RexxJavaScript" ""
178
179   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx" "Extension" ".rex"
180   WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java" "Extension" ".
          ↪ rxj"
181
182   WriteRegStr HKCR "REXXScript" "" "RexxScript (Rexx script) executed by ooRexx"
183   WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$INSTDIR\install\images\windows\
          ↪ oorexx.ico,0"
184   WriteRegStr HKCR "REXXScript\shell" "" "open"
185   WriteRegStr HKCR "REXXScript\shell\edit" "" "Edit"
186   WriteRegStr HKCR "REXXScript\shell\edit\command" "" 'notepad.exe "%1"'
187   WriteRegStr HKCR "REXXScript\shell\open" "" "Run"
188   WriteRegStr HKCR "REXXScript\shell\open\command" "" '"$REXX_HOME\rexx.exe" "%1
          ↪ " %*'
189
190   ; set CLASSPATH and PATH
191   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
192   ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$INSTDIR\bsf-v400-20090910.jar"
193
194   ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$INSTDIR"
195
196   SetRegView 64
197   ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
```

```
      ↪ RuntimeLib"
198   ${StrRep} $2 "$1" "\jvm.dll" ""
199   ${StrRep} $2 "$2" "client" "server"
200   ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$2"
201
202   ; is OpenOffice installed?
203   StrCmp "" $OO_HOME SearchLibreOffice OOFound
204
205   ; yes, so copy necessary files
206   OOFound:
207      KillProcDLL::KillProc "soffice.exe"
208      KillProcDLL::KillProc "soffice.bin"
209      SetOutPath "$OO_HOMEprogram"
210      File "${BSF_ROOT}\install\64\BSF4ooRexx.dll"
211      File "${BSF_ROOT}\BSF.CLS"
212      File "${BSF_ROOT}\UNO.CLS"
213      File "${BSF_ROOT}\UNO_API_info.rxo"
214      File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
215      File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
216      File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
217
218      SetOutPath "$OO_HOMEBasis\share\Scripts\javascript"
219      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
         ↪ HowtoCreateApiInfo"
220
221      SetOutPath "$OO_HOMEBasis\share\Scripts"
222      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
223
224      SetOutPath "$OO_HOMEBasis\share\Scripts\python"
225      File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
226
227      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEprogram\"
228      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEBasis\program\classes\
         ↪ unoil.jar"
229      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\ridl.jar"
230      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\jurt.jar"
231      ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$OO_HOMEURE\java\juh.jar"
232
233      ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$OO_HOMEprogram"
234
235      SetOutPath "$INSTDIR\install"
236      ExecWait '"$OO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
         ↪ .oxt'
237      StrCmp "" $LO_HOME AddOfficeReg LOFound
238
239   SearchLibreOffice:
240      ; is LibreOffice installed?
241      StrCmp "" $LO_HOME Done LOFound
242
243   ; yes, so copy necessary files
244   LOFound:
245      KillProcDLL::KillProc "soffice.exe"
246      KillProcDLL::KillProc "soffice.bin"
247      SetOutPath $LO_HOME"program"
248      File "${BSF_ROOT}\install\64\BSF4ooRexx.dll"
249      File "${BSF_ROOT}\BSF.CLS"
```

```
250       File "${BSF_ROOT}\UNO.CLS"
251       File "${BSF_ROOT}\UNO_API_info.rxo"
252       File "${BSF_ROOT}\UNO_CREATE_INTERFACE_LIST.RXO"
253       File "${BSF_ROOT}\UNO_ReflectionUtil.cls"
254       File "${BSF_ROOT}\UNO_XINTERFACES.RXO"
255
256       SetOutPath "$LO_HOMEBasis\share\Scripts\javascript"
257       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\javascript\
              ↪ HowtoCreateApiInfo"
258
259       SetOutPath "$LO_HOMEBasis\share\Scripts"
260       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\oorexx"
261
262       SetOutPath "$LO_HOMEBasis\share\Scripts\python"
263       File /r "${BSF_ROOT}\install\OOo_ScriptLibraries\python\pythonSamples"
264
265       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEprogram\"
266       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEBasis\program\classes\
              ↪ unoil.jar"
267       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\ridl.jar"
268       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\jurt.jar"
269       ${EnvVarUpdate} $0 "CLASSPATH" "A" "HKLM" "$LO_HOMEURE\java\juh.jar"
270
271       ${EnvVarUpdate} $0 "PATH" "A" "HKLM" "$LO_HOMEprogram"
272
273       SetOutPath "$INSTDIR\install"
274       ExecWait '"$LO_HOMEprogram\unopkg.exe" add --shared ScriptProviderForooRexx
              ↪ .oxt'
275       Goto AddOfficeReg
276
277    AddOfficeReg:
278       WriteRegStr HKCR ".RXO" "" "RexxOOoScript"
279       WriteRegStr HKCR ".RXO" "Content Type" "text/x-rexx-java-ooo"
280       WriteRegStr HKCR ".RXO" "PerceivedType" "text"
281       WriteRegStr HKCR ".RXO\ShellNew" "FileName" "$INSTDIR\Templates\
              ↪ RexxOOoScript (Rexx script using OpenOffice.org) executed by ooRexx.
              ↪ rxo"
282
283       WriteRegStr HKCR "RexxOOoScript" "" "RexxOOoScript (Rexx script using
              ↪ OpenOffice.org/LibreOffice) executed by ooRexx"
284       WriteRegStr HKCR "RexxOOoScript\DefaultIcon" "" "$INSTDIR\install\images\
              ↪ windows\oorexx4ooo.ico,0"
285       WriteRegStr HKCR "RexxOOoScript\shell" "" "open"
286       WriteRegStr HKCR "RexxOOoScript\shell\edit" "" "Edit"
287       WriteRegStr HKCR "RexxOOoScript\shell\edit\command" "" 'notepad.exe "%1"'
288       WriteRegStr HKCR "RexxOOoScript\shell\open" "" "Run"
289       WriteRegStr HKCR "RexxOOoScript\shell\open\command" "" '"$REXX_HOME\rexx.
              ↪ exe" "%1" %*'
290
291       WriteRegStr HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo" "
              ↪ Extension" ".rxo"
292       Goto Done
293
294    Done:
295       ; reload file associations
296       ExecWait "$INSTDIR\install\associationChanged.exe"
```

```
297       MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully reinstalled.
              ↪ "
298   SectionEnd
299
300   ; Add Uninstall and Reinstall Entry to the start menue
301   Section -AdditionalIcons
302     CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Uninstall ${
          ↪ PRODUCT_NAME}.lnk" "$INSTDIR\uninst.exe" "" "$INSTDIR\install\images\
          ↪ windows\bsf4rexx-uninstall.ico"
303     CreateShortCut "$SMPROGRAMS\${PRODUCT_NAME}\Installation\Reinstall ${
          ↪ PRODUCT_NAME}.lnk" "$INSTDIR\${PRODUCT_NAME}_Setup_x64_reinstall.exe" ""
          ↪  "$INSTDIR\install\images\windows\bsf4rexx-reinstall.ico"
304   SectionEnd
305
306   ; create Uninstaller and Uninstall registry entries
307   Section -Post
308     WriteUninstaller "$INSTDIR\uninst.exe"
309     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayName" "
          ↪ $(^Name)"
310     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "
          ↪ UninstallString" "$INSTDIR\uninst.exe"
311     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayIcon" "
          ↪ $INSTDIR\install\images\windows\bsf4oorexx.ico"
312     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "DisplayVersion
          ↪ " "$PRODUCT_VERSION"
313     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoRepair" 0
314     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "NoModify" 0
315     WriteRegStr ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}" "ModifyPath" "
          ↪ $INSTDIR\${PRODUCT_NAME}_Setup_x64_reinstall.exe"
316   SectionEnd
317
318   ; if no silent mode, show success message
319   Function un.onUninstSuccess
320     IfSilent jump_if_silent jump_if_not_silent
321
322     jump_if_not_silent:
323       HideWindow
324       MessageBox MB_ICONINFORMATION|MB_OK "$(^Name) was successfully uninstalled.
              ↪ " IDOK
325
326     jump_if_silent:
327
328   FunctionEnd
329
330   ; if no silent mode, ask user if he want to uninstall
331   Function un.onInit
332     Call un.getBSFVersion
333     IfSilent jump_if_silent jump_if_not_silent
334
335     jump_if_not_silent:
336       MessageBox MB_ICONQUESTION|MB_YESNO|MB_DEFBUTTON2 "Do you want to uninstall
              ↪  $(^Name) and all the components?" IDYES +2
337       Abort
338
339     jump_if_silent:
340
```

```
341 FunctionEnd
342
343 ; undo all changes
344 Section Uninstall
345   SetShellVarContext all
346
347   ; is OpenOffice/LibreOffice installed?
348   Call un.FindOpenOffice
349   Call un.FindLibreOffice
350   StrCmp "" $OO_HOME SearchLibreOffice OOFound
351
352   ; OpenOffice found: so delete files
353   OOFound:
354      KillProcDLL::KillProc "soffice.exe"
355      KillProcDLL::KillProc "soffice.bin"
356
357      RmDir /r "$OO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
358      RmDir /r "$OO_HOMEBasis\share\Scripts\oorexx"
359      RmDir /r "$OO_HOMEBasis\share\Scripts\python\pythonSamples"
360
361      ExecWait "$OO_HOMEprogram\unopkg.exe remove --shared
         ↪ ScriptProviderForooRexx.oxt"
362
363      Delete "$OO_HOMEprogram\BSF4ooRexx.dll"
364      Delete "$OO_HOMEprogram\BSF.CLS"
365      Delete "$OO_HOMEprogram\UNO.CLS"
366      Delete "$OO_HOMEprogram\UNO_API_info.rxo"
367      Delete "$OO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
368      Delete "$OO_HOMEprogram\UNO_ReflectionUtil.cls"
369      Delete "$OO_HOMEprogram\UNO_XINTERFACES.RXO"
370
371      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEprogram\"
372      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEBasis\program\classes
         ↪ \unoil.jar"
373      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\ridl.jar"
374      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\jurt.jar"
375      ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$OO_HOMEURE\java\juh.jar"
376
377      ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$OO_HOMEprogram"
378      StrCmp "" $LO_HOME RemOfficeReg LOFound
379
380   SearchLibreOffice:
381      ; is LibreOffice installed?
382      StrCmp "" $LO_HOME Done LOFound
383
384   ; LibreOffice found: so delete files
385   LOFound:
386      KillProcDLL::KillProc "soffice.exe"
387      KillProcDLL::KillProc "soffice.bin"
388
389      RmDir /r "$LO_HOMEBasis\share\Scripts\javascript\HowtoCreateApiInfo"
390      RmDir /r "$LO_HOMEBasis\share\Scripts\oorexx"
391      RmDir /r "$LO_HOMEBasis\share\Scripts\python\pythonSamples"
392
393      ExecWait "$LO_HOMEprogram\unopkg.exe remove --shared
         ↪ ScriptProviderForooRexx.oxt"
```

```
394
395       Delete "$LO_HOMEprogram\BSF4ooRexx.dll"
396       Delete "$LO_HOMEprogram\BSF.CLS"
397       Delete "$LO_HOMEprogram\UNO.CLS"
398       Delete "$LO_HOMEprogram\UNO_API_info.rxo"
399       Delete "$LO_HOMEprogram\UNO_CREATE_INTERFACE_LIST.RXO"
400       Delete "$LO_HOMEprogram\UNO_ReflectionUtil.cls"
401       Delete "$LO_HOMEprogram\UNO_XINTERFACES.RXO"
402
403       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEprogram\"
404       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEBasis\program\classes
              ↪ \unoil.jar"
405       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\ridl.jar"
406       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\jurt.jar"
407       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$LO_HOMEURE\java\juh.jar"
408
409       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$LO_HOMEprogram"
410       Goto RemOfficeReg
411
412    RemOfficeReg:
413       DeleteRegKey HKCR ".RXO"
414       DeleteRegKey HKCR "RexxOOoScript"
415       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java-ooo"
416       Goto Done
417
418    Done:
419       SetRegView 64 ; change registry mode to 64 bit
420       DeleteRegKey ${PRODUCT_UNINST_ROOT_KEY} "${PRODUCT_UNINST_KEY}"
421       SetRegView 32 ; change registry mode to 32 bit
422       DeleteRegKey HKCR ".JREXX"
423       DeleteRegKey HKCR ".ORX"
424       DeleteRegKey HKCR ".REXX"
425       DeleteRegKey HKCR ".REXXJ"
426       DeleteRegKey HKCR ".RXJ"
427       DeleteRegKey HKCR ".TESTGROUP"
428       DeleteRegKey HKCR ".TESTUNIT"
429       DeleteRegKey HKCR "RexxJavaScript"
430       DeleteRegKey HKCR ".CLS"
431       DeleteRegKey HKCR ".rex"
432       DeleteRegKey HKCU "Software\Microsoft\Windows\CurrentVersion\Explorer\
              ↪ FileExts\.rxj"
433       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx"
434       DeleteRegKey HKCR "Mime\Database\Content Type\text/x-rexx-java"
435
436       WriteRegStr HKCR "REXXScript" "" "ooRexx Rexx Progam"
437       WriteRegStr HKCR "REXXScript\DefaultIcon" "" "$REXX_HOME\rexx.exe,0"
438
439       ; reload file associations
440       ExecWait "$INSTDIR\install\associationChanged.exe"
441
442       RMDir /r "$INSTDIR"
443       RMDir /r "$SMPROGRAMS\${PRODUCT_NAME}"
444       RMDir "$INSTDIR"
445       RMDir "$SMPROGRAMS\${PRODUCT_NAME}"
446
447       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-rexx-engine.jar"
```

```
448       ${un.EnvVarUpdate} $0 "CLASSPATH" "R" "HKLM" "$INSTDIR\bsf-v400-20090910.
              ↪ jar"
449
450       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$INSTDIR"
451
452       SetRegView 64
453       ReadRegStr $1 HKLM "SOFTWARE\JavaSoft\Java Runtime Environment\1.6" "
              ↪ RuntimeLib"
454       ${un.StrRep} $2 "$1" "\jvm.dll" ""
455       ${un.StrRep} $2 "$2" "client" "server"
456       ${un.EnvVarUpdate} $0 "PATH" "R" "HKLM" "$2"
457
458       SetAutoClose true
459   SectionEnd
460
461   ; check if Java and ooRexx is installed
462   ; find OpenOffice/LibreOffice folder if is installed
463   Function .onInit
464       ReadEnvStr $SYSDRIVE SYSTEMDRIVE
465       InitPluginsDir
466
467       Call getBSFVersion
468       GetVersion::WindowsPlatformArchitecture
469       Pop $WINDOWS_ARCHITECTURE ;32 or 64
470       StrCmp $WINDOWS_ARCHITECTURE "64" x64 x32
471
472       x64:
473           Call FindooRexx
474           StrCmp "" $REXX_INSTALLATION_MSG RexxSuccess RexxNoSuccess
475           Goto Done
476       x32:
477           MessageBox MB_OK|MB_ICONEXCLAMATION "You need a 64 bit Windows to start
                  ↪ this installer."
478           Abort
479
480       RexxSuccess:
481           Call FindJava
482           StrCmp "" $JAVA_INSTALLATION_MSG JavaSuccess JavaNoSuccess
483
484       JavaSuccess:
485           ; save reinstaller
486           CopyFiles "$INSTDIR\${PRODUCT_NAME}_Setup_x64_reinstall.exe" "$SYSDRIVE\
                  ↪ "
487           Call FindOpenOffice
488           Call FindLibreOffice
489           Goto Done
490
491       JavaNoSuccess:
492           MessageBox MB_OK|MB_ICONEXCLAMATION $JAVA_INSTALLATION_MSG
493           Goto Error
494
495       RexxNoSuccess:
496           MessageBox MB_OK|MB_ICONEXCLAMATION $REXX_INSTALLATION_MSG
497           Goto Error
498
499       Error:
```

```
500         RMDir /r "$INSTDIR"
501         Abort
502
503     Done:
504 FunctionEnd
505
506 ; read the BSF4ooRexx version by using getBSFVersion.rex
507 Function getBSFVersion
508     Push $0
509
510     SetOutPath "$INSTDIR"
511     File "${SCRIPT_PATH}\getBSFVersion.rex"
512     File "${BSF_ROOT}\BSF.CLS"
513     ExecWait "rexx getBSFVersion.rex"
514
515     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
516     FileRead $0 $PRODUCT_VERSION
517     FileClose $0
518
519     Delete "$INSTDIR\getBSFVersion.rex"
520     Delete "$INSTDIR\BSF.CLS"
521     Delete "$INSTDIR\BSFVersion.txt"
522
523     Pop $0
524 FunctionEnd
525
526 ; read the BSF4ooRexx version by using getBSFVersion.rex
527 ; uninstall function
528 Function un.getBSFVersion
529     Push $0
530
531     SetOutPath "$INSTDIR"
532     File "${SCRIPT_PATH}\getBSFVersion.rex"
533     File "${BSF_ROOT}\BSF.CLS"
534     ExecWait "rexx getBSFVersion.rex"
535
536     FileOpen $0 "$INSTDIR\BSFVersion.txt" r
537     FileRead $0 $PRODUCT_VERSION
538     FileClose $0
539
540     Delete "$INSTDIR\getBSFVersion.rex"
541     Delete "$INSTDIR\BSF.CLS"
542     Delete "$INSTDIR\BSFVersion.txt"
543
544     Pop $0
545 FunctionEnd
546
547 ; read Java version by using java_version.rex
548 Function FindJava
549     Push $0
550     Push $1
551
552     SetOutPath "$INSTDIR"
553     File "${SCRIPT_PATH}\java_version.rex"
554     ExecWait "rexx java_version.rex bitness" $0
555     ExecWait "rexx java_version.rex version" $1
```

```
556     Delete "$INSTDIR\java_version.rex"
557     StrCmp $0 "64" CheckVersion JavaVerNotCorrect
558
559     CheckVersion:
560         StrCmp $1 "1" Done JavaVerNotCorrect
561
562     JavaVerNotCorrect:
563         StrCpy $JAVA_INSTALLATION_MSG "You need Java version 1.4 (64 bit) or
                ↪ newer to run this program."
564
565     Done:
566         Pop $1
567         Pop $0
568 FunctionEnd
569
570 ; read ooRexx version from registry
571 ; and bitness by using rexx_version.rex
572 Function FindooRexx
573     Push $0
574     Push $1
575     Push $2
576     Push $3
577
578     ReadRegStr $REXX_VER HKLM "Software\Microsoft\Windows\CurrentVersion\
            ↪ Uninstall\ooRexx" "DisplayVersion"
579     StrCmp "" "$REXX_VER" RexxNotPresent
580     ReadRegStr $0 HKLM "SYSTEM\CurrentControlSet\Control\Session Manager\
            ↪ Environment" "REXX_HOME"
581     StrCmp "" $0 RexxNotPresent CheckRexxVer
582
583     RexxNotPresent:
584         StrCpy $REXX_INSTALLATION_MSG "ooRexx is not installed on your computer.
                ↪  You need version 4.1.0 (64 bit) or newer to run this program."
585         Goto Done
586
587     CheckRexxVer:
588         GetFullPathName /SHORT $REXX_HOME "$0"
589         StrCpy $0 $REXX_VER 1 0
590         StrCpy $1 $REXX_VER 1 2
591         StrCpy $2 $REXX_VER 1 4
592         StrCpy $REXX_VER "$0$1$2"
593         IntCmp 410 $REXX_VER FoundCorrectRexxVer FoundCorrectRexxVer
                ↪ RexxVerNotCorrect
594
595     FoundCorrectRexxVer:
596         IfFileExists "$REXX_HOME\rexx.exe" CheckRexxBitness RexxNotPresent
597
598     CheckRexxBitness:
599         SetOutPath "$INSTDIR"
600         File "${SCRIPT_PATH}\rexx_version.rex"
601         ExecWait "rexx rexx_version.rex" $3
602         Delete "$INSTDIR\rexx_version.rex"
603         StrCmp $3 "64" Done RexxVerNotCorrect
604
605     RexxVerNotCorrect:
606         StrCpy $REXX_INSTALLATION_MSG "The version of ooRexx installed on your
```

```
607
608     Done:
609         Pop $3
610         Pop $2
611         Pop $1
612         Pop $0
613 FunctionEnd
614
615 ; find OpenOffice folder
616 Function FindOpenOffice
617     Push $0
618
619     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\OpenOffice.
        ↪ org\3" "OFFICEINSTALLLOCATION"
620     StrCmp "" $0 OO_2 Done
621
622     OO_2:
623         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
            ↪ OFFICEINSTALLLOCATION"
624         StrCmp "" $0 Done OOFound
625
626     OOFound:
627         GetFullPathName $OO_HOME "$0"
628         Goto Done
629
630     Done:
631         Pop $0
632 FunctionEnd
633
634 ; find OpenOffice folder (uninstall function)
635 Function un.FindOpenOffice
636     Push $0
637
638     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\OpenOffice.org\Layers\OpenOffice.
        ↪ org\3" "OFFICEINSTALLLOCATION"
639     StrCmp "" $0 OO_2 Done
640
641     OO_2:
642         ReadRegStr $0 HKLM "SOFTWARE\OpenOffice.org\Layers\OpenOffice.org\3" "
            ↪ OFFICEINSTALLLOCATION"
643         StrCmp "" $0 Done OOFound
644
645     OOFound:
646         GetFullPathName $OO_HOME "$0"
647         Goto Done
648
649     Done:
650         Pop $0
651 FunctionEnd
652
653 ; find LibreOffice folder
654 Function FindLibreOffice
655     Push $0
656
```

```
657     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice\3"
            ↪ "OFFICEINSTALLLOCATION"
658     StrCmp "" $0 LO_2 Done
659
660     LO_2:
661         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
662         StrCmp "" $0 Done LOFound
663
664     LOFound:
665         GetFullPathName $LO_HOME "$0"
666         Goto Done
667
668     Done:
669         Pop $0
670 FunctionEnd
671
672 ; find LibreOffice folder (uninstall function)
673 Function un.FindLibreOffice
674     Push $0
675
676     ReadRegStr $0 HKLM "SOFTWARE\WoW6432Node\LibreOffice\Layers_\LibreOffice\3"
            ↪ "OFFICEINSTALLLOCATION"
677     StrCmp "" $0 LO_2 Done
678
679     LO_2:
680         ReadRegStr $0 HKLM "SOFTWARE\LibreOffice\Layers_\LibreOffice\3" "
                ↪ OFFICEINSTALLLOCATION"
681         StrCmp "" $0 Done LOFound
682
683     LOFound:
684         GetFullPathName $LO_HOME "$0"
685         Goto Done
686
687     Done:
688         Pop $0
689 FunctionEnd
```

Listing 39: The NSIS Reinstall Script for 64 bit Windows

7.2.5 Java Version Check Script

The following listing contains an ooRexx script that returns the version of the currently installed Java. If the bitness parameter is given while executing the script, it will return 64 if Java runs in 64 bit mode.

```
#!/usr/bin/rexx
/* Name: java_version.rex
   Purpose: Returns the Java version and bitness
   Author: Juergen Hesse, Manuel Paar
   Version: 1.0

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     WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
     implied.
     See the License for the specific language governing permissions and
     limitations under the License.
   -------------------------------------------------------------------
*/

if arg(1) = "bitness" then exit (getJavaBitness())

exit (getJavaVersion())

::routine getJavaVersion
   "java -version 2>&1 | rxqueue"
   java_ver=0
   do while queued()>0
      parse pull 'java version "' val1 "." val2 "."
      if val1>=1 & val2>=4 then java_ver=1
   end
   return java_ver

::routine getJavaBitness
   "java -version 2>&1 | rxqueue"
   bitness=0
   do while queued()>0
      parse pull "Java HotSpot(TM) " val " "
      if val="64-Bit" then bitness=64
   end
   return bitness
```

Listing 40: Java version check script.

7.2.6 ooRexx Bitness Check Script

The following listing contains an ooRexx script that returns the bitness of the currently installed ooRexx.

```
#!/usr/bin/rexx
/* Name: rexx_version.rex
   Purpose: Returns the ooRexx bitness
   Author: Juergen Hesse, Manuel Paar
   Version: 1.0

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     WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
     implied.
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     limitations under the License.
   -------------------------------------------------------------------
*/

exit(getRexxBitness())


::routine getRexxBitness

   "rexx -v 2>&1 | rxqueue"
   bitness=0
   do while queued()>0
      parse pull hint ":" val .
      if hint="Addressing Mode" then bitness=val
   end
   return bitness
```

Listing 41: ooRexx bitness check script.

7.2.7 Get BSF Version Script

The following listing contains an ooRexx script that returns the version of the currently installed BSF4ooRexx.

```
#!/usr/bin/rexx
/* Name: getBSFVersion.rex
   Purpose: Returns the current version of BSF4ooRexx
   Author: Juergen Hesse, Manuel Paar
   Version: 1.0

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     limitations under the License.
   -------------------------------------------------------------------
*/

CHAROUT("BSFVersion.txt", getVersion(),1)

::routine getVersion
  'find /n ".bsf4rexx~version" BSF.CLS 2>&1 | rxqueue'
  do while queued()>0
    parse pull rest '~version="' version '"'
    if version <> "" then return version
  end
  return version
```

Listing 42: The GetBSFVersion script.

7.2.8 cmd File for Automatically Creating the Windows Installers

The following listing contains a batch script which creates 32 and 64 bit installers and reinstallers by using the NSIS scripts shown in chapters 7.2.1, 7.2.2, 7.2.3 and 7.2.4.

```
rem Name: createInstaller.cmd
rem Purpose: Creates 32 and 64 bit installer for Windows
rem Author: Juergen Hesse, Manuel Paar
rem Version: 1.0
rem
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rem    limitations under the License.
rem -------------------------------------------------------------------


set prog="C:\Program Files (x86)\NSIS\makensis.exe"

%prog% BSF4ooRexx_x32_reinstall.nsi
%prog% BSF4ooRexx_x32.nsi
%prog% BSF4ooRexx_x64_reinstall.nsi
%prog% BSF4ooRexx_x64.nsi
del BSF4ooRexx_Setup_x32_reinstall.exe
del BSF4ooRexx_Setup_x64_reinstall.exe
```

Listing 43: The cmd file for automatically creating the Windows Installers.

7.3 Debian Linux Source Codes

Author: Jürgen Hesse

The following chapters contain the source codes for the Debian Linux package.

7.3.1 Debian Package `control` File

The following listing contains the `control` file for building the BSF4ooRexx Debian package. It is described in detail in chapter 3.2.3.

```
1  Package: bsf4oorexx
2  Version: 406.20110822
3  Section: Interpreted Computer Languages
4  Priority: optional
5  Homepage: http://sourceforge.net/projects/bsf4oorexx
6  Maintainer: BSF4ooRexx <bsf4oorexx-devel@lists.sourceforge.net>
7  Architecture: all
8  Description: Bean Scripting Framework for Open Object Rexx (BSF4ooRexx)
9   BSF4ooRexx version for ooRexx 4.1.0 and later exploiting the new ooRexx kernel
       ↪ (e.g. allowing implementing Java methods in ooRexx, callbacks from Java
       ↪ to ooRexx). [Camouflages Java as the dynamically typed, caseless and easy
       ↪  to use ooRexx.]
10  .
11  Special note for users of Ubuntu 11.04: If you want to use the LibreOffice
       ↪ version from the repository, you should invoke the command "sudo apt-get
       ↪ install libreoffice libreoffice-java-common" from the command line. If
       ↪ you then experience any problems working with LibreOffice, you need to
       ↪ remove the .libreoffice folder from your home directory. When the program
       ↪  starts up the next time, the folder is rebuilt and everything works well
       ↪  again.
12 Depends: oorexx (>= 4.1.0), openjdk-6-jre
13 Recommends: openoffice.org (>= 3.2.0), openoffice.org-calc (>= 3.2.0),
      ↪ openoffice.org-draw (>= 3.2.0), openoffice.org-impress (>= 3.2.0),
      ↪ openoffice.org-writer (>= 3.2.0)
```

Listing 44: The Debian package control file.

7.3.2 Debian Package `preinst` Script

The following listing contains the preinstallation script. It is used to remove an old existing version of the BSF4ooRexx folder before installation.

```
#!/bin/sh

# Name: preinst
# Purpose: BSF4ooRexx deb package preinstallation script.
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    limitations under the License.
# -------------------------------------------------------------------


sudo rm -rf /opt/BSF4ooRexx
```

Listing 45: The Debian preinst file.

7.3.3 Debian Package `postinst` Script

The following listing contains the postinstallation script. It is used to run the setup script after the BSF4ooRexx files have been moved to the file system.

```
#!/bin/sh

# Name: postinst
# Purpose: BSF4ooRexx deb package post installation script.
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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# -------------------------------------------------------------------

cd /tmp/BSF4ooRexxInstall/bsf4oorexx/install/
sudo rexx /tmp/BSF4ooRexxInstall/bsf4oorexx/install/setupAllAndRun.rex install /
   ↪ @q:
```

Listing 46: The Debian package postinst file.

7.3.4 Debian Package `prerm` Script

The following listing contains the `prerm` script. It is used to remove the BSF4ooRexx files from the filesystem before the Debian package itself is removed.

```
#!/bin/sh

# Name: prerm
# Purpose: BSF4ooRexx deb package prerm script.
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    limitations under the License.
# -------------------------------------------------------------------

cd /opt/BSF4ooRexx/install
sudo rexx /opt/BSF4ooRexx/install/setupAllAndRun.rex uninstall /@q:
```

Listing 47: The Debian package prerm file.

7.3.5 Shell Script for Automatically Creating the Debian Package

The following listing contains a shell script which creates the Debian package from the BSF4ooRexx distribution files.

```
#!/bin/sh

# Name: makeDeb.sh
# Purpose: Shell script for creating the BSF4ooRexx Debian package.
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
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#    limitations under the License.
# -------------------------------------------------------------------


BUILD_DIR=../build
SRC_DIR=../src
BSF4OOREXX_SRC_DIR=BSF4ooRexx_src
PACKAGE_BASE_NAME=bsf4oorexx
BSF4OOREXX_VERSION=`rexx getBSFversion.rex > /dev/null && cat BSFVersion.txt`
PACKAGE_NAME=$PACKAGE_BASE_NAME-$BSF4OOREXX_VERSION

DEBIAN_DIR=$BUILD_DIR/$PACKAGE_NAME/DEBIAN
OPT_DIR=$BUILD_DIR/$PACKAGE_NAME/tmp
CONTROL_FILE=$DEBIAN_DIR/control

echo
echo "Cleaning build directory..."
echo

rm -rf $BUILD_DIR/$PACKAGE_NAME*

echo
echo "Building .deb package..."
echo

echo
echo "Creating directory structure..."
echo
```

```
50
51 mkdir -p $BUILD_DIR/$PACKAGE_NAME/DEBIAN
52 mkdir -p $BUILD_DIR/$PACKAGE_NAME/tmp/BSF4ooRexxInstall
53
54 echo
55 echo "Copying package information..."
56 echo
57
58 echo "Package: $PACKAGE_BASE_NAME" > $CONTROL_FILE
59 echo "Version: $BSF4OOREXX_VERSION" >> $CONTROL_FILE
60
61 cat control >> $CONTROL_FILE
62 cp postinst $DEBIAN_DIR/
63 cp preinst $DEBIAN_DIR/
64 cp prerm $DEBIAN_DIR/
65 cp copyright $DEBIAN_DIR/
66
67 echo
68 echo "Copying package files..."
69 echo
70
71 cp -r $BSF4OOREXX_SRC_DIR/* $OPT_DIR/BSF4ooRexxInstall
72
73 echo
74 echo "Building package file..."
75 echo
76
77 dpkg-deb --build $BUILD_DIR/$PACKAGE_NAME
78 chmod 777 $BUILD_DIR/$PACKAGE_NAME.deb
79
80 echo
81 echo "Done."
82 echo
```

Listing 48: Shell script for creating the Debian Package.

7.4 Red Hat Linux Source Codes

Author: Jürgen Hesse

The following chapters contain the source codes for the Red Hat Linux package.

7.4.1 SPECS File for Red Hat based Linux Distributions

The following listing contains the SPECS file for creating a RPM package from the BSF4ooRexx distribution files. It is described in detail in chapter 4.2.3.

```
1  Name:           bsf4oorexx
2  Version:        406
3  Release:        20110822
4  Summary:        Bean Scripting Framework for Open Object Rexx (BSF4ooRexx)
5
6  Group:          Development/Tools
7  License:        Apache Software License 2.0
8  URL:            http://sourceforge.net/projects/bsf4oorexx/
9  #Source0:
10 BuildRoot:      %{_tmppath}/%{name}-%{version}-%{release}-root-%(%{__id_u} -n)
11
12 Autoreq:        0
13 BuildRequires:  unzip
14 Requires:       oorexx >= 4.1.0
15 Requires:       java-1.6.0-openjdk >= 1.6.0
16
17 %description
18 BSF4Rexx version for ooRexx 4.1.0 and later, exploiting the new ooRexx kernel (e
       ↪ .g. allowing implementation of Java methods in ooRexx, callbacks from Java
       ↪  to ooRexx). [Camouflages Java as the dynamically typed, caseless and easy
       ↪  to use ooRexx.]
19
20 %prep
21 unzip -d %{_builddir}/BSF4ooRexxInstall %{_sourcedir}/BSF4ooRexx_install
22
23 %build
24
25 %install
26 rm -rf %{buildroot}
27 mkdir -p %{buildroot}/tmp/
28 cp -r BSF4ooRexxInstall %{buildroot}/tmp/
29
30 %post
31 rm -rf /opt/BSF4ooRexx
32 cp -R %{_sourcedir}/BSF4ooRexxInstall /tmp/
33 #cd /tmp/BSF4ooRexxInstall/bsf4oorexx/install/
34 rexx /tmp/BSF4ooRexxInstall/bsf4oorexx/install/setupAllAndRun.rex install /@q:
```

```
35
36 %preun
37 #cd /opt/BSF4ooRexx
38 rexx /opt/BSF4ooRexx/install/setupAllAndRun.rex uninstall /@q:
39
40 %clean
41
42 %files
43 %defattr(-,root,root,-)
44 %doc
45 /tmp/*
46
47 %changelog
```

Listing 49: The RPM package's SPECS file.

7.4.2 Shell Script for Automatically Creating the RPM Package

The following listing contains a shell script which creates the RPM package from the BSF4ooRexx distribution files.

```
#!/bin/sh

# Name: makeRPM.sh
# Purpose: Shell script for creating the BSF4ooRexx RPM package.
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    limitations under the License.
# -------------------------------------------------------------------

BSF4OOREXXZIPDIR=../Downloads
BSF4OOREXXZIPFILE=BSF4ooRexx_install.zip
BSF4OOREXXDISTRIBUTION=$BSF4OOREXXZIPDIR/$BSF4OOREXXZIPFILE

SPECSFILE=SPECS/bsf4oorexx.spec

echo
echo "Cleaning build directory..."
echo

rm -r BUILD/BSF4ooRexxInstall

echo
echo "Copying source files..."
echo

cp $BSF4OOREXXDISTRIBUTION SOURCES/

echo
echo "Extracting sources..."
echo

unzip SOURCES/$BSF4OOREXXZIPFILE -d /tmp/bsftemp/

echo
```

```
50 echo "Retrieving version information..."
51 echo
52
53 rexx getBSFversion.rex
54 BSFVERSION=`cat BSFVersion.txt`
55 BSFRELEASE=`cat BSFRelease.txt`
56
57 echo
58 echo "Building .spec file..."
59 echo
60
61 echo "Name:           bsf4oorexx" > $SPECSFILE
62 echo "Version:        $BSFVERSION" >> $SPECSFILE
63 echo "Release:        $BSFRELEASE" >> $SPECSFILE
64 cat bsf4oorexx.spec.template >> $SPECSFILE
65
66 echo
67 echo "Building RPM package..."
68 echo
69
70 rpmbuild -ba $SPECSFILE
71
72 echo
73 echo "Done."
74 echo
```

Listing 50: Shell script for creating the RPM package.

7.5 Mac OS X Source Codes

Authors: Jürgen Hesse, Manuel Paar

The following chapters contain the source codes for the Mac OS X installer.

7.5.1 The Mac OS X Menu Bar

The following listing contains a Java program, which, when executed, shows a menu bar from which the user can access the BSF4ooRexx components just like they are available under Windows and Linux.

```
1  /* Name: BSF4ooRexx.java
2     Purpose: Menu bar for ooRexx with BSF4ooRexx
3     Author: Manuel Paar, Juergen Hesse
4     Version: 1.0
5
6     ------------------- Apache Version 2.0 license --------------------
7        Copyright (C) 2011 Manuel Paar, Juergen Hesse
8
9        Licensed under the Apache License, Version 2.0 (the "License");
10       you may not use this file except in compliance with the License.
11       You may obtain a copy of the License at
12
13           http://www.apache.org/licenses/LICENSE-2.0
14
15       Unless required by applicable law or agreed to in writing, software
16       distributed under the License is distributed on an "AS IS" BASIS,
17       WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
18       implied.
19       See the License for the specific language governing permissions and
20       limitations under the License.
21    -----------------------------------------------------------------
22 */
23
24 import java.awt.*;
25 import java.awt.event.*;
26 import javax.swing.*;
27 import javax.imageio.*;
28
29 public class BSF4ooRexx implements Runnable, ActionListener
30 {
31   private JFrame frame;
32   private JMenuBar menuBar;
33   private JMenu ooRexxMenu;
34   private JMenu bsfMenu;
35   private JMenu installMenu;
```

```
36    private JMenuItem licenceMenuItem;
37    private JMenuItem ooRexxMenuItem;
38    private JMenuItem docMenuItem;
39    private JMenuItem ooRexxSamplesMenuItem;
40    private JMenuItem infoMenuItem;
41    private JMenuItem downloadMenuItem;
42    private JMenuItem uninstallMenuItem;
43    private JMenuItem reinstallMenuItem;
44    private JMenuItem samplesMenuItem;
45    private JMenuItem utilMenuItem;
46    private JMenuItem ooRexxTryMenuItem;
47
48    private ImageIcon reinstallIcon = null;
49    private ImageIcon uninstallIcon = null;
50
51    public static void main(String[] args)
52    {
53      // Enable the Apple-style
54      System.setProperty("apple.laf.useScreenMenuBar", "true");
55      System.setProperty("com.apple.mrj.application.apple.menu.about.name", "
          ↪ ooRexx Menu Bar");
56      SwingUtilities.invokeLater(new BSF4ooRexx());
57    }
58
59    public void run()
60    {
61      try {
62        // Set the dock icon
63        com.apple.eawt.Application.getApplication().setDockIconImage(
64           ImageIO.read(this.getClass().getClassLoader().getResource("oorexx.png")
               ↪ ));
65        // Set the uninstall and reinstall icons
66        reinstallIcon = new ImageIcon(this.getClass().getClassLoader().getResource
              ↪ ("BSF4RexxReinstall.png"));
67        uninstallIcon = new ImageIcon(this.getClass().getClassLoader().getResource
              ↪ ("BSF4RexxUninstall.png"));
68      } catch (java.io.IOException e) {
69          System.out.println(e.getMessage());
70      }
71
72      frame = new JFrame("ooRexx");
73      menuBar = new JMenuBar();
74
75      ooRexxMenu = new JMenu("Open Object Rexx");
76
77      // Create the License menu item
78      licenceMenuItem = new JMenuItem("Licence");
79      licenceMenuItem.setAccelerator(KeyStroke.getKeyStroke('L', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
80      licenceMenuItem.addActionListener(this);
81      ooRexxMenu.add(licenceMenuItem);
82
83      // Create the ooRexx Homepage menu item
84      ooRexxMenuItem = new JMenuItem("ooRexx Home Page");
85      ooRexxMenuItem.setAccelerator(KeyStroke.getKeyStroke('O', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
```

```
86      ooRexxMenuItem.addActionListener(this);
87      ooRexxMenu.add(ooRexxMenuItem);
88
89      // Create the ooRexxTry menu item
90      ooRexxTryMenuItem = new JMenuItem("ooRexxTry");
91      ooRexxTryMenuItem.setAccelerator(KeyStroke.getKeyStroke('T', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
92      ooRexxTryMenuItem.addActionListener(this);
93      ooRexxMenu.add(ooRexxTryMenuItem);
94
95      // Create the Documentation menu item
96      docMenuItem = new JMenuItem("Documentation");
97      docMenuItem.setAccelerator(KeyStroke.getKeyStroke('D', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
98      docMenuItem.addActionListener(this);
99      ooRexxMenu.add(docMenuItem);
100
101     // Create the ooRexx Samples menu item
102     ooRexxSamplesMenuItem = new JMenuItem("ooRexx Samples");
103     ooRexxSamplesMenuItem.setAccelerator(KeyStroke.getKeyStroke('S', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
104     ooRexxSamplesMenuItem.addActionListener(this);
105     ooRexxMenu.add(ooRexxSamplesMenuItem);
106
107
108     bsfMenu = new JMenu("BSF4ooRexx");
109
110     // Create the Information menu item
111     infoMenuItem = new JMenuItem("Information");
112     infoMenuItem.setAccelerator(KeyStroke.getKeyStroke('I', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
113     infoMenuItem.addActionListener(this);
114     bsfMenu.add(infoMenuItem);
115
116     // Create the BSF4ooRexx Download Page menu item
117     downloadMenuItem = new JMenuItem("BSF4ooRexx Download Page");
118     downloadMenuItem.setAccelerator(KeyStroke.getKeyStroke('B', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
119     downloadMenuItem.addActionListener(this);
120     bsfMenu.add(downloadMenuItem);
121
122     // Create the Utilities menu item
123     utilMenuItem = new JMenuItem("Utilities");
124     utilMenuItem.setAccelerator(KeyStroke.getKeyStroke('U', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
125     utilMenuItem.addActionListener(this);
126     bsfMenu.add(utilMenuItem);
127
128     // Create the BSF4ooRexx Samples menu item
129     samplesMenuItem = new JMenuItem("BSF4ooRexx Samples");
130     samplesMenuItem.setAccelerator(KeyStroke.getKeyStroke('E', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
131     samplesMenuItem.addActionListener(this);
132     bsfMenu.add(samplesMenuItem);
133
134
```

```
135     installMenu = new JMenu("Installation");
136
137     // Create the Uninstall menu item
138     uninstallMenuItem = new JMenuItem("Uninstall ooRexx with BSF4ooRexx");
139     uninstallMenuItem.setAccelerator(KeyStroke.getKeyStroke('X', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
140     uninstallMenuItem.addActionListener(this);
141
142     // Create the Reinstall menu item
143     reinstallMenuItem = new JMenuItem("Reinstall ooRexx with BSF4ooRexx");
144     reinstallMenuItem.setAccelerator(KeyStroke.getKeyStroke('R', Toolkit.
            ↪ getDefaultToolkit().getMenuShortcutKeyMask()));
145     reinstallMenuItem.addActionListener(this);
146     installMenu.add(uninstallMenuItem);
147     installMenu.add(reinstallMenuItem);
148
149     menuBar.add(ooRexxMenu);
150     menuBar.add(bsfMenu);
151     menuBar.add(installMenu);
152
153     frame.setJMenuBar(menuBar);
154
155     // Show only the menu bar and no window
156     frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
157     frame.setSize(0, 0);
158     frame.setLocation(Short.MIN_VALUE, Short.MIN_VALUE);
159     frame.setUndecorated(true);
160     frame.setVisible(true);
161   }
162
163   public void actionPerformed(ActionEvent ev)
164   {
165     Object source = ev.getSource();
166
167     try {
168         // On-Click events for the menu items
169     if (source == licenceMenuItem)
170             Runtime.getRuntime().exec("open /opt/ooRexx/CPLv1.0.txt");
171     else if (source == ooRexxMenuItem)
172             Runtime.getRuntime().exec("open http://www.oorexx.org/");
173         else if (source == ooRexxTryMenuItem)
174                 Runtime.getRuntime().exec("/Applications/ooRexx.app/Contents/
                    ↪ MacOS/run_ooRexxTry.sh");
175     else if (source == docMenuItem)
176             Runtime.getRuntime().exec("open /Applications/ooRexx.app/Contents/
                ↪ documentation/ooRexx/");
177     else if (source == ooRexxSamplesMenuItem)
178             Runtime.getRuntime().exec("open /Applications/ooRexx.app/Contents/
                ↪ samples/ooRexx/");
179     else if (source == infoMenuItem)
180             Runtime.getRuntime().exec("open /Applications/ooRexx.app/Contents/
                ↪ documentation/BSF4ooRexx/");
181         else if (source == downloadMenuItem)
182             Runtime.getRuntime().exec("open http://sourceforge.net/projects/
                ↪ bsf4oorexx/");
183         else if (source == utilMenuItem)
```

```
184             Runtime.getRuntime().exec("open /Applications/ooRexx.app/Contents/
                    ↪ utilities/");
185         else if (source == samplesMenuItem)
186             Runtime.getRuntime().exec("open /Applications/ooRexx.app/Contents/
                    ↪ samples/BSF4ooRexx/");
187         else if (source == uninstallMenuItem) {
188             if (JOptionPane.showConfirmDialog(null, "Do you really want to
                    ↪ uninstall?", "Uninstall ooRexx with BSF4ooRexx", JOptionPane.
                    ↪ OK_CANCEL_OPTION, JOptionPane.QUESTION_MESSAGE, uninstallIcon)
                    ↪  == 0) {
189             Runtime.getRuntime().exec("sh /Applications/ooRexx.app/Contents/
                    ↪ Resources/uninstall.sh uninstall");
190             System.exit(0);
191         }
192     }
193         else if (source == reinstallMenuItem) {
194             if (JOptionPane.showConfirmDialog(null, "Do you really want to
                    ↪ reinstall?", "Reinstall ooRexx with BSF4ooRexx", JOptionPane.
                    ↪ OK_CANCEL_OPTION, JOptionPane.QUESTION_MESSAGE, reinstallIcon)
                    ↪  == 0) {
195             Runtime.getRuntime().exec("sh /Applications/ooRexx.app/Contents/
                    ↪ Resources/uninstall.sh reinstall");
196             System.exit(0);
197         }
198     }
199
200     } catch (java.io.IOException e) {
201         System.out.println(e.getMessage());
202     }
203   }
204 }
```

Listing 51: The ooRexx menu bar.

7.5.2 The Mac OS X Installer's 32 bit Install Script

The following listing contains the installation script for Mac OS X if 32 bit ooRexx and Java are used. It is used in chapter 5.3.3.

```
#!/bin/sh
# Name: BSF4ooRexx_install_postflight.sh
# Purpose: Install script for ooRexx and BSF4ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
# ------------------- Apache Version 2.0 license --------------------
#    Copyright (C) 2011 Juergen Hesse, Manuel Paar
#
#    Licensed under the Apache License, Version 2.0 (the "License");
#    you may not use this file except in compliance with the License.
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#
#          http://www.apache.org/licenses/LICENSE-2.0
#
#    Unless required by applicable law or agreed to in writing, software
#    distributed under the License is distributed on an "AS IS" BASIS,
#    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
#    implied.
#    See the License for the specific language governing permissions and
#    limitations under the License.
# -------------------------------------------------------------------
VER=4
REL=0
MOD=3
PREFIX=/opt/ooRexx

echo $SCRIPT_NAME: linking files
echo $SCRIPT_NAME: linking libraries to /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.$VER.$REL.$MOD.dylib   /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.$VER.dylib       /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.a             /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.dylib         /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.la            /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.$VER.$REL.$MOD.dylib     /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.$VER.dylib         /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.a               /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.dylib           /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.la              /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.$VER.$REL.$MOD.dylib  /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.$VER.dylib      /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.a            /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.dylib        /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.la           /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.$VER.$REL.$MOD.dylib /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.$VER.dylib     /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.a           /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.dylib       /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.la          /usr/lib
```

```
50 ln -sf $PREFIX/lib/ooRexx/librxmath.$VER.$REL.$MOD.dylib   /usr/lib
51 ln -sf $PREFIX/lib/ooRexx/librxmath.$VER.dylib       /usr/lib
52 ln -sf $PREFIX/lib/ooRexx/librxmath.a             /usr/lib
53 ln -sf $PREFIX/lib/ooRexx/librxmath.dylib         /usr/lib
54 ln -sf $PREFIX/lib/ooRexx/librxmath.la            /usr/lib
55 ln -sf $PREFIX/lib/ooRexx/librxregexp.$VER.$REL.$MOD.dylib /usr/lib
56 ln -sf $PREFIX/lib/ooRexx/librxregexp.$VER.dylib     /usr/lib
57 ln -sf $PREFIX/lib/ooRexx/librxregexp.a           /usr/lib
58 ln -sf $PREFIX/lib/ooRexx/librxregexp.dylib       /usr/lib
59 ln -sf $PREFIX/lib/ooRexx/librxregexp.la          /usr/lib
60 ln -sf $PREFIX/lib/ooRexx/librxsock.$VER.$REL.$MOD.dylib   /usr/lib
61 ln -sf $PREFIX/lib/ooRexx/librxsock.$VER.dylib       /usr/lib
62 ln -sf $PREFIX/lib/ooRexx/librxsock.a             /usr/lib
63 ln -sf $PREFIX/lib/ooRexx/librxsock.dylib         /usr/lib
64 ln -sf $PREFIX/lib/ooRexx/librxsock.la            /usr/lib
65 ln -sf $PREFIX/lib/ooRexx/librxunixsys.$VER.$REL.$MOD.dylib   /usr/lib
66 ln -sf $PREFIX/lib/ooRexx/librxunixsys.$VER.dylib       /usr/lib
67 ln -sf $PREFIX/lib/ooRexx/librxunixsys.a             /usr/lib
68 ln -sf $PREFIX/lib/ooRexx/librxunixsys.dylib         /usr/lib
69 ln -sf $PREFIX/lib/ooRexx/librxunixsys.la            /usr/lib
70
71 echo $SCRIPT_NAME: linking programs to /usr/bin
72 ln -sf $PREFIX/bin/csvStream.cls    /usr/bin
73 ln -sf $PREFIX/bin/mime.cls /usr/bin
74 ln -sf $PREFIX/bin/oorexx-config    /usr/bin
75 ln -sf $PREFIX/bin/rexx          /usr/bin
76 ln -sf $PREFIX/bin/rexx.cat      /usr/bin
77 ln -sf $PREFIX/bin/rexx.img      /usr/bin
78 ln -sf $PREFIX/bin/rexxc         /usr/bin
79 ln -sf $PREFIX/bin/rexximage     /usr/bin
80 ln -sf $PREFIX/bin/rxapi         /usr/bin
81 ln -sf $PREFIX/bin/rxftp.cls     /usr/bin
82 ln -sf $PREFIX/bin/rxqueue       /usr/bin
83 ln -sf $PREFIX/bin/rxregexp.cls  /usr/bin
84 ln -sf $PREFIX/bin/rxsubcom      /usr/bin
85 ln -sf $PREFIX/bin/smtp.cls /usr/bin
86 ln -sf $PREFIX/bin/socket.cls   /usr/bin
87 ln -sf $PREFIX/bin/socket.cls    /usr/bin
88 ln -sf $PREFIX/bin/streamsocket.cls /usr/bin
89 ln -sf $PREFIX/share/ooRexx/rexxtry.rex /usr/bin
90
91 echo $SCRIPT_NAME: linking man pages to /usr/share/man/man1
92 ln -sf $PREFIX/share/man/man1/oorexx-config.1 /usr/share/man/man1
93 ln -sf $PREFIX/share/man/man1/rexx.1          /usr/share/man/man1
94 ln -sf $PREFIX/share/man/man1/rexxc.1         /usr/share/man/man1
95 ln -sf $PREFIX/share/man/man1/rxmigrate.1     /usr/share/man/man1
96 ln -sf $PREFIX/share/man/man1/rxqueue.1       /usr/share/man/man1
97 ln -sf $PREFIX/share/man/man1/rxsubcom.1      /usr/share/man/man1
98
99 echo $SCRIPT_NAME: linking headers to /usr/include
100 ln -sf $PREFIX/include/oorexxapi.h        /usr/include
101 ln -sf $PREFIX/include/oorexxerrors.h     /usr/include
102 ln -sf $PREFIX/include/rexx.h             /usr/include
103 ln -sf $PREFIX/include/rexxapidefs.h      /usr/include
104 ln -sf $PREFIX/include/rexxapitypes.h     /usr/include
105 ln -sf $PREFIX/include/rexxplatformapis.h /usr/include
```

```
106 ln -sf $PREFIX/include/rexxplatformdefs.h /usr/include
107
108 echo $SCRIPT_NAME:  linking Launch Daemon to /Library/LaunchDaemons
109 ln -sf $PREFIX/bin/org.rexxla.oorexx.rxapid.plist /Library/LaunchDaemons
110
111 echo $SCRIPT_NAME:  stop/start rxapi Daemon
112 launchctl unload -w /Library/LaunchDaemons/org.rexxla.oorexx.rxapid.plist
113 killall rxapi
114 launchctl load -w /Library/LaunchDaemons/org.rexxla.oorexx.rxapid.plist
115
116 echo $SCRIPT_NAME: linking complete
117
118 BSF4OOREXX_FRAMEWORK=/opt/BSF4ooRexx
119
120 BIN_DIR=/usr/bin
121 JAVA_LIB_DIR=/usr/lib/java
122
123 OOO_SCRIPTLIB_SRC=$BSF4OOREXX_FRAMEWORK/Libraries/OOo_ScriptLibraries
124
125 OOO_APP_DIR=/Applications/OpenOffice.org.app
126 OOO_SCRIPTLIB_DIR=$OOO_APP_DIR/Contents/basis-link/share/Scripts
127 OOO_UNO_PATH=$OOO_APP_DIR/Contents/program
128 OOO_JAVA_PATH=$OOO_APP_DIR/Contents/basis-link/ure-link/share/java
129 OOO_UNOIL_PATH=$OOO_APP_DIR/Contents/basis-link/program/classes
130
131 LO_APP_DIR=/Applications/LibreOffice.app
132 LO_SCRIPTLIB_DIR=$LO_APP_DIR/Contents/basis-link/share/Scripts
133 LO_UNO_PATH=$LO_APP_DIR/Contents/program
134 LO_JAVA_PATH=$LO_APP_DIR/Contents/basis-link/ure-link/share/java
135 LO_UNOIL_PATH=$LO_APP_DIR/Contents/basis-link/program/classes
136
137 echo ""
138 echo "Creating symbolic links in $BIN_DIR..."
139 echo ""
140
141 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/BSF.CLS
       ↪ $BIN_DIR
142 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/BSF_OnTheFly.cls
       ↪ $BIN_DIR
143 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/rexxj2.sh
       ↪ $BIN_DIR
144 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/rexxj2.sh
       ↪ $BIN_DIR/rexxj.sh
145 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/rgf_util2.rex
       ↪ $BIN_DIR
146 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO.CLS
       ↪ $BIN_DIR
147 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_XINTERFACES.RXO
       ↪ $BIN_DIR
148 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_ReflectionUtil.cls
       ↪ $BIN_DIR
149 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_CREATE_INTERFACE_LIST.RXO
       ↪ $BIN_DIR
150 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_API_info.rxo
       ↪ $BIN_DIR
151
```

```
152 echo ""
153 echo "Creating symbolic links in $JAVA_LIB_DIR..."
154 echo ""
155
156 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Libraries/libBSF4ooRexx.dylib $JAVA_LIB_DIR
157 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Libraries/libBSF4ooRexx.dylib /usr/lib
158
159 echo ""
160 echo "Creating symbolic links in $BSF4OOREXX_FRAMEWORK..."
161 echo ""
162
163 sudo ln -sf /Applications/ooRexx.app/Contents/samples/BSF4ooRexx
    ↪ $BSF4OOREXX_FRAMEWORK/Samples
164 sudo ln -sf /Applications/ooRexx.app/Contents/documentation/BSF4ooRexx
    ↪ $BSF4OOREXX_FRAMEWORK/Information
165 sudo ln -sf /Applications/ooRexx.app/Contents/utilities $BSF4OOREXX_FRAMEWORK/
    ↪ Utilities
166
167 sudo mkdir -p $BSF4OOREXX_FRAMEWORK/Install
168 sudo ln -sf /Applications/ooRexx.app/Contents/Resources/one_click_uninstall.sh
    ↪ $BSF4OOREXX_FRAMEWORK/Install/uninstall.sh
169 sudo ln -sf /Applications/ooRexx.app/Contents/Resources/one_click_reinstall.sh
    ↪ $BSF4OOREXX_FRAMEWORK/Install/reinstall.sh
170
171 OOORESULT=`lipo -info /Applications/OpenOffice.org.app/Contents/program/soffice
    ↪ | grep i386;`
172
173 if [[ -e $OOO_APP_DIR && ! -z $OOORESULT ]]
174     then
175         echo ""
176         echo "Adding OpenOffice JARs to CLASSPATH..."
177         echo ""
178
179         OOO_JUH=$OOO_JAVA_PATH/juh.jar
180     OOO_JURT=$OOO_JAVA_PATH/jurt.jar
181     OOO_RIDL=$OOO_JAVA_PATH/ridl.jar
182
183         sudo echo "export CLASSPATH=\$CLASSPATH:$OOO_UNO_PATH:$OOO_JUH:$OOO_JURT
            ↪ :$OOO_RIDL" >> /etc/profile
184         sudo echo "export CLASSPATH=\$CLASSPATH:$OOO_UNOIL_PATH/unoil.jar" >> /
            ↪ etc/profile
185     sudo $OOO_UNO_PATH/unopkg add --shared $BSF4OOREXX_FRAMEWORK/Libraries/
        ↪ ScriptProviderForooRexx.oxt
186 fi
187
188 LORESULT=`lipo -info /Applications/LibreOffice.app/Contents/program/soffice |
    ↪ grep i386;`
189
190 if [[ -e $LO_APP_DIR && ! -z $LORESULT ]]
191     then
192         echo ""
193         echo "Adding LibreOffice JARs to CLASSPATH..."
194         echo ""
195
196         LO_JUH=$LO_JAVA_PATH/juh.jar
197     LO_JURT=$LO_JAVA_PATH/jurt.jar
```

```
198     LO_RIDL=$LO_JAVA_PATH/ridl.jar
199
200         sudo echo "export CLASSPATH=\$CLASSPATH:$LO_UNO_PATH:$LO_JUH:$LO_JURT:
                ↪ $LO_RIDL" >> /etc/profile
201         sudo echo "export CLASSPATH=\$CLASSPATH:$LO_UNOIL_PATH/unoil.jar" >> /
                ↪ etc/profile
202     sudo $LO_UNO_PATH/unopkg add --shared $BSF4OOREXX_FRAMEWORK/Libraries/
            ↪ ScriptProviderForooRexx.oxt
203 fi
204
205 echo ""
206 echo "Adding BSF4ooRexx JARs to CLASSPATH..."
207 echo ""
208
209 sudo echo "export CLASSPATH=\$CLASSPATH:$BSF4OOREXX_FRAMEWORK/Classes/bsf-rexx-
        ↪ engine.jar" >> /etc/profile
210 sudo echo "export CLASSPATH=\$CLASSPATH:$BSF4OOREXX_FRAMEWORK/Classes/bsf-v400
        ↪ -20090910.jar" >> /etc/profile
211
212 if [[ -e $OOO_APP_DIR && ! -z $OOORESULT ]]
213     then
214     echo ""
215     echo "Installing OOo macros..."
216     echo ""
217
218     sudo cp -r $OOO_SCRIPTLIB_SRC/javascript/* $OOO_SCRIPTLIB_DIR/javascript
219     sudo cp -r $OOO_SCRIPTLIB_SRC/python/* $OOO_SCRIPTLIB_DIR/python
220     sudo cp -r $OOO_SCRIPTLIB_SRC/oorexx $OOO_SCRIPTLIB_DIR/
221     # wu_tools muss per hand ins Framework kopiert werden!
222     sudo cp $BSF4OOREXX_FRAMEWORK/Commands/UNO_API_info.rxo $OOO_SCRIPTLIB_DIR/
            ↪ oorexx/wu_tools/
223 fi
224
225 if [[ -e $LO_APP_DIR && ! -z $LORESULT ]]
226     then
227     echo ""
228     echo "Installing LibreOffice macros..."
229     echo ""
230
231     sudo cp -r $OOO_SCRIPTLIB_SRC/javascript/* $LO_SCRIPTLIB_DIR/javascript
232     sudo cp -r $OOO_SCRIPTLIB_SRC/python/* $LO_SCRIPTLIB_DIR/python
233     sudo cp -r $OOO_SCRIPTLIB_SRC/oorexx $LO_SCRIPTLIB_DIR/
234     # wu_tools muss per hand ins Framework kopiert werden!
235     sudo cp $BSF4OOREXX_FRAMEWORK/Commands/UNO_API_info.rxo $LO_SCRIPTLIB_DIR/
            ↪ oorexx/wu_tools/
236 fi
237
238 echo ""
239 echo "Cleaning up..."
240 echo ""
241
242 sudo /System/Library/Frameworks/CoreServices.framework/Versions/Current/
        ↪ Frameworks/LaunchServices.framework/Versions/Current/Support/lsregister -
        ↪ kill -r -domain system -domain local -domain
```

Listing 52: The Mac OS X Installer's postflight script.

7.5.3 The Mac OS X Installer's 64 bit Install Script

The following listing contains the installation script for Mac OS X if 64 bit ooRexx and Java are used.

```
#!/bin/sh
# Name: BSF4ooRexx_install_postflight.sh
# Purpose: Install script for ooRexx and BSF4ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    limitations under the License.
# -------------------------------------------------------------------
VER=4
REL=0
MOD=3
PREFIX=/opt/ooRexx

echo $SCRIPT_NAME: linking files
echo $SCRIPT_NAME: linking libraries to /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.$VER.$REL.$MOD.dylib   /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.$VER.dylib       /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.a             /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.dylib         /usr/lib
ln -sf $PREFIX/lib/ooRexx/libhostemu.la            /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.$VER.$REL.$MOD.dylib     /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.$VER.dylib         /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.a               /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.dylib           /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexx.la              /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.$VER.$REL.$MOD.dylib  /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.$VER.dylib      /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.a            /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.dylib        /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxapi.la           /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.$VER.$REL.$MOD.dylib /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.$VER.dylib     /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.a           /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.dylib       /usr/lib
ln -sf $PREFIX/lib/ooRexx/librexxutil.la          /usr/lib
```

```
50 ln -sf $PREFIX/lib/ooRexx/librxmath.$VER.$REL.$MOD.dylib   /usr/lib
51 ln -sf $PREFIX/lib/ooRexx/librxmath.$VER.dylib       /usr/lib
52 ln -sf $PREFIX/lib/ooRexx/librxmath.a             /usr/lib
53 ln -sf $PREFIX/lib/ooRexx/librxmath.dylib         /usr/lib
54 ln -sf $PREFIX/lib/ooRexx/librxmath.la            /usr/lib
55 ln -sf $PREFIX/lib/ooRexx/librxregexp.$VER.$REL.$MOD.dylib /usr/lib
56 ln -sf $PREFIX/lib/ooRexx/librxregexp.$VER.dylib     /usr/lib
57 ln -sf $PREFIX/lib/ooRexx/librxregexp.a           /usr/lib
58 ln -sf $PREFIX/lib/ooRexx/librxregexp.dylib       /usr/lib
59 ln -sf $PREFIX/lib/ooRexx/librxregexp.la          /usr/lib
60 ln -sf $PREFIX/lib/ooRexx/librxsock.$VER.$REL.$MOD.dylib   /usr/lib
61 ln -sf $PREFIX/lib/ooRexx/librxsock.$VER.dylib       /usr/lib
62 ln -sf $PREFIX/lib/ooRexx/librxsock.a             /usr/lib
63 ln -sf $PREFIX/lib/ooRexx/librxsock.dylib         /usr/lib
64 ln -sf $PREFIX/lib/ooRexx/librxsock.la            /usr/lib
65 ln -sf $PREFIX/lib/ooRexx/librxunixsys.$VER.$REL.$MOD.dylib   /usr/lib
66 ln -sf $PREFIX/lib/ooRexx/librxunixsys.$VER.dylib       /usr/lib
67 ln -sf $PREFIX/lib/ooRexx/librxunixsys.a             /usr/lib
68 ln -sf $PREFIX/lib/ooRexx/librxunixsys.dylib         /usr/lib
69 ln -sf $PREFIX/lib/ooRexx/librxunixsys.la            /usr/lib
70
71 echo $SCRIPT_NAME: linking programs to /usr/bin
72 ln -sf $PREFIX/bin/csvStream.cls    /usr/bin
73 ln -sf $PREFIX/bin/mime.cls /usr/bin
74 ln -sf $PREFIX/bin/oorexx-config    /usr/bin
75 ln -sf $PREFIX/bin/rexx          /usr/bin
76 ln -sf $PREFIX/bin/rexx.cat      /usr/bin
77 ln -sf $PREFIX/bin/rexx.img      /usr/bin
78 ln -sf $PREFIX/bin/rexxc         /usr/bin
79 ln -sf $PREFIX/bin/rexximage     /usr/bin
80 ln -sf $PREFIX/bin/rxapi         /usr/bin
81 ln -sf $PREFIX/bin/rxftp.cls     /usr/bin
82 ln -sf $PREFIX/bin/rxqueue       /usr/bin
83 ln -sf $PREFIX/bin/rxregexp.cls  /usr/bin
84 ln -sf $PREFIX/bin/rxsubcom      /usr/bin
85 ln -sf $PREFIX/bin/smtp.cls /usr/bin
86 ln -sf $PREFIX/bin/socket.cls   /usr/bin
87 ln -sf $PREFIX/bin/socket.cls    /usr/bin
88 ln -sf $PREFIX/bin/streamsocket.cls /usr/bin
89 ln -sf $PREFIX/share/ooRexx/rexxtry.rex /usr/bin
90
91 echo $SCRIPT_NAME: linking man pages to /usr/share/man/man1
92 ln -sf $PREFIX/share/man/man1/oorexx-config.1 /usr/share/man/man1
93 ln -sf $PREFIX/share/man/man1/rexx.1          /usr/share/man/man1
94 ln -sf $PREFIX/share/man/man1/rexxc.1         /usr/share/man/man1
95 ln -sf $PREFIX/share/man/man1/rxmigrate.1     /usr/share/man/man1
96 ln -sf $PREFIX/share/man/man1/rxqueue.1       /usr/share/man/man1
97 ln -sf $PREFIX/share/man/man1/rxsubcom.1      /usr/share/man/man1
98
99 echo $SCRIPT_NAME: linking headers to /usr/include
100 ln -sf $PREFIX/include/oorexxapi.h        /usr/include
101 ln -sf $PREFIX/include/oorexxerrors.h     /usr/include
102 ln -sf $PREFIX/include/rexx.h             /usr/include
103 ln -sf $PREFIX/include/rexxapidefs.h      /usr/include
104 ln -sf $PREFIX/include/rexxapitypes.h     /usr/include
105 ln -sf $PREFIX/include/rexxplatformapis.h /usr/include
```

```
106 ln -sf $PREFIX/include/rexxplatformdefs.h /usr/include
107
108 echo $SCRIPT_NAME:  linking Launch Daemon to /Library/LaunchDaemons
109 ln -sf $PREFIX/bin/org.rexxla.oorexx.rxapid.plist /Library/LaunchDaemons
110
111 echo $SCRIPT_NAME:  stop/start rxapi Daemon
112 launchctl unload -w /Library/LaunchDaemons/org.rexxla.oorexx.rxapid.plist
113 killall rxapi
114 launchctl load -w /Library/LaunchDaemons/org.rexxla.oorexx.rxapid.plist
115
116 echo $SCRIPT_NAME: linking complete
117
118 BSF4OOREXX_FRAMEWORK=/opt/BSF4ooRexx
119
120 BIN_DIR=/usr/bin
121 JAVA_LIB_DIR=/usr/lib/java
122
123 OOO_SCRIPTLIB_SRC=$BSF4OOREXX_FRAMEWORK/Libraries/OOo_ScriptLibraries
124
125 OOO_APP_DIR=/Applications/OpenOffice.org.app
126 OOO_SCRIPTLIB_DIR=$OOO_APP_DIR/Contents/basis-link/share/Scripts
127 OOO_UNO_PATH=$OOO_APP_DIR/Contents/program
128 OOO_JAVA_PATH=$OOO_APP_DIR/Contents/basis-link/ure-link/share/java
129 OOO_UNOIL_PATH=$OOO_APP_DIR/Contents/basis-link/program/classes
130
131 LO_APP_DIR=/Applications/LibreOffice.app
132 LO_SCRIPTLIB_DIR=$LO_APP_DIR/Contents/basis-link/share/Scripts
133 LO_UNO_PATH=$LO_APP_DIR/Contents/program
134 LO_JAVA_PATH=$LO_APP_DIR/Contents/basis-link/ure-link/share/java
135 LO_UNOIL_PATH=$LO_APP_DIR/Contents/basis-link/program/classes
136
137 echo ""
138 echo "Creating symbolic links in $BIN_DIR..."
139 echo ""
140
141 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/BSF.CLS
       ↪ $BIN_DIR
142 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/BSF_OnTheFly.cls
       ↪ $BIN_DIR
143 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/rexxj2.sh
       ↪ $BIN_DIR
144 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/rexxj2.sh
       ↪ $BIN_DIR/rexxj.sh
145 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/rgf_util2.rex
       ↪ $BIN_DIR
146 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO.CLS
       ↪ $BIN_DIR
147 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_XINTERFACES.RXO
       ↪ $BIN_DIR
148 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_ReflectionUtil.cls
       ↪ $BIN_DIR
149 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_CREATE_INTERFACE_LIST.RXO
       ↪ $BIN_DIR
150 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Commands/UNO_API_info.rxo
       ↪ $BIN_DIR
151
```

```
152 echo ""
153 echo "Creating symbolic links in $JAVA_LIB_DIR..."
154 echo ""
155
156 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Libraries/libBSF4ooRexx.dylib $JAVA_LIB_DIR
157 sudo ln -sf $BSF4OOREXX_FRAMEWORK/Libraries/libBSF4ooRexx.dylib /usr/lib
158
159 echo ""
160 echo "Creating symbolic links in $BSF4OOREXX_FRAMEWORK..."
161 echo ""
162
163 sudo ln -sf /Applications/ooRexx.app/Contents/samples/BSF4ooRexx
    ↪ $BSF4OOREXX_FRAMEWORK/Samples
164 sudo ln -sf /Applications/ooRexx.app/Contents/documentation/BSF4ooRexx
    ↪ $BSF4OOREXX_FRAMEWORK/Information
165 sudo ln -sf /Applications/ooRexx.app/Contents/utilities $BSF4OOREXX_FRAMEWORK/
    ↪ Utilities
166
167 sudo mkdir -p $BSF4OOREXX_FRAMEWORK/Install
168 sudo ln -sf /Applications/ooRexx.app/Contents/Resources/one_click_uninstall.sh
    ↪ $BSF4OOREXX_FRAMEWORK/Install/uninstall.sh
169 sudo ln -sf /Applications/ooRexx.app/Contents/Resources/one_click_reinstall.sh
    ↪ $BSF4OOREXX_FRAMEWORK/Install/reinstall.sh
170
171 OOORESULT=`lipo -info /Applications/OpenOffice.org.app/Contents/program/soffice
    ↪ | grep x86_64;`
172
173 if [[ -e $OOO_APP_DIR && ! -z $OOORESULT ]]
174     then
175         echo ""
176         echo "Adding OpenOffice JARs to CLASSPATH..."
177         echo ""
178
179         OOO_JUH=$OOO_JAVA_PATH/juh.jar
180     OOO_JURT=$OOO_JAVA_PATH/jurt.jar
181     OOO_RIDL=$OOO_JAVA_PATH/ridl.jar
182
183         sudo echo "export CLASSPATH=\$CLASSPATH:$OOO_UNO_PATH:$OOO_JUH:$OOO_JURT
            ↪ :$OOO_RIDL" >> /etc/profile
184         sudo echo "export CLASSPATH=\$CLASSPATH:$OOO_UNOIL_PATH/unoil.jar" >> /
            ↪ etc/profile
185     sudo $OOO_UNO_PATH/unopkg add --shared $BSF4OOREXX_FRAMEWORK/Libraries/
        ↪ ScriptProviderForooRexx.oxt
186 fi
187
188 LORESULT=`lipo -info /Applications/LibreOffice.app/Contents/program/soffice |
    ↪ grep x86_64;`
189
190 if [[ -e $LO_APP_DIR && ! -z $LORESULT ]]
191     then
192         echo ""
193         echo "Adding LibreOffice JARs to CLASSPATH..."
194         echo ""
195
196         LO_JUH=$LO_JAVA_PATH/juh.jar
197     LO_JURT=$LO_JAVA_PATH/jurt.jar
```

```
198     LO_RIDL=$LO_JAVA_PATH/ridl.jar
199
200         sudo echo "export CLASSPATH=\$CLASSPATH:$LO_UNO_PATH:$LO_JUH:$LO_JURT:
                ↪ $LO_RIDL" >> /etc/profile
201         sudo echo "export CLASSPATH=\$CLASSPATH:$LO_UNOIL_PATH/unoil.jar" >> /
                ↪ etc/profile
202     sudo $LO_UNO_PATH/unopkg add --shared $BSF4OOREXX_FRAMEWORK/Libraries/
            ↪ ScriptProviderForooRexx.oxt
203 fi
204
205 echo ""
206 echo "Adding BSF4ooRexx JARs to CLASSPATH..."
207 echo ""
208
209 sudo echo "export CLASSPATH=\$CLASSPATH:$BSF4OOREXX_FRAMEWORK/Classes/bsf-rexx-
        ↪ engine.jar" >> /etc/profile
210 sudo echo "export CLASSPATH=\$CLASSPATH:$BSF4OOREXX_FRAMEWORK/Classes/bsf-v400
        ↪ -20090910.jar" >> /etc/profile
211
212 if [[ -e $OOO_APP_DIR && ! -z $OOORESULT ]]
213     then
214     echo ""
215     echo "Installing OOo macros..."
216     echo ""
217
218     sudo cp -r $OOO_SCRIPTLIB_SRC/javascript/* $OOO_SCRIPTLIB_DIR/javascript
219     sudo cp -r $OOO_SCRIPTLIB_SRC/python/* $OOO_SCRIPTLIB_DIR/python
220     sudo cp -r $OOO_SCRIPTLIB_SRC/oorexx $OOO_SCRIPTLIB_DIR/
221     # wu_tools muss per hand ins Framework kopiert werden!
222     sudo cp $BSF4OOREXX_FRAMEWORK/Commands/UNO_API_info.rxo $OOO_SCRIPTLIB_DIR/
            ↪ oorexx/wu_tools/
223 fi
224
225 if [[ -e $LO_APP_DIR && ! -z $LORESULT ]]
226     then
227     echo ""
228     echo "Installing LibreOffice macros..."
229     echo ""
230
231     sudo cp -r $OOO_SCRIPTLIB_SRC/javascript/* $LO_SCRIPTLIB_DIR/javascript
232     sudo cp -r $OOO_SCRIPTLIB_SRC/python/* $LO_SCRIPTLIB_DIR/python
233     sudo cp -r $OOO_SCRIPTLIB_SRC/oorexx $LO_SCRIPTLIB_DIR/
234     # wu_tools muss per hand ins Framework kopiert werden!
235     sudo cp $BSF4OOREXX_FRAMEWORK/Commands/UNO_API_info.rxo $LO_SCRIPTLIB_DIR/
            ↪ oorexx/wu_tools/
236 fi
237
238 echo ""
239 echo "Cleaning up..."
240 echo ""
241
242 sudo /System/Library/Frameworks/CoreServices.framework/Versions/Current/
        ↪ Frameworks/LaunchServices.framework/Versions/Current/Support/lsregister -
        ↪ kill -r -domain system -domain local -domain
```

Listing 53: The Mac OS X Installer's postflight script.

7.5.4 The Mac OS X Installer's Uninstall Script

The following listing contains the uninstall script for Mac OS X. It is described in detail in chapter 5.1.3.

```
#!/bin/sh

# Name: uninstall.sh
# Purpose: Uninstall script for ooRexx and BSF4ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
#    implied.
#    See the License for the specific language governing permissions and
#    limitations under the License.
# -------------------------------------------------------------------

# First uninstalls ooRexx and BSF4ooRexx and then reinstalls them
if [ $1 = "reinstall" ]
    then
        osascript -e "do shell script \"cp -r /Applications/ooRexx.app/Contents/
            ↪ sources /tmp/; cp /Applications/ooRexx.app/Contents/Resources/
            ↪ ooRexx_uninstall_preflight.sh /tmp/; cp /Applications/ooRexx.app/
            ↪ Contents/Resources/BSF4ooRexx_uninstall_preflight.sh /tmp/; sh /
            ↪ tmp/ooRexx_uninstall_preflight.sh reinstall\" with administrator
            ↪ privileges"
fi

# Just uninstalls ooRexx and BSF4ooRexx
if [ $1 = "uninstall" ]
    then osascript -e "do shell script \"cp /Applications/ooRexx.app/Contents/
        ↪ Resources/ooRexx_uninstall_preflight.sh /tmp/; cp /Applications/ooRexx
        ↪ .app/Contents/Resources/BSF4ooRexx_uninstall_preflight.sh /tmp/; sh /
        ↪ tmp/ooRexx_uninstall_preflight.sh uninstall\" with administrator
        ↪ privileges"
fi
```

Listing 54: The Mac OS X uninstall script.

7.5.5 The Mac OS X Installer's 32 bit BSF4ooRexx Uninstall Script

The following listing contains the 32 bit uninstall preflight script for Mac OS X. It is described in detail in chapter 5.1.3.

```
#!/bin/sh

# Name: BSF4ooRexx_uninstall_preflight.sh
# Purpose: Uninstall script for BSF4ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    you may not use this file except in compliance with the License.
#    You may obtain a copy of the License at
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#
#    Unless required by applicable law or agreed to in writing, software
#    distributed under the License is distributed on an "AS IS" BASIS,
#    WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
#    implied.
#    See the License for the specific language governing permissions and
#    limitations under the License.
# -------------------------------------------------------------------

BSF4OOREXX_FRAMEWORK=/opt/BSF4ooRexx

BIN_DIR=/usr/bin
JAVA_LIB_DIR=/usr/lib/java

OOO_APP_DIR=/Applications/OpenOffice.org.app
OOO_SCRIPTLIB_DIR=$OOO_APP_DIR/Contents/basis-link/share/Scripts
OOO_UNO_PATH=$OOO_APP_DIR/Contents/program
OOO_JAVA_PATH=$OOO_APP_DIR/Contents/basis-link/ure-link/share/java
OOO_UNOIL_PATH=$OOO_APP_DIR/Contents/basis-link/program/classes

LO_APP_DIR=/Applications/LibreOffice.app
LO_SCRIPTLIB_DIR=$LO_APP_DIR/Contents/basis-link/share/Scripts
LO_UNO_PATH=$LO_APP_DIR/Contents/program
LO_JAVA_PATH=$LO_APP_DIR/Contents/basis-link/ure-link/share/java
LO_UNOIL_PATH=$LO_APP_DIR/Contents/basis-link/program/classes

echo ""
echo "Removing executable files from $BIN_DIR..."
echo ""

sudo rm $BIN_DIR/BSF.CLS
sudo rm $BIN_DIR/BSF_OnTheFly.cls
sudo rm $BIN_DIR/rexxj2.sh
sudo rm $BIN_DIR/rexxj.sh
```

```
50 sudo rm $BIN_DIR/rgf_util2.rex
51 sudo rm $BIN_DIR/UNO.CLS
52 sudo rm $BIN_DIR/UNO_API_info.rxo
53 sudo rm $BIN_DIR/UNO_CREATE_INTERFACE_LIST.RXO
54 sudo rm $BIN_DIR/UNO_ReflectionUtil.cls
55 sudo rm $BIN_DIR/UNO_XINTERFACES.RXO
56
57 echo ""
58 echo "Removing framework directory $BSF4OOREXX_FRAMEWORK..."
59 echo ""
60
61 sudo rm -rf $BSF4OOREXX_FRAMEWORK
62
63 echo ""
64 echo "Removing dynamic link library from $JAVA_LIB_DIR..."
65 echo ""
66
67 sudo rm $JAVA_LIB_DIR/libBSF4ooRexx.dylib
68 sudo rm /usr/lib/libBSF4ooRexx.dylib
69
70 echo ""
71 echo "Removing CLASSPATH settings for OpenOffice..."
72 echo ""
73
74 OOORESULT=`lipo -info /Applications/OpenOffice.org.app/Contents/program/soffice
      ↪ | grep i386;`
75
76 if [[ -e $OOO_APP_DIR && ! -z $OOORESULT ]]
77     then
78         sudo grep -vx "export CLASSPATH=\$CLASSPATH:/Applications/OpenOffice.org
              ↪ .app/Contents/program:/Applications/OpenOffice.org.app/Contents/
              ↪ basis-link/ure-link/share/java/juh.jar:/Applications/OpenOffice.
              ↪ org.app/Contents/basis-link/ure-link/share/java/jurt.jar:/
              ↪ Applications/OpenOffice.org.app/Contents/basis-link/ure-link/share
              ↪ /java/ridl.jar
79 export CLASSPATH=\$CLASSPATH:/Applications/OpenOffice.org.app/Contents/basis-
      ↪ link/program/classes/unoil.jar" /etc/profile > /tmp/profile.tmp
80         sudo $OOO_UNO_PATH/unopkg remove --shared ScriptProviderForooRexx.oxt
81         sudo rm -rf $OOO_SCRIPTLIB_DIR/javascript/HowtoCreateApiInfo
82         sudo rm -rf $OOO_SCRIPTLIB_DIR/python/pythonSamples
83         sudo rm -rf $OOO_SCRIPTLIB_DIR/oorexx/ooRexx_examples
84         sudo rm -rf $OOO_SCRIPTLIB_DIR/oorexx/wu_tools
85 fi
86
87 sudo cp /tmp/profile.tmp /etc/profile
88 sudo rm /tmp/profile.tmp
89
90 echo ""
91 echo "Removing CLASSPATH settings for LibreOffice..."
92 echo ""
93
94 LORESULT=`lipo -info /Applications/LibreOffice.app/Contents/program/soffice |
      ↪ grep i386;`
95
96 if [[ -e $LO_APP_DIR && ! -z $LORESULT ]]
97     then
```

```
98          sudo grep -vx "export CLASSPATH=\$CLASSPATH:/Applications/LibreOffice.
               ↪ app/Contents/program:/Applications/LibreOffice.app/Contents/basis-
               ↪ link/ure-link/share/java/juh.jar:/Applications/LibreOffice.app/
               ↪ Contents/basis-link/ure-link/share/java/jurt.jar:/Applications/
               ↪ LibreOffice.app/Contents/basis-link/ure-link/share/java/ridl.jar
99  export CLASSPATH=\$CLASSPATH:/Applications/LibreOffice.app/Contents/basis-link/
       ↪ program/classes/unoil.jar" /etc/profile > /tmp/profile.tmp
100         sudo $LO_UNO_PATH/unopkg remove --shared ScriptProviderForooRexx.oxt
101         sudo rm -rf $LO_SCRIPTLIB_DIR/javascript/HowtoCreateApiInfo
102         sudo rm -rf $LO_SCRIPTLIB_DIR/python/pythonSamples
103         sudo rm -rf $LO_SCRIPTLIB_DIR/oorexx/ooRexx_examples
104         sudo rm -rf $LO_SCRIPTLIB_DIR/oorexx/wu_tools
105 fi
106
107 sudo cp /tmp/profile.tmp /etc/profile
108 sudo rm /tmp/profile.tmp
109
110 echo ""
111 echo "Removing CLASSPATH settings for BSF4ooRexx..."
112 echo ""
113
114 sudo grep -vx "export CLASSPATH=\$CLASSPATH:/opt/BSF4ooRexx/Classes/bsf-rexx-
       ↪ engine.jar
115 export CLASSPATH=\$CLASSPATH:/opt/BSF4ooRexx/Classes/bsf-v400-20090910.jar" /etc
       ↪ /profile > /tmp/profile.tmp
116
117 sudo cp /tmp/profile.tmp /etc/profile
118 sudo rm /tmp/profile.tmp
119
120 echo ""
121 echo "Removing Application Bundle..."
122 echo ""
123
124 sudo rm -rf /Applications/ooRexx.app
125
126 echo ""
127 echo "Removing the ooRexx directory..."
128 echo ""
129
130 sudo rm -rf /opt/ooRexx
131
132
133 echo ""
134 echo "Cleaning up..."
135 echo ""
136
137 sudo /System/Library/Frameworks/CoreServices.framework/Versions/Current/
       ↪ Frameworks/LaunchServices.framework/Versions/Current/Support/lsregister -
       ↪ kill -r -domain system -domain local -domain
138
139 sudo rm -rf /tmp/ooRexx_uninstall_preflight.sh
140
141 if [ $1 = "uninstall" ]
142     then
143         osascript -e 'tell app "System Events" to display dialog "ooRexx and
               ↪ BSF4ooRexx have successfully been removed!" buttons "OK" default
```

```
            ↪ button 1 with title "Success!" with icon 1'
fi

if [ $1 = "reinstall" ]
    then
        sudo sh /tmp/sources/reinstall.sh
        sudo cp -r /tmp/sources /Applications/ooRexx.app/Contents/
        sudo rm -rf /tmp/sources
fi

sudo rm $0
```

Listing 55: The BSF4ooRexx uninstall script.

7.5.6 The Mac OS X Installer's 64 bit BSF4ooRexx Uninstall Script

The following listing contains the 64 bit uninstall preflight script for Mac OS X.

```
#!/bin/sh

# Name: BSF4ooRexx_uninstall_preflight.sh
# Purpose: Uninstall script for BSF4ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    limitations under the License.
# -------------------------------------------------------------------

BSF4OOREXX_FRAMEWORK=/opt/BSF4ooRexx

BIN_DIR=/usr/bin
JAVA_LIB_DIR=/usr/lib/java

OOO_APP_DIR=/Applications/OpenOffice.org.app
OOO_SCRIPTLIB_DIR=$OOO_APP_DIR/Contents/basis-link/share/Scripts
OOO_UNO_PATH=$OOO_APP_DIR/Contents/program
OOO_JAVA_PATH=$OOO_APP_DIR/Contents/basis-link/ure-link/share/java
OOO_UNOIL_PATH=$OOO_APP_DIR/Contents/basis-link/program/classes

LO_APP_DIR=/Applications/LibreOffice.app
LO_SCRIPTLIB_DIR=$LO_APP_DIR/Contents/basis-link/share/Scripts
LO_UNO_PATH=$LO_APP_DIR/Contents/program
LO_JAVA_PATH=$LO_APP_DIR/Contents/basis-link/ure-link/share/java
LO_UNOIL_PATH=$LO_APP_DIR/Contents/basis-link/program/classes

echo ""
echo "Removing executable files from $BIN_DIR..."
echo ""

sudo rm $BIN_DIR/BSF.CLS
sudo rm $BIN_DIR/BSF_OnTheFly.cls
sudo rm $BIN_DIR/rexxj2.sh
sudo rm $BIN_DIR/rexxj.sh
sudo rm $BIN_DIR/rgf_util2.rex
sudo rm $BIN_DIR/UNO.CLS
```

```
52 sudo rm $BIN_DIR/UNO_API_info.rxo
53 sudo rm $BIN_DIR/UNO_CREATE_INTERFACE_LIST.RXO
54 sudo rm $BIN_DIR/UNO_ReflectionUtil.cls
55 sudo rm $BIN_DIR/UNO_XINTERFACES.RXO
56
57 echo ""
58 echo "Removing framework directory $BSF4OOREXX_FRAMEWORK..."
59 echo ""
60
61 sudo rm -rf $BSF4OOREXX_FRAMEWORK
62
63 echo ""
64 echo "Removing dynamic link library from $JAVA_LIB_DIR..."
65 echo ""
66
67 sudo rm $JAVA_LIB_DIR/libBSF4ooRexx.dylib
68 sudo rm /usr/lib/libBSF4ooRexx.dylib
69
70 echo ""
71 echo "Removing CLASSPATH settings for OpenOffice..."
72 echo ""
73
74 OOORESULT=`lipo -info /Applications/OpenOffice.org.app/Contents/program/soffice
    ↪ | grep x86_64;`
75
76 if [[ -e $OOO_APP_DIR && ! -z $OOORESULT ]]
77     then
78         sudo grep -vx "export CLASSPATH=\$CLASSPATH:/Applications/OpenOffice.org
              ↪ .app/Contents/program:/Applications/OpenOffice.org.app/Contents/
              ↪ basis-link/ure-link/share/java/juh.jar:/Applications/OpenOffice.
              ↪ org.app/Contents/basis-link/ure-link/share/java/jurt.jar:/
              ↪ Applications/OpenOffice.org.app/Contents/basis-link/ure-link/share
              ↪ /java/ridl.jar
79 export CLASSPATH=\$CLASSPATH:/Applications/OpenOffice.org.app/Contents/basis-
    ↪ link/program/classes/unoil.jar" /etc/profile > /tmp/profile.tmp
80         sudo $OOO_UNO_PATH/unopkg remove --shared ScriptProviderForooRexx.oxt
81         sudo rm -rf $OOO_SCRIPTLIB_DIR/javascript/HowtoCreateApiInfo
82         sudo rm -rf $OOO_SCRIPTLIB_DIR/python/pythonSamples
83         sudo rm -rf $OOO_SCRIPTLIB_DIR/oorexx/ooRexx_examples
84         sudo rm -rf $OOO_SCRIPTLIB_DIR/oorexx/wu_tools
85 fi
86
87 sudo cp /tmp/profile.tmp /etc/profile
88 sudo rm /tmp/profile.tmp
89
90 echo ""
91 echo "Removing CLASSPATH settings for LibreOffice..."
92 echo ""
93
94 LORESULT=`lipo -info /Applications/LibreOffice.app/Contents/program/soffice |
    ↪ grep x86_64;`
95
96 if [[ -e $LO_APP_DIR && ! -z $LORESULT ]]
97     then
98         sudo grep -vx "export CLASSPATH=\$CLASSPATH:/Applications/LibreOffice.
              ↪ app/Contents/program:/Applications/LibreOffice.app/Contents/basis-
```

```
            ↪ link/ure-link/share/java/juh.jar:/Applications/LibreOffice.app/
            ↪ Contents/basis-link/ure-link/share/java/jurt.jar:/Applications/
            ↪ LibreOffice.app/Contents/basis-link/ure-link/share/java/ridl.jar
99  export CLASSPATH=\$CLASSPATH:/Applications/LibreOffice.app/Contents/basis-link/
        ↪ program/classes/unoil.jar" /etc/profile > /tmp/profile.tmp
100         sudo $LO_UNO_PATH/unopkg remove --shared ScriptProviderForooRexx.oxt
101         sudo rm -rf $LO_SCRIPTLIB_DIR/javascript/HowtoCreateApiInfo
102         sudo rm -rf $LO_SCRIPTLIB_DIR/python/pythonSamples
103         sudo rm -rf $LO_SCRIPTLIB_DIR/oorexx/ooRexx_examples
104         sudo rm -rf $LO_SCRIPTLIB_DIR/oorexx/wu_tools
105 fi
106
107 sudo cp /tmp/profile.tmp /etc/profile
108 sudo rm /tmp/profile.tmp
109
110 echo ""
111 echo "Removing CLASSPATH settings for BSF4ooRexx..."
112 echo ""
113
114 sudo grep -vx "export CLASSPATH=\$CLASSPATH:/opt/BSF4ooRexx/Classes/bsf-rexx-
        ↪ engine.jar
115 export CLASSPATH=\$CLASSPATH:/opt/BSF4ooRexx/Classes/bsf-v400-20090910.jar" /etc
        ↪ /profile > /tmp/profile.tmp
116
117 sudo cp /tmp/profile.tmp /etc/profile
118 sudo rm /tmp/profile.tmp
119
120 echo ""
121 echo "Removing Application Bundle..."
122 echo ""
123
124 sudo rm -rf /Applications/ooRexx.app
125
126 echo ""
127 echo "Removing the ooRexx directory..."
128 echo ""
129
130 sudo rm -rf /opt/ooRexx
131
132
133 echo ""
134 echo "Cleaning up..."
135 echo ""
136
137 sudo /System/Library/Frameworks/CoreServices.framework/Versions/Current/
        ↪ Frameworks/LaunchServices.framework/Versions/Current/Support/lsregister -
        ↪ kill -r -domain system -domain local -domain
138
139 sudo rm -rf /tmp/ooRexx_uninstall_preflight.sh
140
141 if [ $1 = "uninstall" ]
142     then
143         osascript -e 'tell app "System Events" to display dialog "ooRexx and
            ↪ BSF4ooRexx have successfully been removed!" buttons "OK" default
            ↪ button 1 with title "Success!" with icon 1'
144 fi
```

```
145
146 if [ $1 = "reinstall" ]
147     then
148         sudo sh /tmp/sources/reinstall.sh
149         sudo cp -r /tmp/sources /Applications/ooRexx.app/Contents/
150         sudo rm -rf /tmp/sources
151 fi
152
153 sudo rm $0
```

Listing 56: The BSF4ooRexx uninstall script.

7.5.7 The Mac OS X Installer's ooRexx Uninstall Script

The following listing contains uninstall script for ooRexx. It is described in detail in chapter 5.1.3.

```
#!/bin/sh

# Name: ooRexx_uninstall_preflight.sh
# Purpose: Uninstall script for ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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# -------------------------------------------------------------------

echo $SCRIPT_NAME: Removing links from /usr/lib.
rm -f /usr/lib/libhostemu.*
rm -f /usr/lib/librexx.*
rm -f /usr/lib/librexxapi.*
rm -f /usr/lib/librexxutil.*
rm -f /usr/lib/librxmath.*
rm -f /usr/lib/librxregexp.*
rm -f /usr/lib/librxsock.*
rm -f /usr/lib/librxunixsys.*

echo $SCRIPT_NAME: Removing links from /usr/bin.
rm -f /usr/bin/csvStream.cls
rm -f /usr/bin/mime.cls
rm -f /usr/bin/oorexx-config
rm -f /usr/bin/org.rexxla.oorexx.rxapid.plist
rm -f /usr/bin/rexx
rm -f /usr/bin/rexx.cat
rm -f /usr/bin/rexx.img
rm -f /usr/bin/rexxc
rm -f /usr/bin/rexximage
rm -f /usr/bin/rexxtry.rex
rm -f /usr/bin/rxdelipc
rm -f /usr/bin/rxftp.cls
rm -f /usr/bin/rxmigrate
rm -f /usr/bin/rxqueue
```

```
50 rm -f /usr/bin/rxregexp.cls
51 rm -f /usr/bin/rxsubcom
52 rm -f /usr/bin/smtp.cls
53 rm -f /usr/bin/socket.cls
54 rm -f /usr/bin/streamsocket.cls
55
56 echo $SCRIPT_NAME: Removing man pages from /usr/share/man/man1
57 rm -f /usr/share/man/man1/oorexx-config.1
58 rm -f /usr/share/man/man1/rexx.1
59 rm -f /usr/share/man/man1/rexxc.1
60 rm -f /usr/share/man/man1/rxmigrate.1
61 rm -f /usr/share/man/man1/rxqueue.1
62 rm -f /usr/share/man/man1/rxsubcom.1
63
64 echo $SCRIPT_NAME: Removing headers from /usr/include
65 rm -f /usr/include/oorexxapi.h
66 rm -f /usr/include/oorexxerrors.h
67 rm -f /usr/include/rexx.h
68 rm -f /usr/include/rexxapidefs.h
69 rm -f /usr/include/rexxapitypes.h
70 rm -f /usr/include/rexxplatformapis.h
71 rm -f /usr/include/rexxplatformdefs.h
72
73 echo $SCRIPT_NAME: Shutting down rxapi daemon
74 launchctl unload -w /Library/LaunchDaemons/org.rexxla.oorexx.rxapid.plist
75 killall rxapi
76
77 echo $SCRIPT_NAME: Removing rexx LaunchDeamon
78 rm -f /Library/LaunchDeamons/org.rexxla.oorexx.rxapid.plist
79 rm -f /usr/bin/rxapi
80
81 echo $SCRIPT_NAME: Removal of prior version of ooRexx complete.
82
83 sh /tmp/BSF4ooRexx_uninstall_preflight.sh $1
```

Listing 57: The Installer's ooRexx uninstall script.

7.5.8 The Mac OS X Installer's Reinstall Script

The following listing contains the reinstall script for Mac OS X. It is described in detail in chapter 5.1.3.

```
#!/bin/sh

# Name: reinstall.sh
# Purpose: Reinstall script for ooRexx and BSF4ooRexx
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    limitations under the License.
# -------------------------------------------------------------------

SOURCES_DIR=/tmp/sources

BSF4OOREXX_FRAMEWORK=/opt/BSF4ooRexx

APP_PATH=/Applications

echo ""
echo "Reinstalling ooRexx..."
echo ""

sudo cp -r $SOURCES_DIR/ooRexx /opt/

echo ""
echo "Reinstalling BSF4ooRexx..."
echo ""

sudo cp -r $SOURCES_DIR/BSF4ooRexx /opt/

echo ""
echo "Reinstalling Application..."
echo ""

sudo cp -r $SOURCES_DIR/ooRexx.app $APP_PATH/

echo ""
```

```
50 echo "Running scripts..."
51 echo ""
52
53 sudo sh $SOURCES_DIR/BSF4ooRexx_install_postflight.sh
54
55 osascript -e 'tell app "System Events" to display dialog "ooRexx and BSF4ooRexx
    ↪ have successfully been reinstalled!" buttons "OK" default button 1 with
    ↪ title "Success!" with icon 1'
```

Listing 58: The Mac OS X reinstall script.

7.5.9 Makefile for Automatically Creating the Mac OS X Packages

The following listing contains the Makefile for Mac OS X.

```
# Name: Makefile
# Purpose: Makefile for the BSF4ooRexx MacOSX packages.
# Author: Juergen Hesse, Manuel Paar
# Version: 1.0
#
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#    implied.
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#    limitations under the License.
# -------------------------------------------------------------------

ICEBERG32 = ~/Projekte/Iceberg\ opt
ICEBERG64 = ~/Projekte/Iceberg\ opt\ 64

ICEBERG_SOURCES32 = $(ICEBERG32)/sources
ICEBERG_SOURCES64 = $(ICEBERG64)/sources

ICEBERG_PROJECTNAME = ooRexxWithBSF4ooRexx

# Define source directories
SRC_BSF = BSF4ooRexx_install
SRC_DYLIB = $(SRC_BSF)/bsf4oorexx/install
SRC_REXXJ2 = .

SRC_OOREXX32 = ooRex-4-1-1-macosx-32-20110605
SRC_OOREXX64 = ooRex-4-1-1-macosx-64-20110605

# Define destination directories
DST = tmp/BSF4ooRexx
CLASSES = $(DST)/Classes
COMMANDS = $(DST)/Commands
LIBRARIES = $(DST)/Libraries

default:
    @echo
    @echo "Usage:"
    @echo "make {build32|build64}"
    @echo
    @echo "build32:"
    @echo "   Builds the framework with 32 bit libraries."
```

```
52      @echo "build64:"
53      @echo "   Builds the framework with 64 bit libraries."
54      @echo
55      @echo
56
57  build:
58      @echo "-----------------------------------"
59      @echo "Building..."
60      @echo "-----------------------------------"
61      @echo
62      @echo "Creating directory structure..."
63      @echo
64
65      mkdir -p $(CLASSES)
66      mkdir -p $(COMMANDS)
67      mkdir -p $(LIBRARIES)
68
69      @echo
70      @echo "Copying classes (jars)..."
71      @echo
72
73      cp $(SRC_BSF)/bsf4oorexx/bsf-rexx-engine.jar   $(CLASSES)/
74      cp $(SRC_BSF)/bsf4oorexx/bsf-v400-20090910.jar $(CLASSES)/
75
76      @echo
77      @echo "Copying executable files..."
78      @echo
79
80      cp $(SRC_BSF)/bsf4oorexx/BSF.CLS                          $(COMMANDS)/
81      cp $(SRC_BSF)/bsf4oorexx/BSF_OnTheFly.cls                 $(COMMANDS)/
82      cp $(SRC_BSF)/bsf4oorexx/rgf_util2.rex                    $(COMMANDS)/
83      cp $(SRC_BSF)/bsf4oorexx/UNO.CLS                          $(COMMANDS)/
84      cp $(SRC_BSF)/bsf4oorexx/UNO_API_info.rxo                 $(COMMANDS)/
85      cp $(SRC_BSF)/bsf4oorexx/UNO_CREATE_INTERFACE_LIST.RXO    $(COMMANDS)/
86      cp $(SRC_BSF)/bsf4oorexx/UNO_ReflectionUtil.cls           $(COMMANDS)/
87      cp $(SRC_BSF)/bsf4oorexx/UNO_XINTERFACES.RXO              $(COMMANDS)/
88
89      @echo
90      @echo "Copying OpenOffice libraries..."
91      @echo
92
93      cp $(SRC_BSF)/bsf4oorexx/install/ScriptProviderForooRexx.oxt $(LIBRARIES)/
94      cp -r $(SRC_BSF)/bsf4oorexx/install/OOo_ScriptLibraries $(LIBRARIES)/
95
96  build32: build
97
98      @echo
99      @echo "Copying 32 bit files..."
100     @echo
101
102     cp $(SRC_DYLIB)/32/libBSF4ooRexx.dylib $(LIBRARIES)/libBSF4ooRexx.dylib
103     cp $(SRC_REXXJ2)/rexxj2_32.sh $(COMMANDS)/rexxj2.sh
104
105     @echo
106     @echo "Copying 32 bit ooRexx..."
107     @echo
```

```
108
109     cp -r $(SRC_OOREXX32)/* $(ICEBERG_SOURCES32)/ooRexx/
110
111     @echo
112     @echo "Copying BSF4ooRexx..."
113     @echo
114
115     cp -r $(DST) $(ICEBERG_SOURCES32)/
116     rm -rf $(DST)
117
118     @echo
119     @echo "Creating the installer package..."
120     @echo
121
122     freeze -v $(ICEBERG32)/$(ICEBERG_PROJECTNAME)/$(ICEBERG_PROJECTNAME).
           ↪ packproj
123
124     @echo
125     @echo "Done."
126     @echo
127
128
129 build64: build
130
131     @echo
132     @echo "Copying 64 bit files..."
133     @echo
134
135     cp $(SRC_DYLIB)/64/libBSF4ooRexx.dylib $(LIBRARIES)/libBSF4ooRexx.dylib
136     cp $(SRC_REXXJ2)/rexxj2_64.sh $(COMMANDS)/rexxj2.sh
137
138     @echo
139     @echo "Copying 64 bit ooRexx..."
140     @echo
141
142     cp -r $(SRC_OOREXX64)/* $(ICEBERG_SOURCES64)/ooRexx/
143
144     @echo
145     @echo "Copying BSF4ooRexx..."
146     @echo
147
148     cp -r $(DST) $(ICEBERG_SOURCES64)/
149     rm -rf $(DST)
150
151     @echo
152     @echo "Creating the installer package..."
153     @echo
154
155     freeze -v $(ICEBERG64)/$(ICEBERG_PROJECTNAME)/$(ICEBERG_PROJECTNAME).
           ↪ packproj
156
157     @echo
158     @echo "Done."
159     @echo
```

Listing 59: The Mac OS X Makefile.

7.6 List of Figures

1 HM NSIS Editor. 14
2 The BSF4ooRexx setup icon. 15
3 HM NIS Editor's "compile script" button. 23
4 Welcome dialog of the BSF4ooRexx installer for Windows. 24
5 Licence agreement of the BSF4ooRexx installer for Windows. 24
6 Installing window of the BSF4ooRexx installer for Windows. 25
7 Finish dialog of the BSF4ooRexx installer for Windows. 25
8 Uninstall message of the BSF4ooRexx reinstaller for Windows. 26
9 Installing window of the BSF4ooRexx reinstaller for Windows. 27
10 Finish dialog of the BSF4ooRexx reinstaller for Windows. 27
11 Uninstall message of the BSF4ooRexx uninstaller for Windows. 28
12 Uninstalling window of the BSF4ooRexx uninstaller for Windows. 28
13 Finish message of the BSF4ooRexx uninstaller for Windows. 29
14 Debian package structure. 32
15 Installing the Debian package by using the Ubuntu Software Center. 37
16 Authentication for the installation. 38
17 Installation process. 39
18 The Installed-Software-Section after installation. 40
19 Reinstalling BSF4ooRexx from the menu. 41
20 Reinstalling BSF4ooRexx using Ubuntu Software Center. 42
21 Uninstalling the package. 43
22 Deinstallation progress. 43
23 Deinstallation finished. 44
24 The structure of the rpmbuild directory. 48
25 Installing the RPM package by double-clicking. 54
26 Authentication for the installation. 55
27 Installation progress. 55

28 The Installed-Software-Section after installation. 56
29 Reinstalling BSF4ooRexx from the menu. 57
30 Uninstalling the package. 58
31 Authentication for uninstalling the BSF4ooRexx package. 58
32 Deinstallation progress. 59
33 The sources folder. 61
34 The `BSF4ooRexx` folder. 62
35 The `ooRexx` folder. 63
36 The contents of `ooRexx.app`. 64
37 The Iceberg main window. 67
38 The Iceberg project settings. 69
39 The Iceberg project's document settings. 71
40 The Iceberg project's script settings. 72
41 Defining the installer's requirements. 73
42 The Iceberg project's plugin settings. 74
43 The Iceberg project's file settings. 75
44 Welcome dialog of the ooRexx and BSF4ooRexx installer for Mac OS X. . . 77
45 Licence agreement of the ooRexx and BSF4ooRexx installer for Mac OS X. . 77
46 Installation target of the ooRexx and BSF4ooRexx installer for Mac OS X. . . 78
47 Installation summary of the ooRexx and BSF4ooRexx installer for Mac OS X. 78
48 Installing window of the ooRexx and BSF4ooRexx installer for Mac OS X. . . 79
49 Finish dialog of the ooRexx and BSF4ooRexx installer for Mac OS X. 79
50 Reinstall message of the ooRexx and BSF4ooRexx reinstaller for Mac OS X. 80
51 Finish dialog of the ooRexx and BSF4ooRexx reinstaller for Mac OS X. . . . 80
52 Uninstall message of the ooRexx and BSF4ooRexx uninstaller for Mac OS X. 80
53 Finish dialog of the ooRexx and BSF4ooRexx uninstaller for Mac OS X. . . . 81

7.7 List of Tables

1 Predefined variables (macros) for accessing the RPM build directory structure, as described in [21]. 48

2 Stages and their functions, as described in [21]. 52

7.8 List of Listings

1 Usage of `OutFile` and `InstallDir` in NSIS. 11
2 The section `Uninstall`. 12
3 A NSIS callback-function. 12
4 Global variables in NSIS. 12
5 Storing auxiliary variables in NSIS. 13
6 Removing auxiliary variables in NSIS. 13
7 NSIS include- and compressor settings. 14
8 Definition of constants in NSIS. 15
9 Defining the installation wizard's pages. 15
10 Defining the uninstaller's progress page. 15
11 Setting the installer's language. 16
12 Defining additional attributes. 16
13 The `onInit` function. 16
14 The `FindJava` function. 18
15 The "`BSF4ooRexx`" `SEC01` section (selection). 18
16 The `Uninstall` section. 21
17 The `JavaSuccess` function. 22
18 Uninstalling the existing version of BSF4ooRexx. 22
19 The original line. 22
20 The replacement for the original line. 23
21 Installing BSF4ooRexx in silent mode. 26
22 Reinstalling BSF4ooRexx in silent mode. 28
23 Uninstalling BSF4ooRexx in silent mode. 29
24 The control file. 32
25 Making the `postinst` script executable. 36
26 Building the Debian package. 36
27 Installing the package from the command line. 40

28 Uninstalling the package from the command line. 44
29 Installing the required utilities. 47
30 Setup of the basic package structure. 47
31 Creating a basic spec file. 49
32 The bsf4oorexx.spec file. 49
33 Building the RPM package. 53
34 Installing the package from the command line. 56
35 Uninstalling the package from the command line. 59
36 The NSIS Installation Script for 32bit Windows. 85
37 The NSIS Installation Script for 64bit Windows. 100
38 The NSIS Reinstall Script for 32bit Windows. 115
39 The NSIS Reinstall Script for 64 bit Windows 130
40 Java version check script. 144
41 ooRexx bitness check script. 145
42 The GetBSFVersion script. 146
43 The cmd file for automatically creating the Windows Installers. 147
44 The Debian package control file. 148
45 The Debian preinst file. 149
46 The Debian package postinst file. 150
47 The Debian package prerm file. 151
48 Shell script for creating the Debian Package. 152
49 The RPM package's SPECS file. 154
50 Shell script for creating the RPM package. 156
51 The ooRexx menu bar. 158
52 The Mac OS X Installer's postflight script. 163
53 The Mac OS X Installer's postflight script. 168
54 The Mac OS X uninstall script. 173
55 The BSF4ooRexx uninstall script. 174

56 The BSF4ooRexx uninstall script. 178
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